CMPT 732-G200 Practices for Visual Computing

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SVD

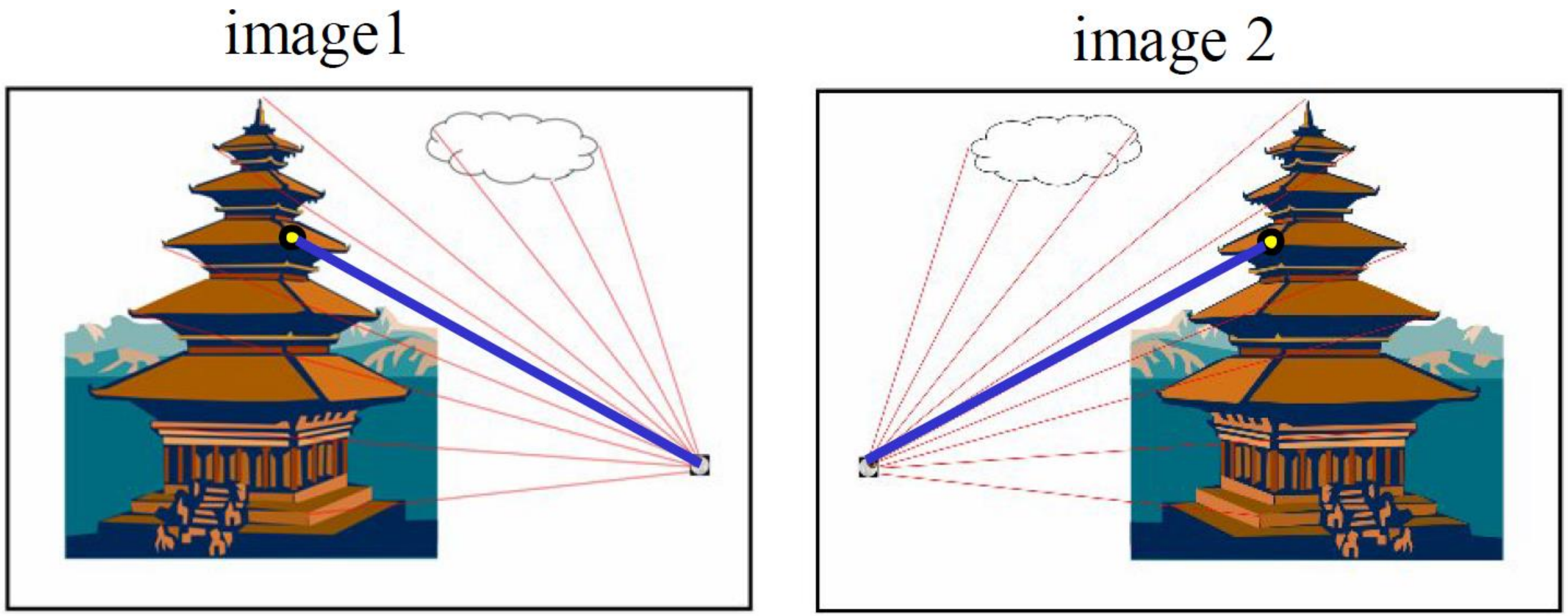
- We learned about Singular Value Decomposition in the previous class.
- We will learn about one of the applications in which SVD comes handy.

Eight Point Algorithm

- We will learn about an algorithm that is used in 3D reconstruction.

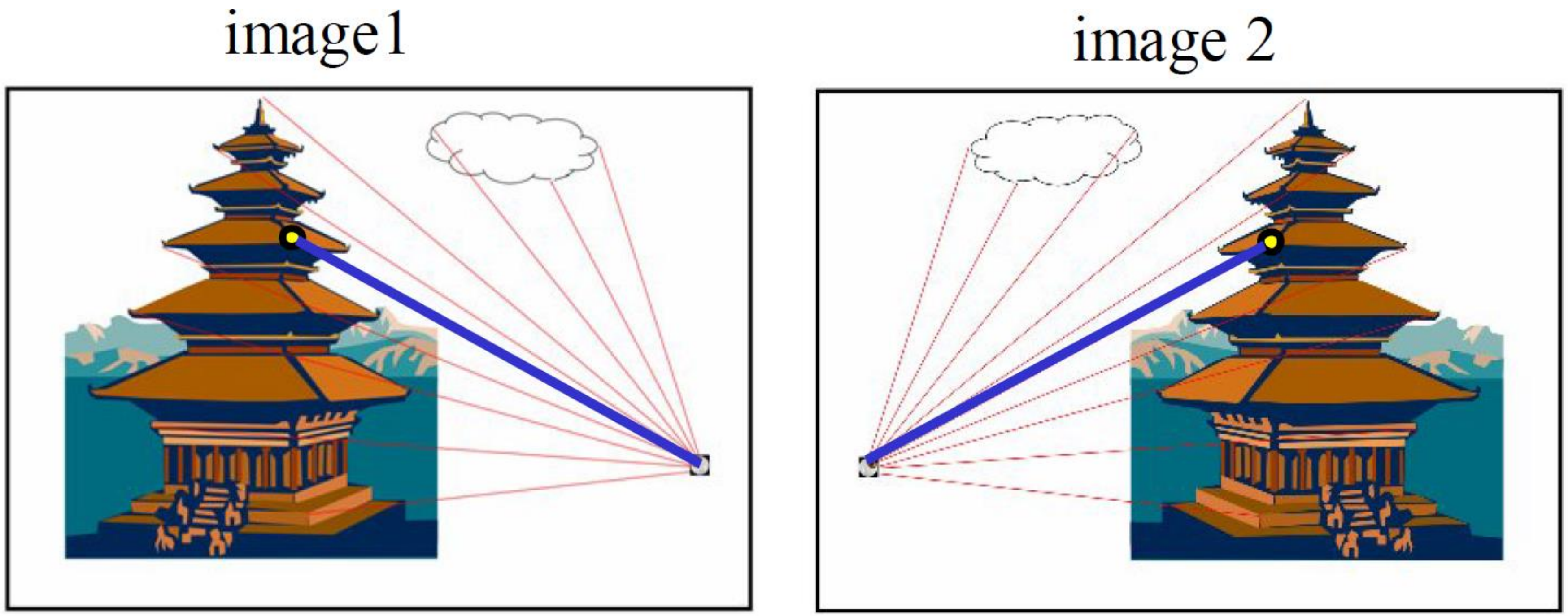
Motivation

- Having two images viewed at two different locations, we are looking for point correspondence.



Motivation

- Any two images showing an overlapping view of the world can be treated as a stereo pair.



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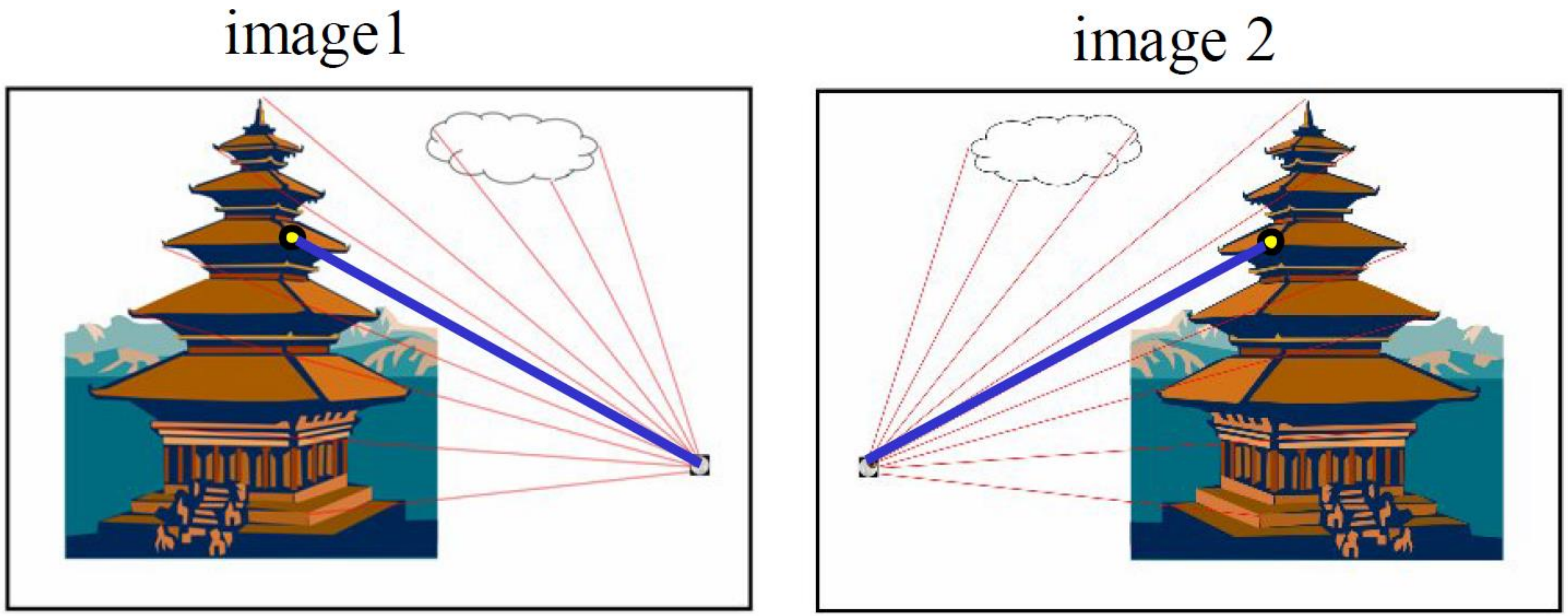
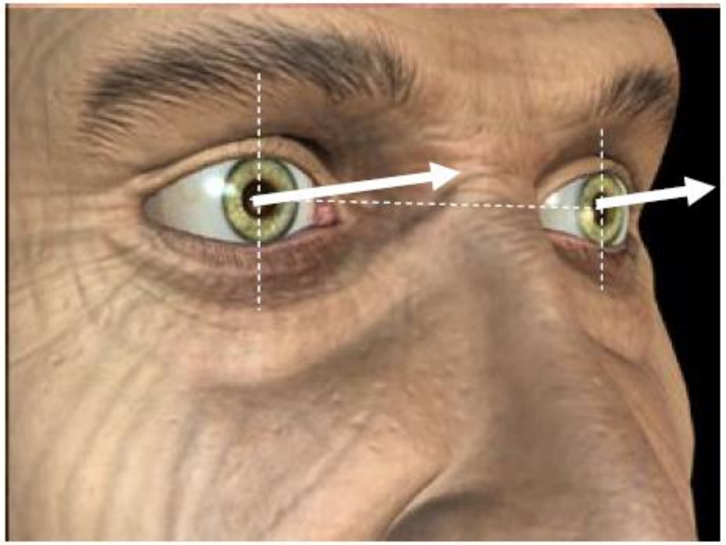
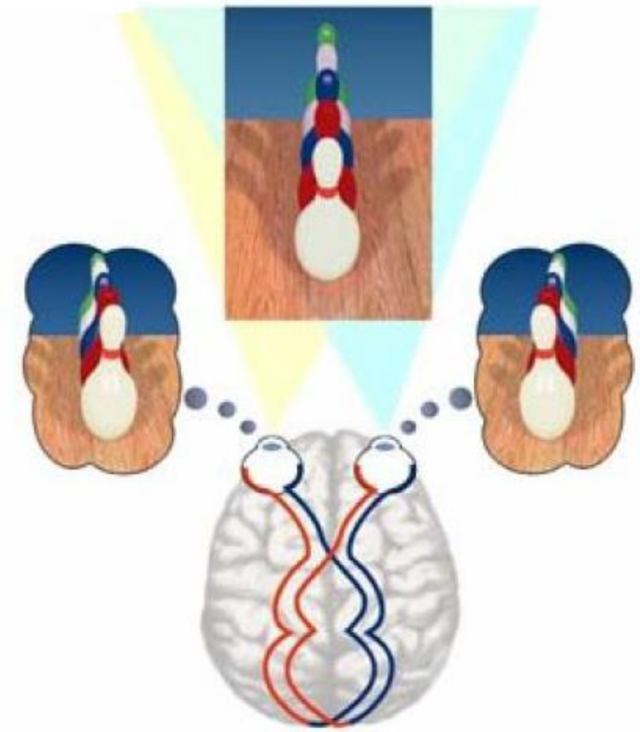


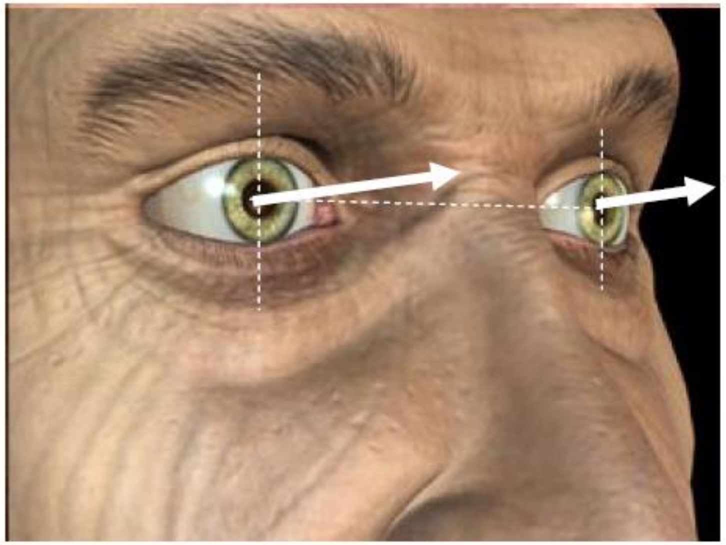
Stereo Vision

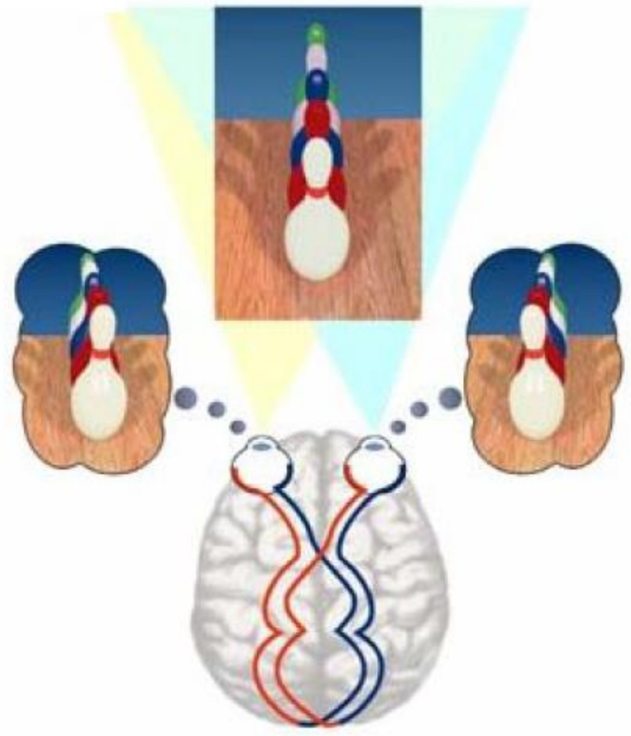
- Our vision works based on a stereo pair image.

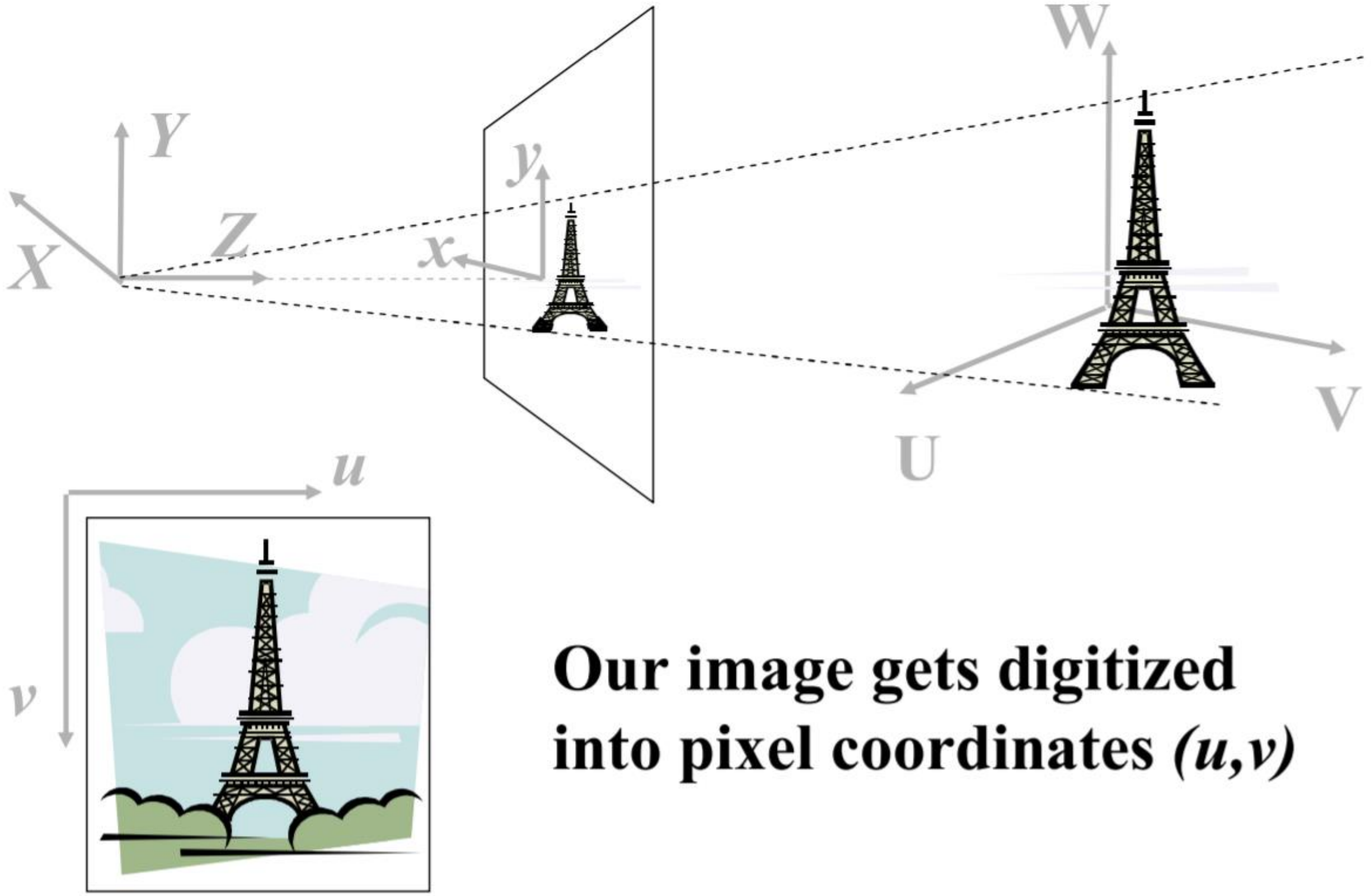
Stereo Vision

- Depth can be inferred by more than two images.





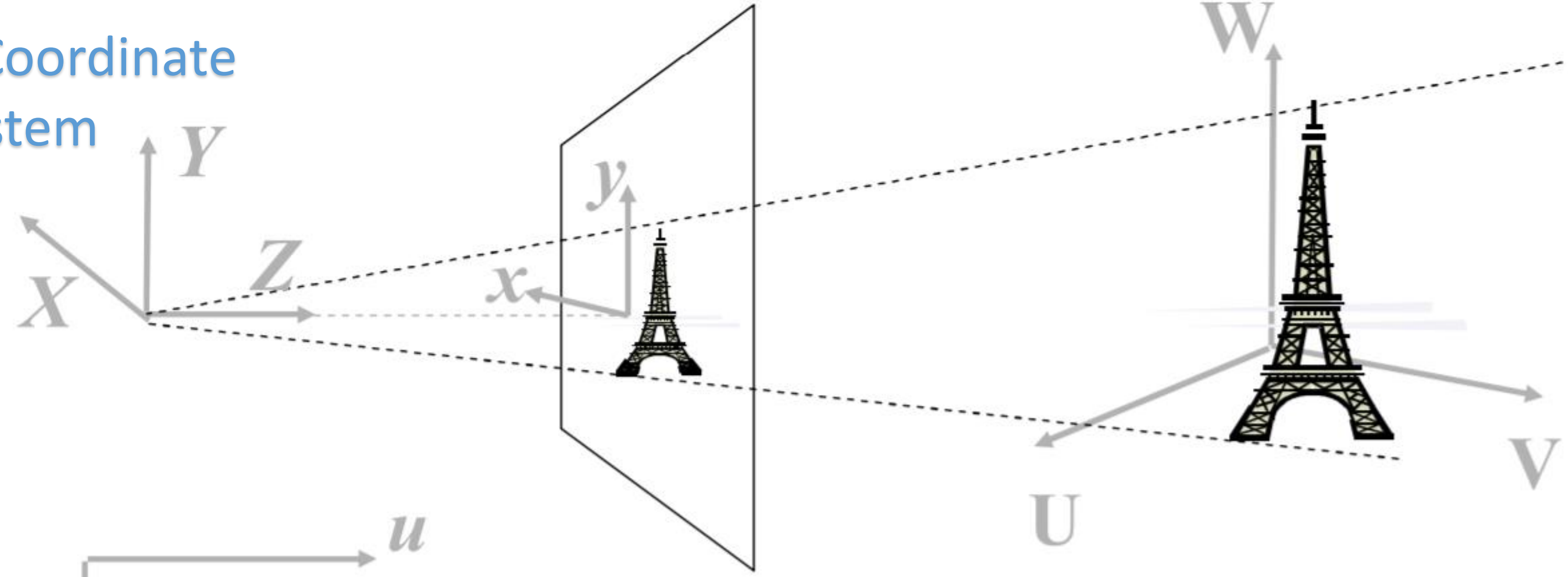
Review of Coordinate Transformations

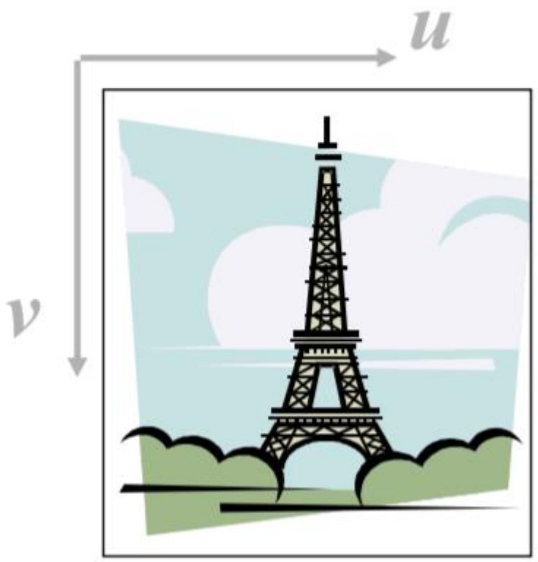


Our image gets digitized into pixel coordinates *(u,v)*

Review of Coordinate Transformations

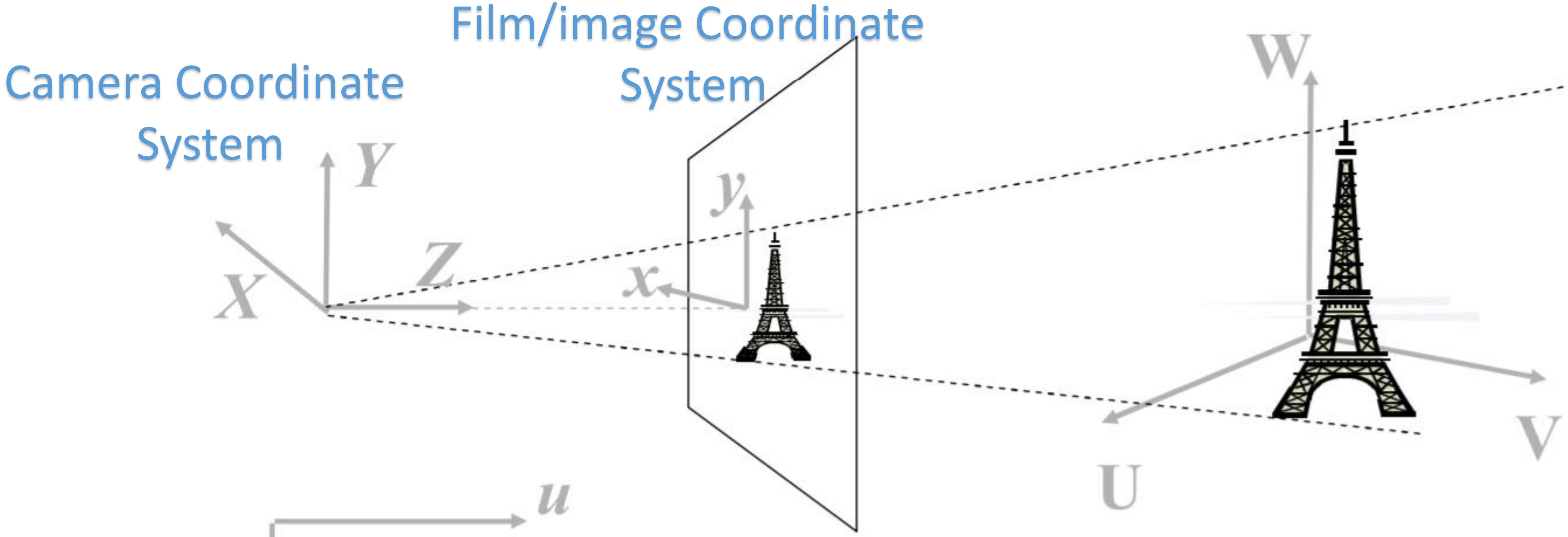
Camera Coordinate System

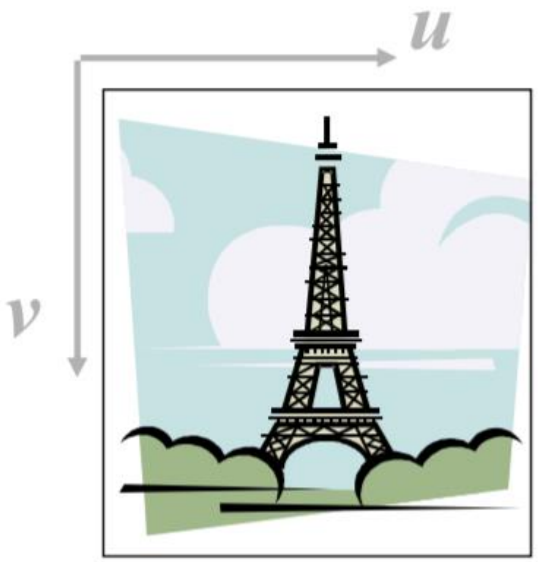




Our image gets digitized into pixel coordinates *(u,v)*

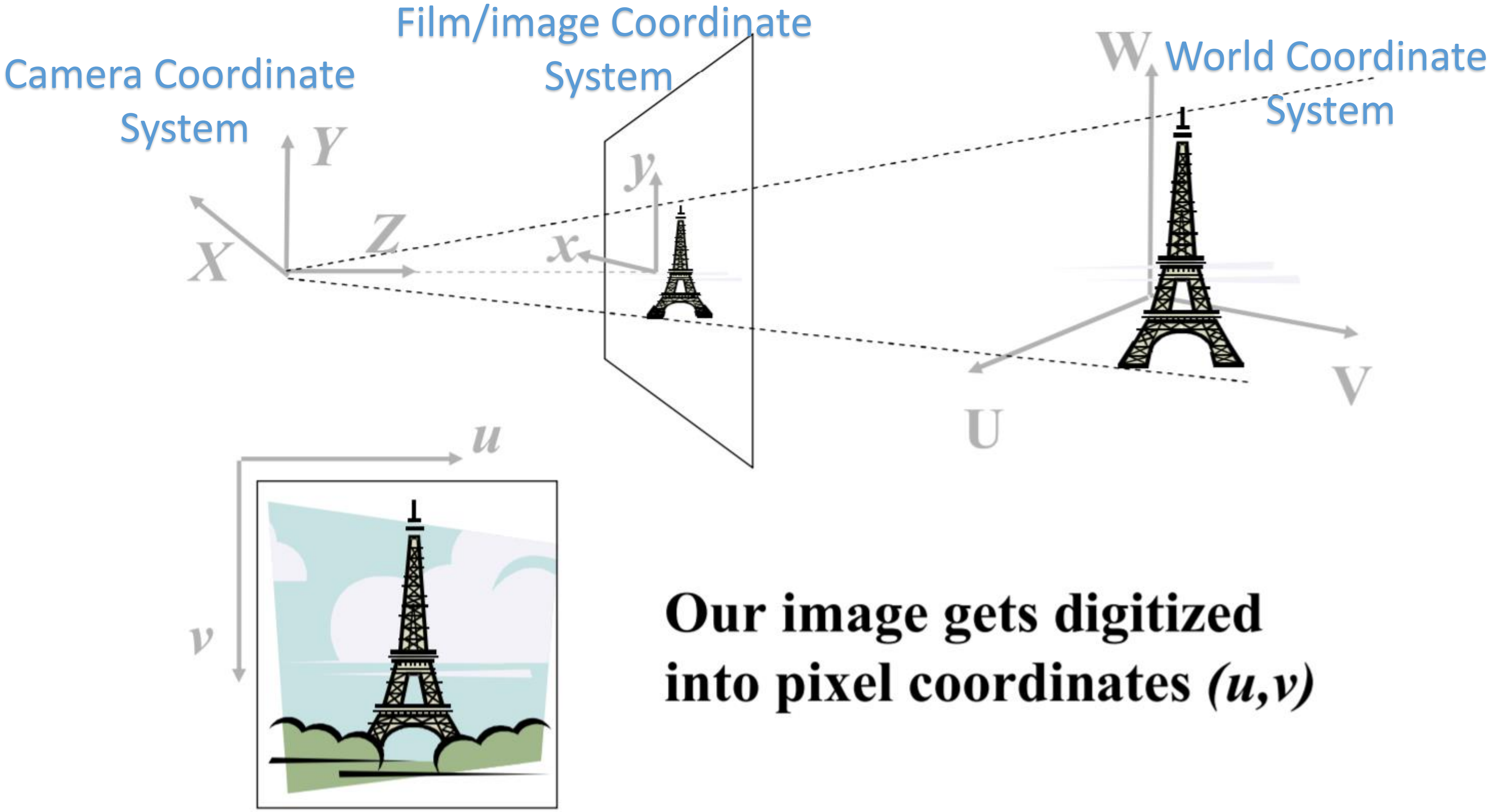
Review of Coordinate Transformations





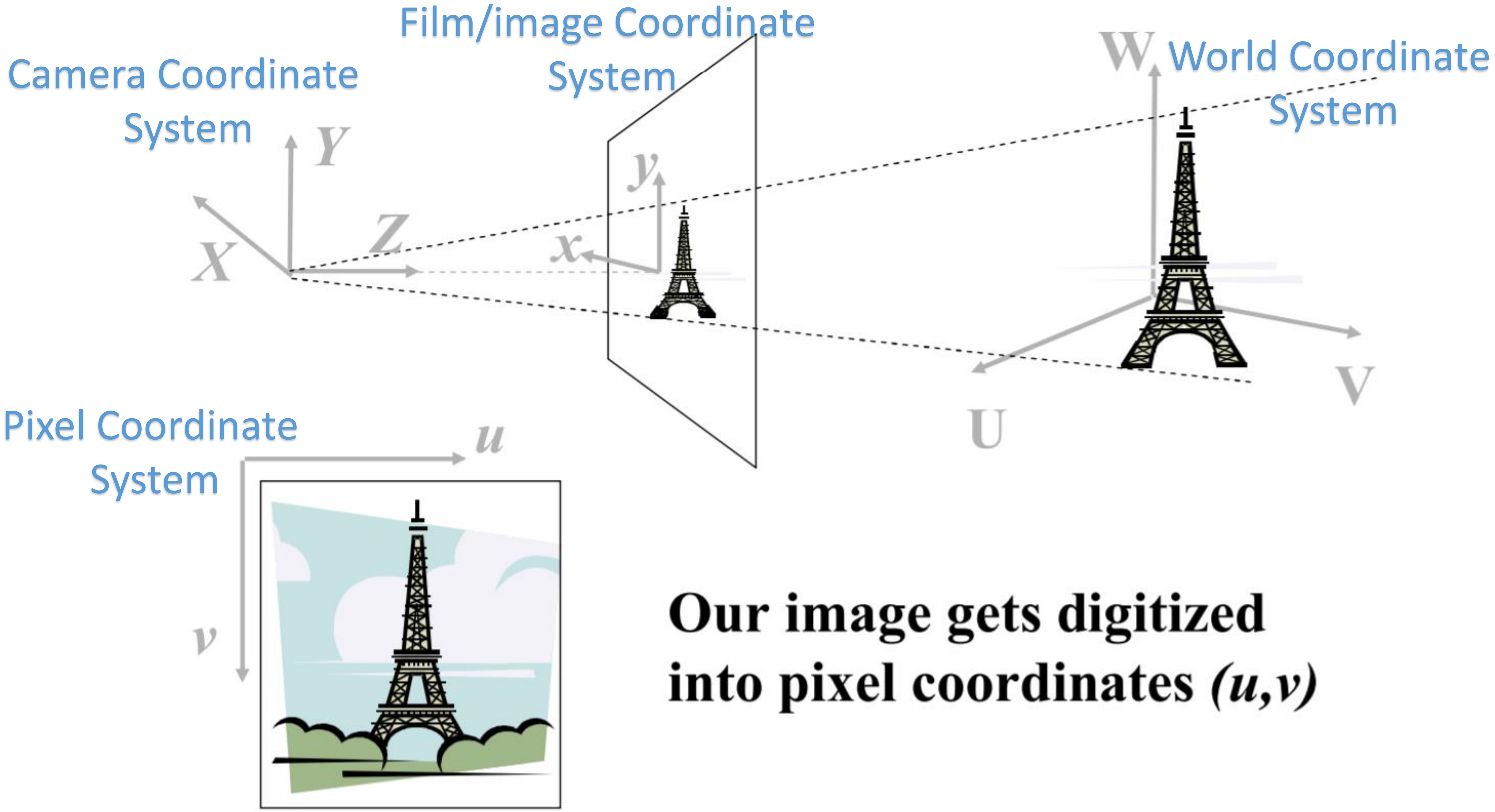
Our image gets digitized into pixel coordinates *(u,v)*

Review of Coordinate Transformations



Our image gets digitized into pixel coordinates *(u,v)*

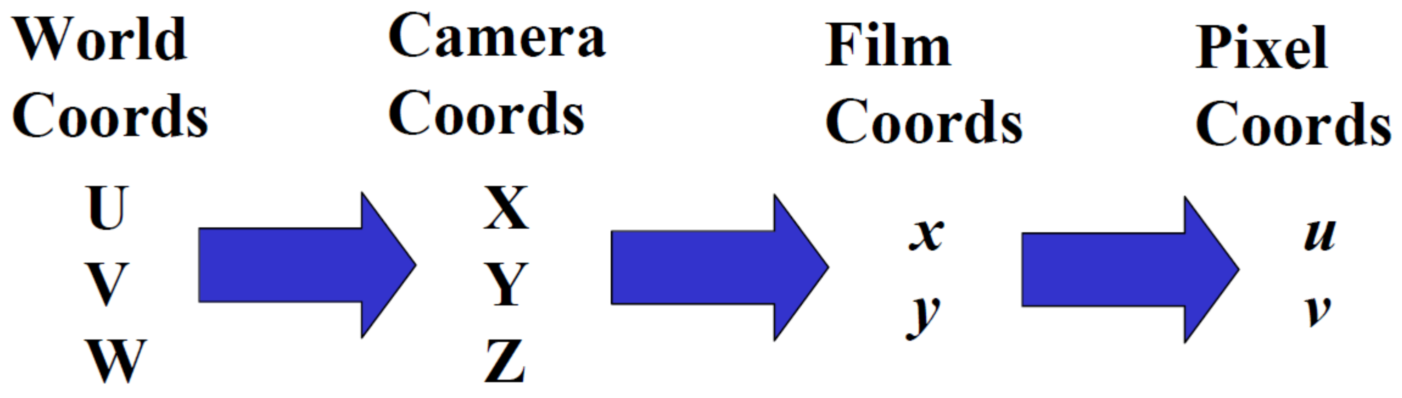
Review of Coordinate Transformations



Our image gets digitized into pixel coordinates *(u,v)*

Review of Coordinate Transformations

- The process of transforming between coordinate systems.



Review of Coordinate Transformations

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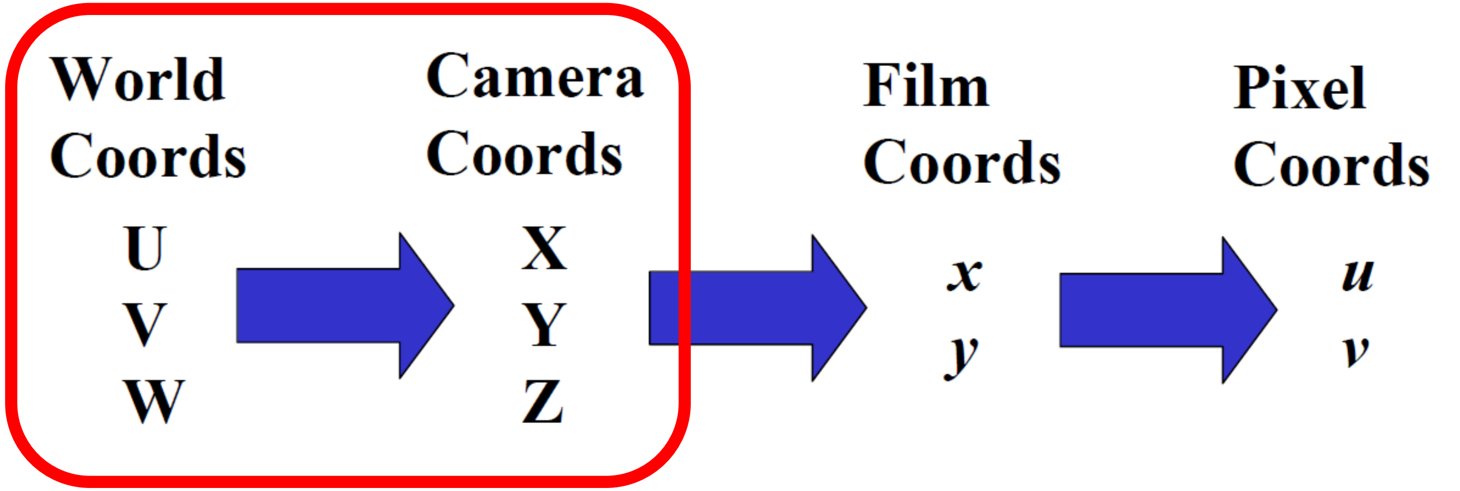
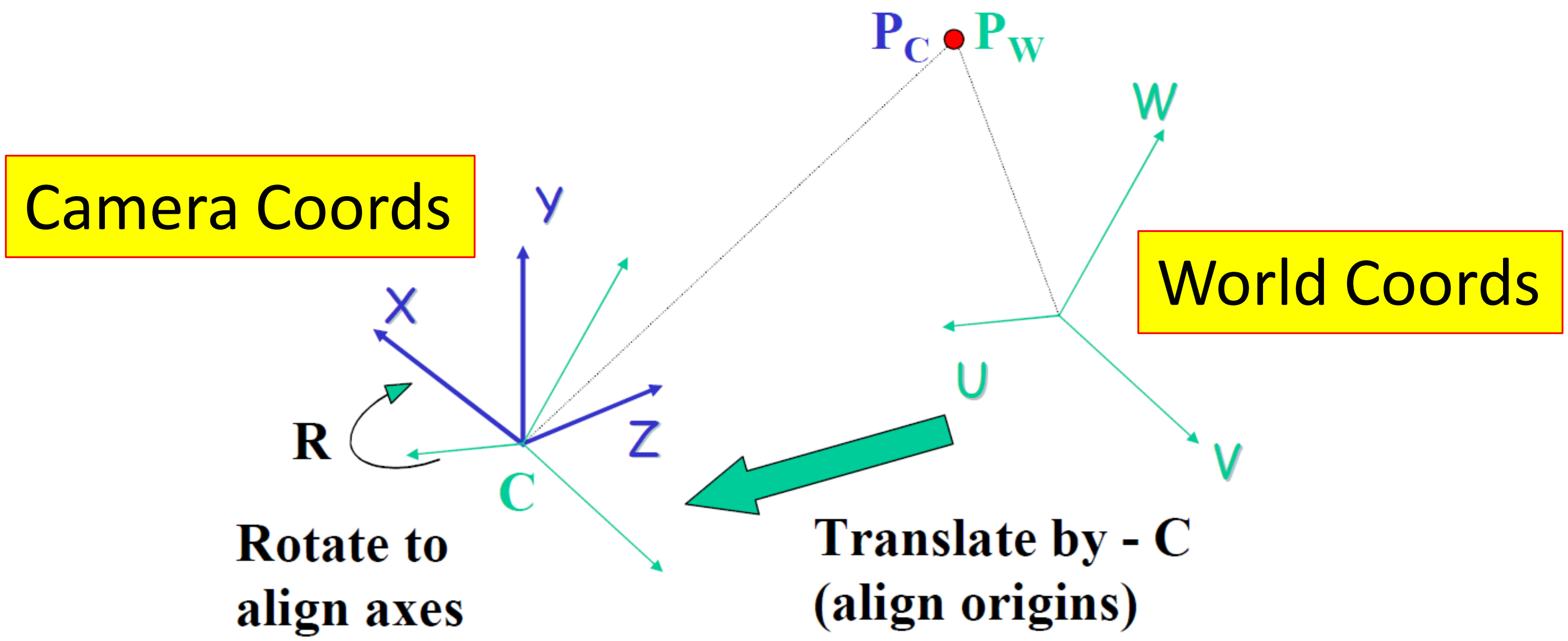


World to Camera Transformation

- Two coordinate systems can be aligned by a simple rotation and translation.

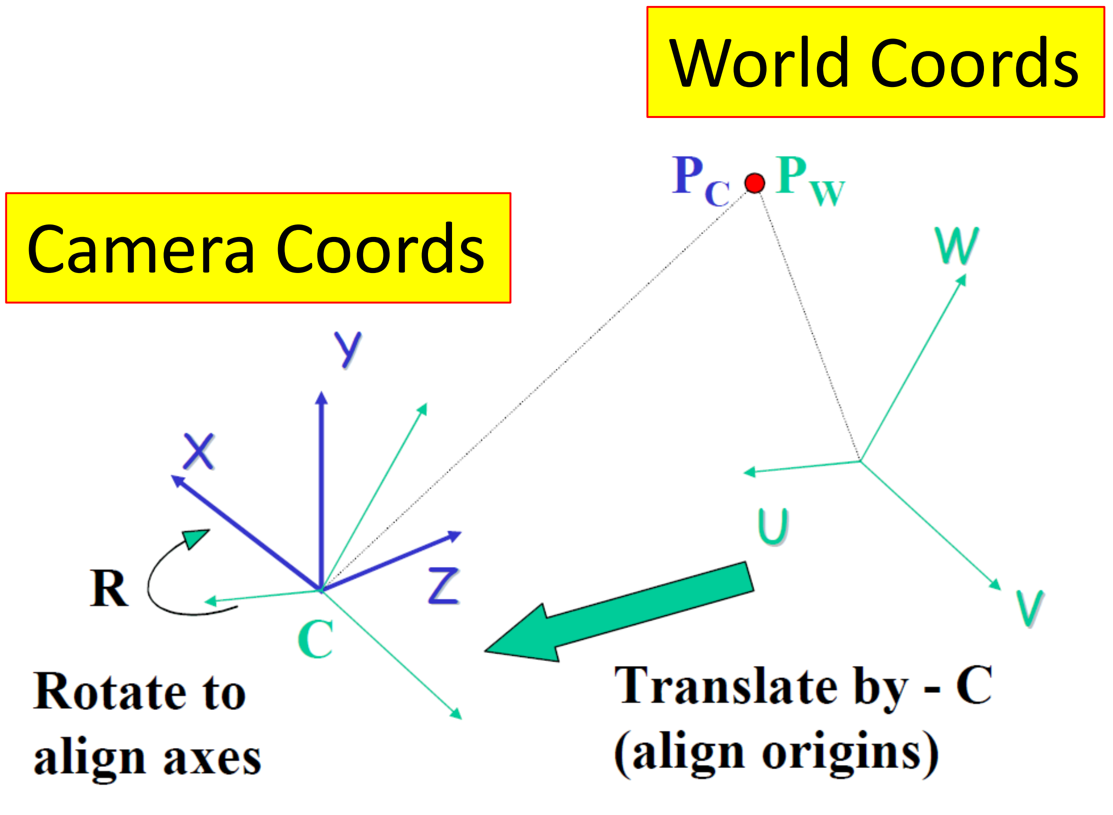


$$P_C = R\,(P_W - C)$$

World to Camera Transformation

- Matrix representation

$$\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & -c_x \\ 0 & 1 & 0 & -c_y \\ 0 & 0 & 1 & -c_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} U \\ V \\ W \\ 1 \end{pmatrix}$$

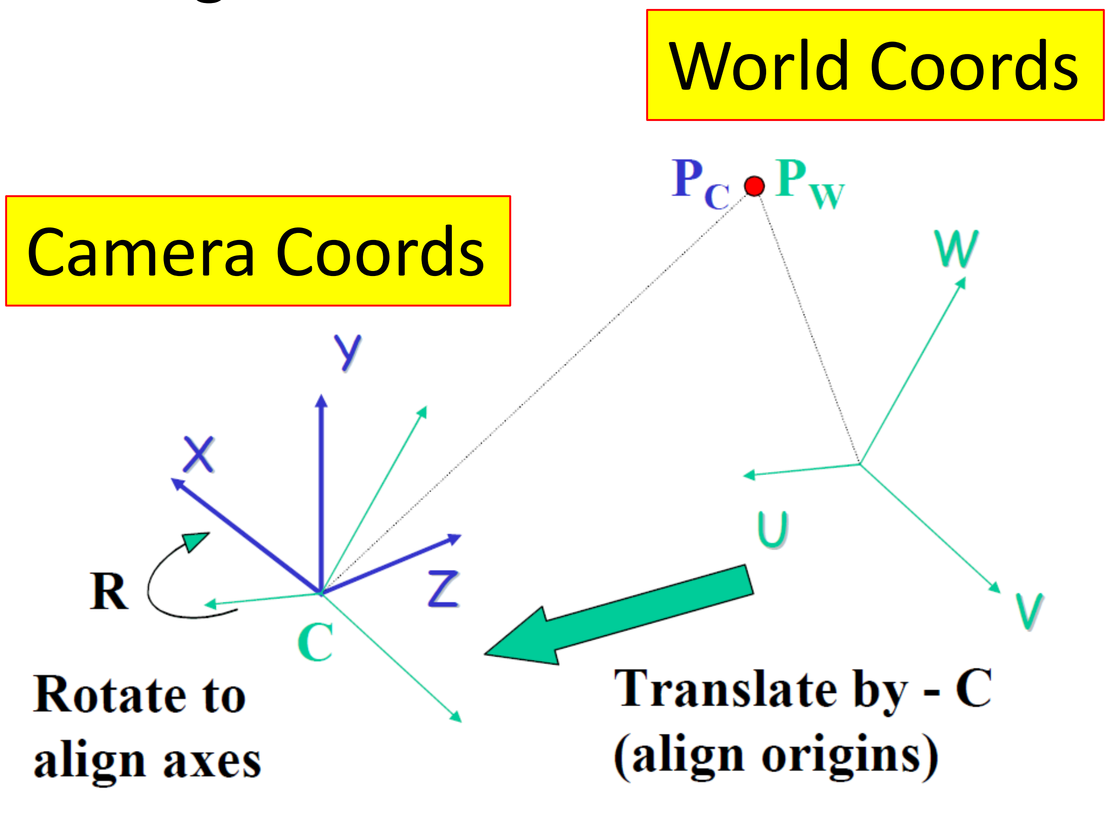


$$P_C = R\,(P_W - C)$$

World to Camera Transformation

- Translation is obvious (difference between origins of two coordinate systems)

$$\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & -c_x \\ 0 & 1 & 0 & -c_y \\ 0 & 0 & 1 & -c_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} U \\ V \\ W \\ 1 \end{pmatrix}$$

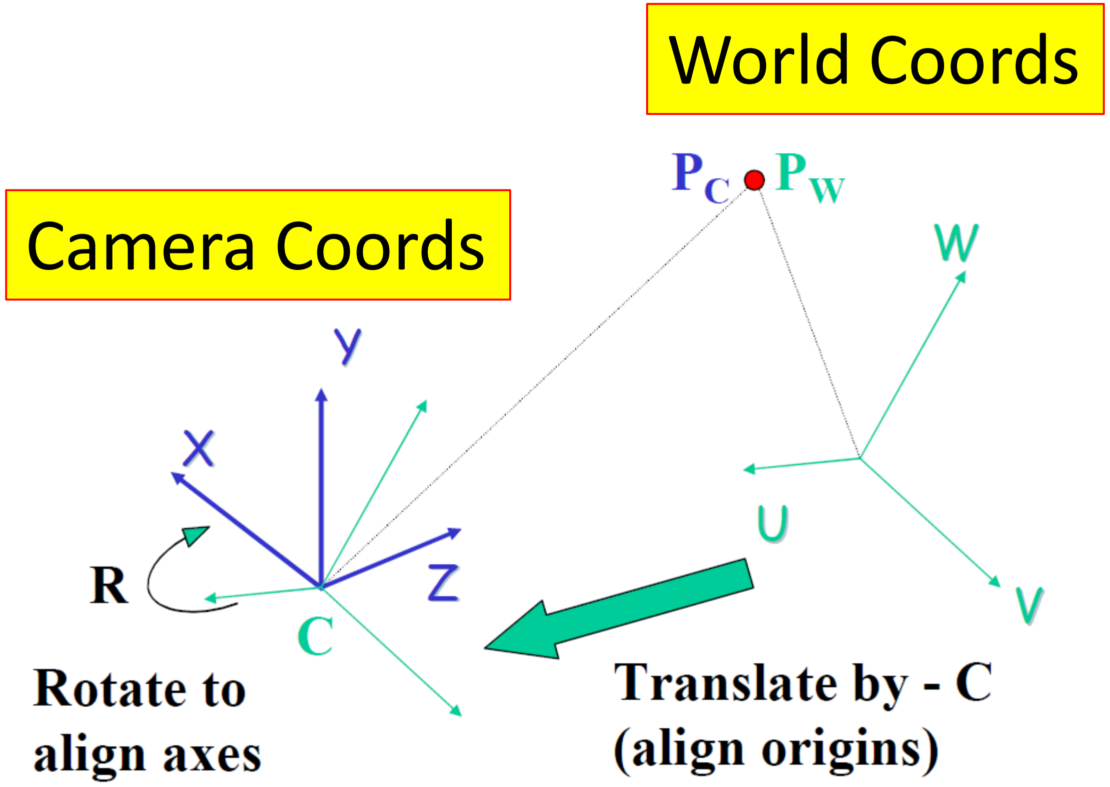


$$P_C = R\,(P_W - C)$$

World to Camera Transformation

- Let's assume translation is zero, just focus on rotation.

$$\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cancel{\begin{pmatrix} 1 & 0 & 0 & -c_x \\ 0 & 1 & 0 & -c_y \\ 0 & 0 & 1 & -c_z \\ 0 & 0 & 0 & 1 \end{pmatrix}} \begin{pmatrix} U \\ V \\ W \\ 1 \end{pmatrix}$$



$$P_C = R\,(P_W - C)$$

World to Camera Transformation

- As a result, we will have

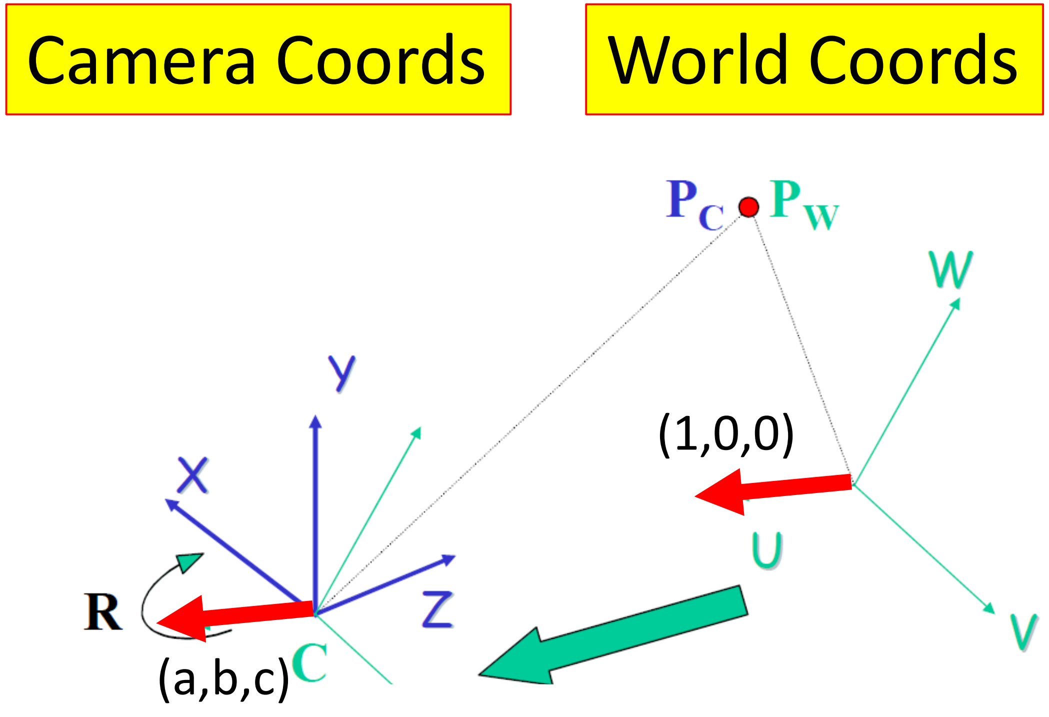
$$\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} U \\ V \\ W \\ 1 \end{pmatrix}$$

World to Camera Transformation

- We find the corresponding vector to world U axis (1,0,0) in camera coordinate system (a,b,c):

$$\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} U \\ V \\ W \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \\ \mathbf{c} \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{0} \\ 1 \end{pmatrix}$$



World to Camera Transformation

- This determines the first column of R matrix:

$$\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} U \\ V \\ W \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \\ \mathbf{c} \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{0} \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \\ \mathbf{c} \\ 1 \end{pmatrix} = \begin{pmatrix} \mathbf{a} & r_{12} & r_{13} & 0 \\ \mathbf{b} & r_{22} & r_{23} & 0 \\ \mathbf{c} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{0} \\ 1 \end{pmatrix}$$

World to Camera Transformation

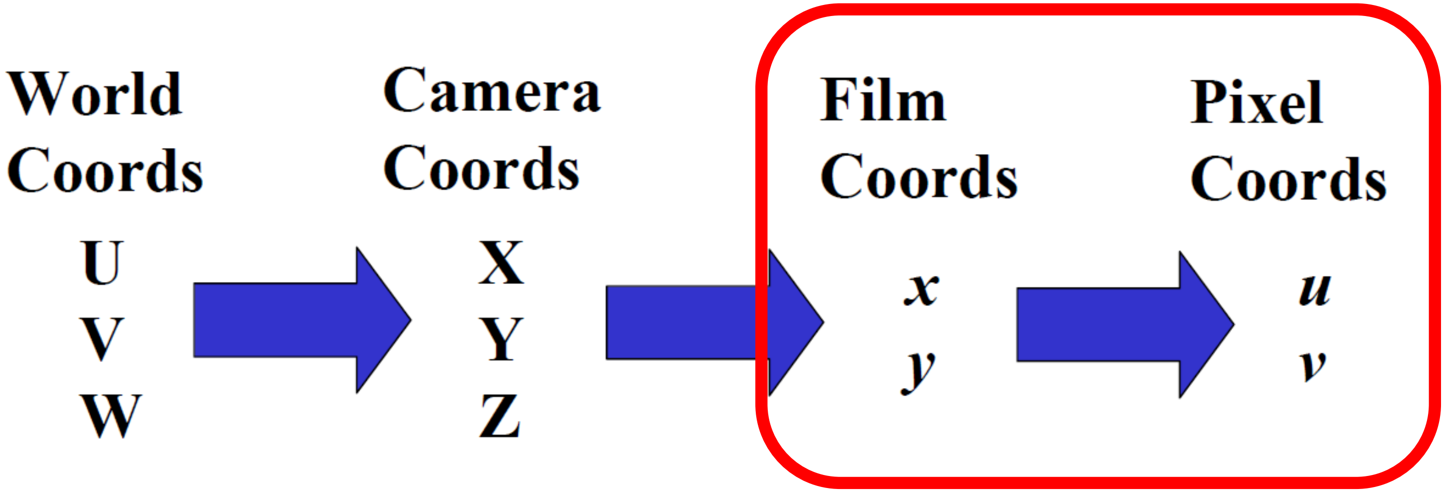
- We do similar thing for other axis: y and z and we determine R:

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} U \\ V \\ W \\ 1 \end{bmatrix}$$

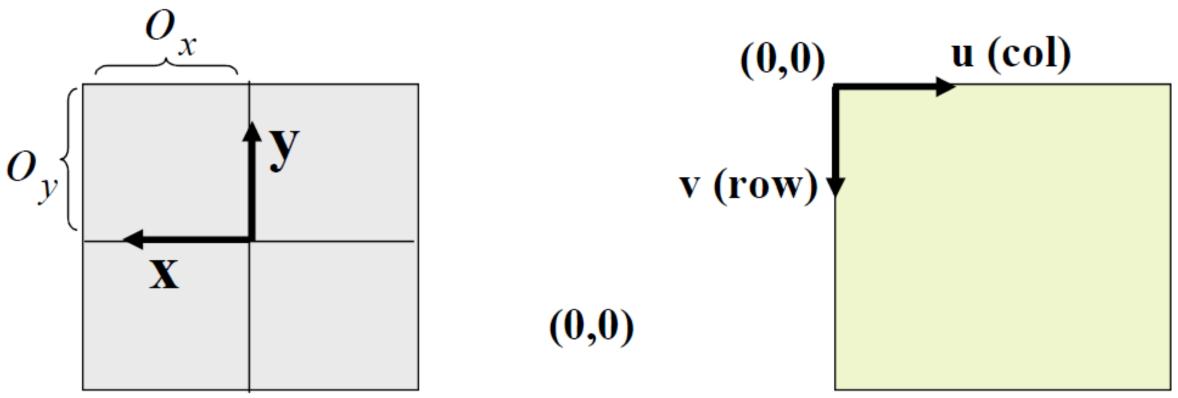
(1,0,0) (0,1,0) (0,0,1)

Review of Coordinate Transformations

- The process of transforming between coordinate systems.



Film to Pixel Coordinate

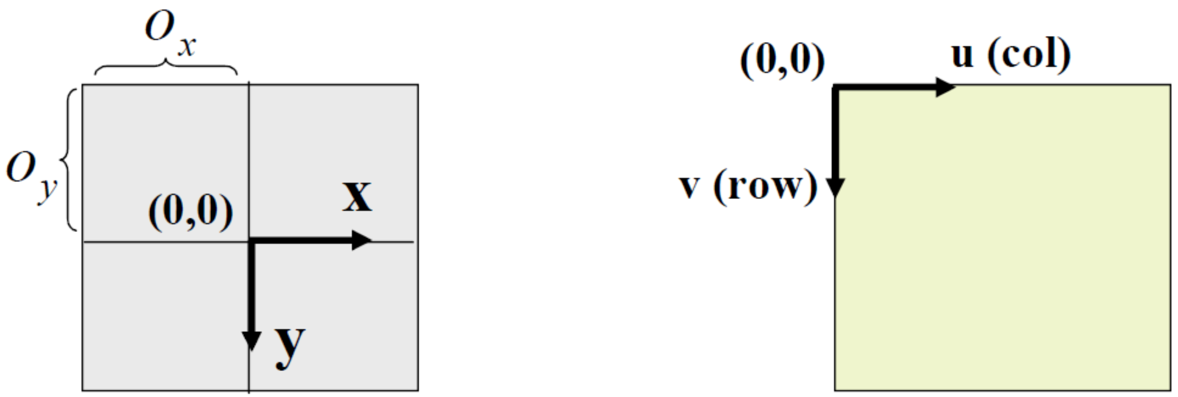


$$u = -x + O_x$$

$$v = -y + O_y$$

Film to Pixel Coordinate

- Depending on the coordinate systems, the transformation can be different.



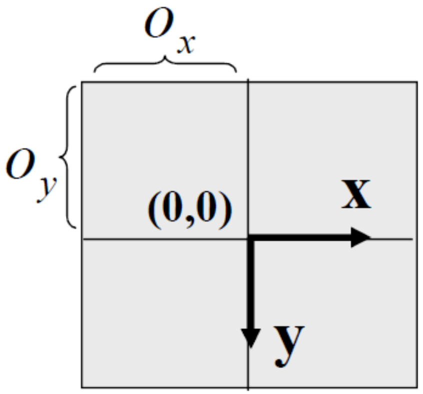
$$u = x + O_x$$

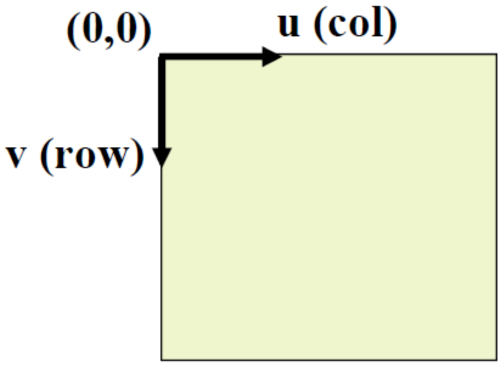
$$v = y + O_y$$

Film to Pixel Coordinate

- If we have aspect ratio $\frac{s_y}{s_x}$, the formula changes to the following to give you the pixel index:

$$u = \frac{1}{s_x} x + O_x$$
$$v = \frac{1}{s_y} y + O_y$$



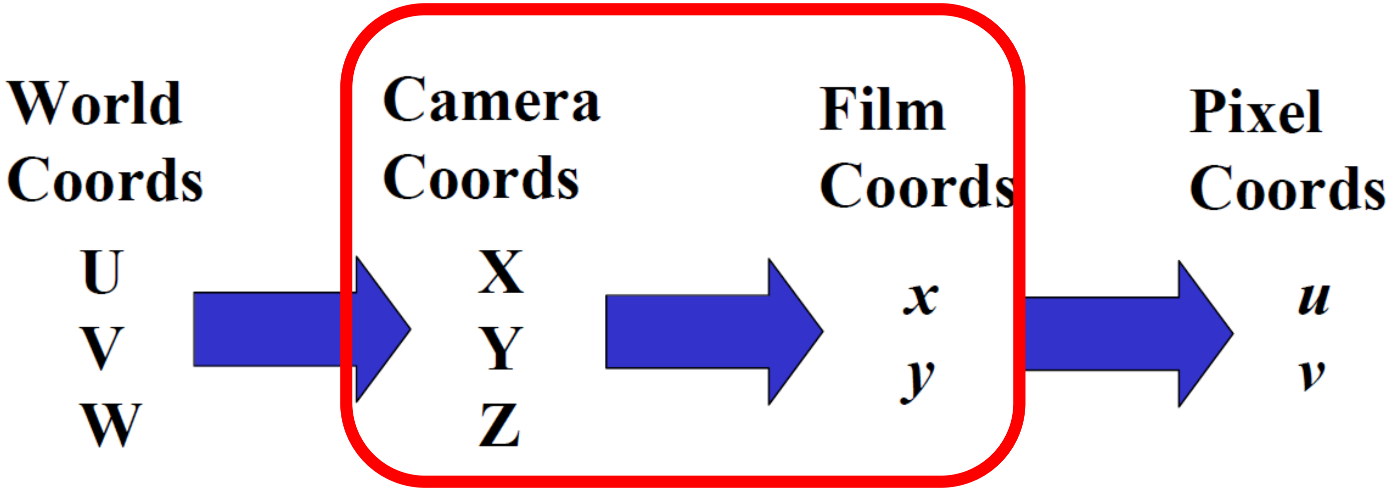


Film to Pixel Coordinate

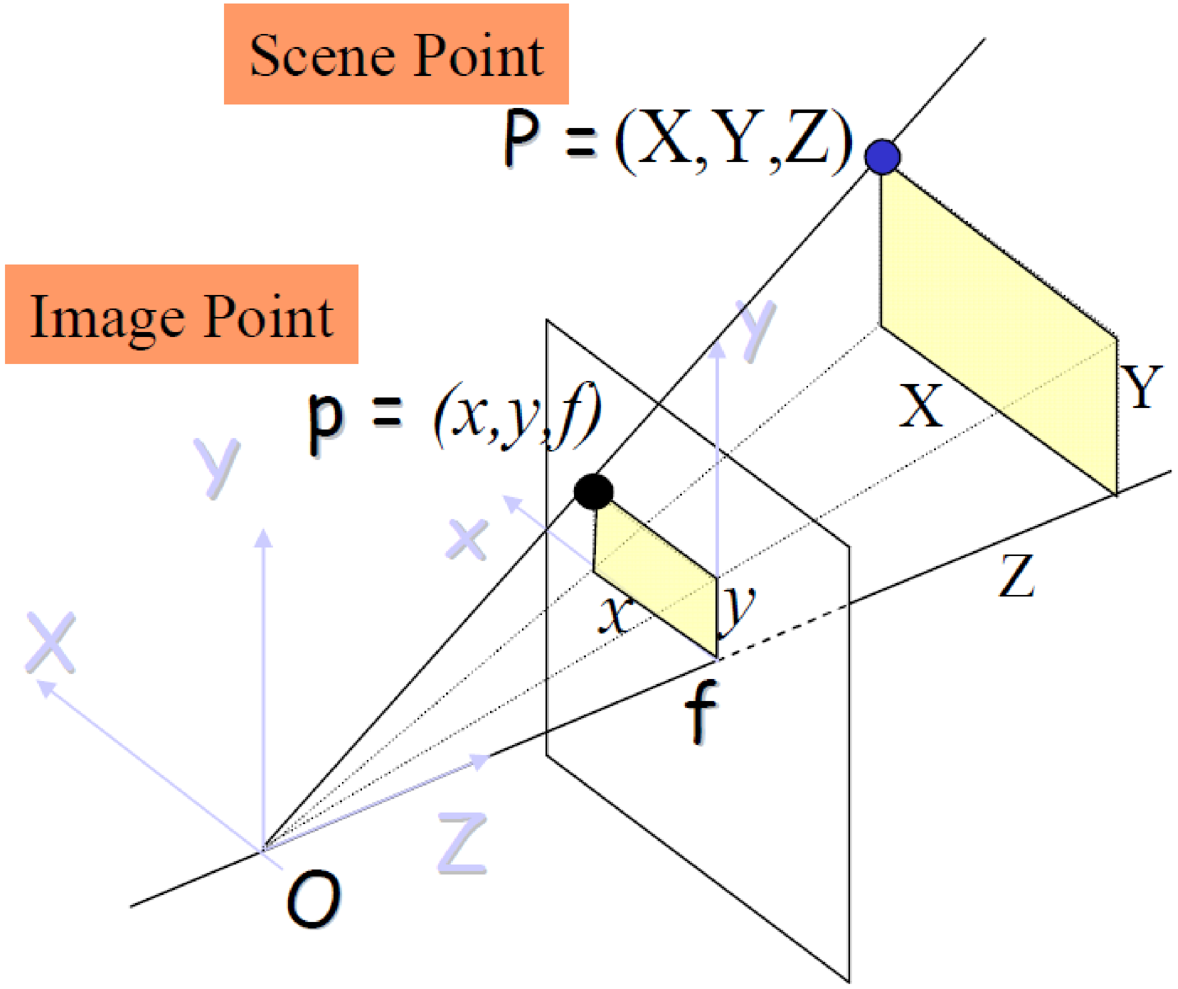
- This whole transformation can be written in matrix form: M.
- Exercise: find the matrix.

Review of Coordinate Transformations

- The process of transforming between coordinate systems.

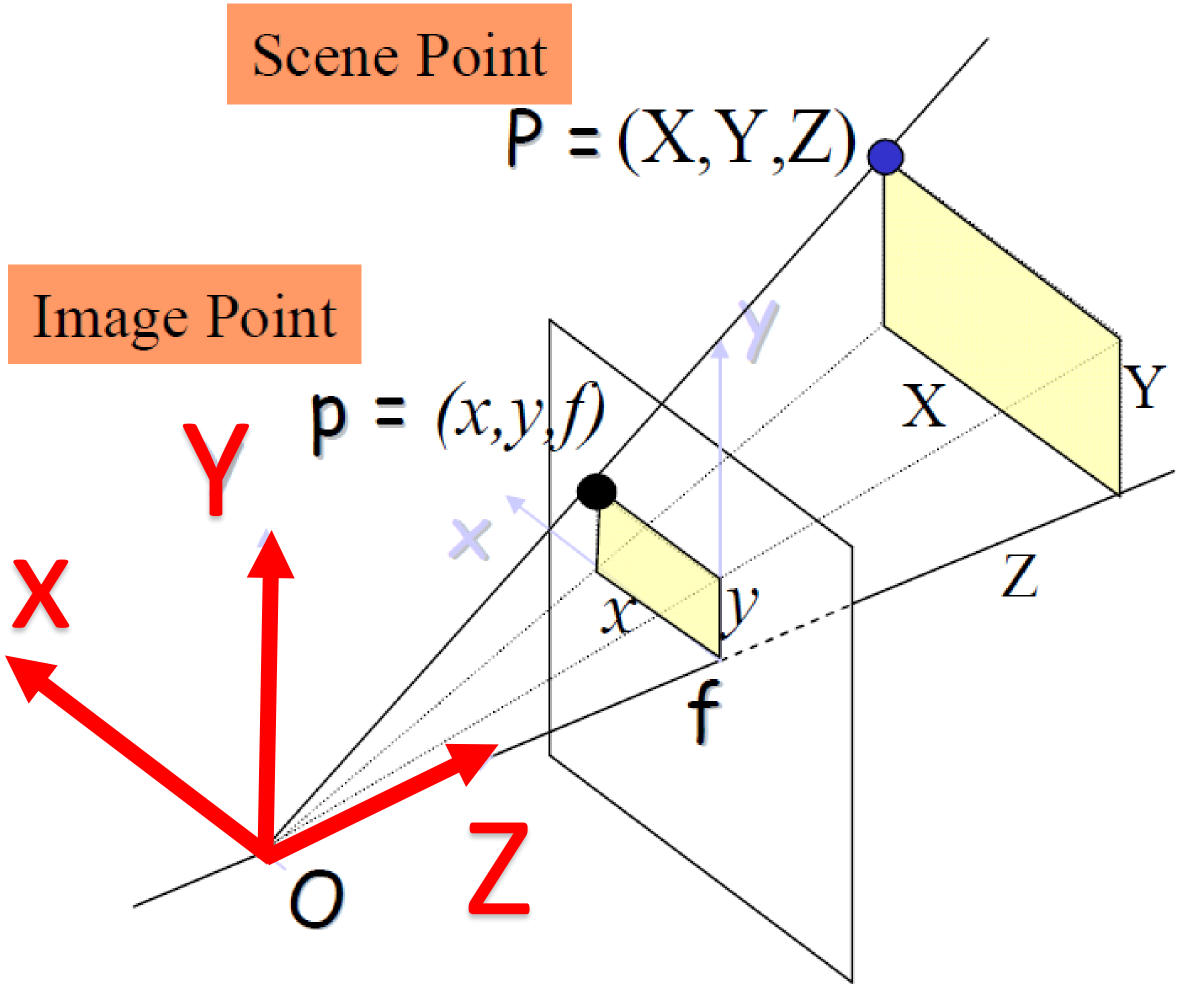


Perspective Projection



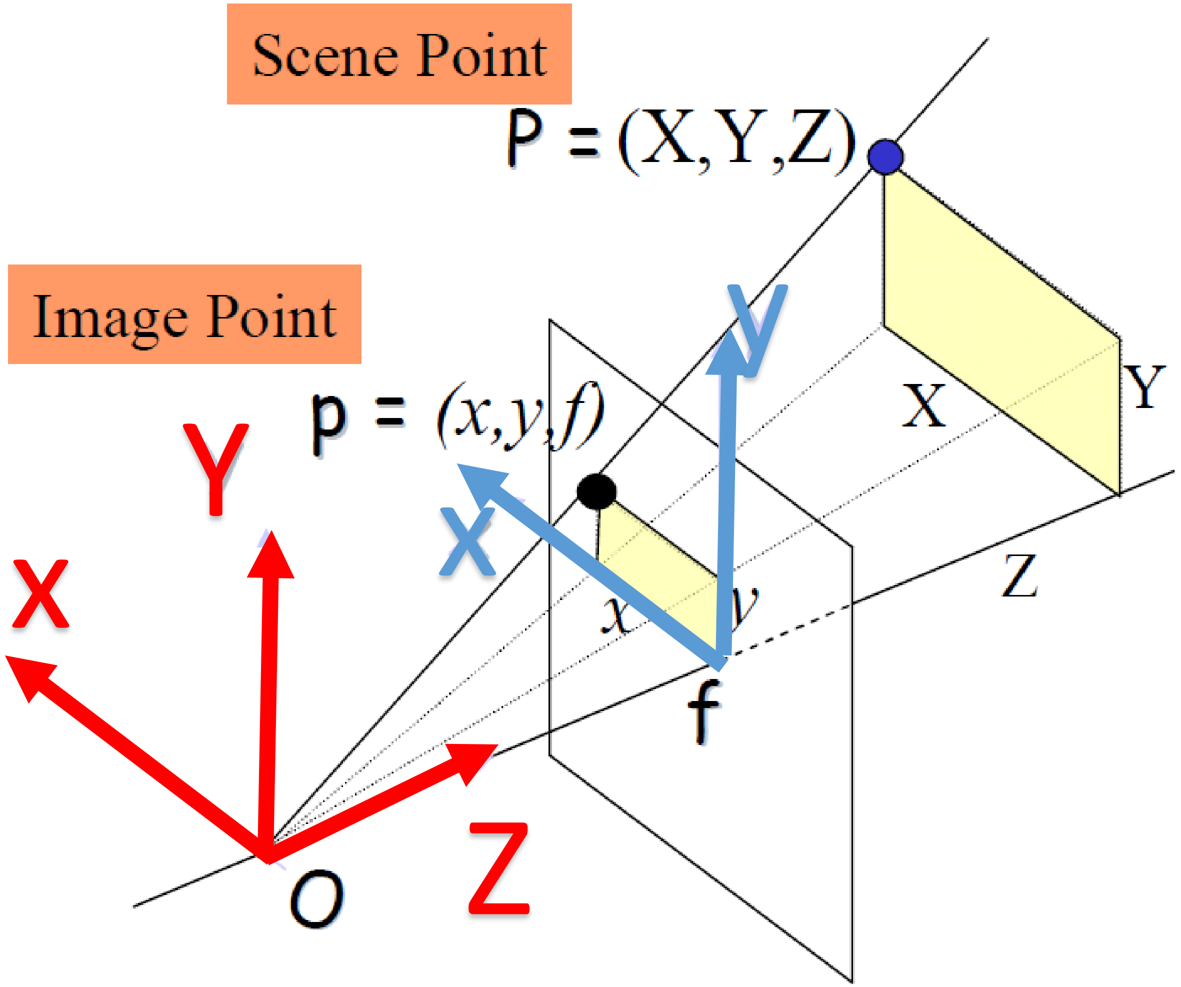
Perspective Projection

- Camera has a coordinate system



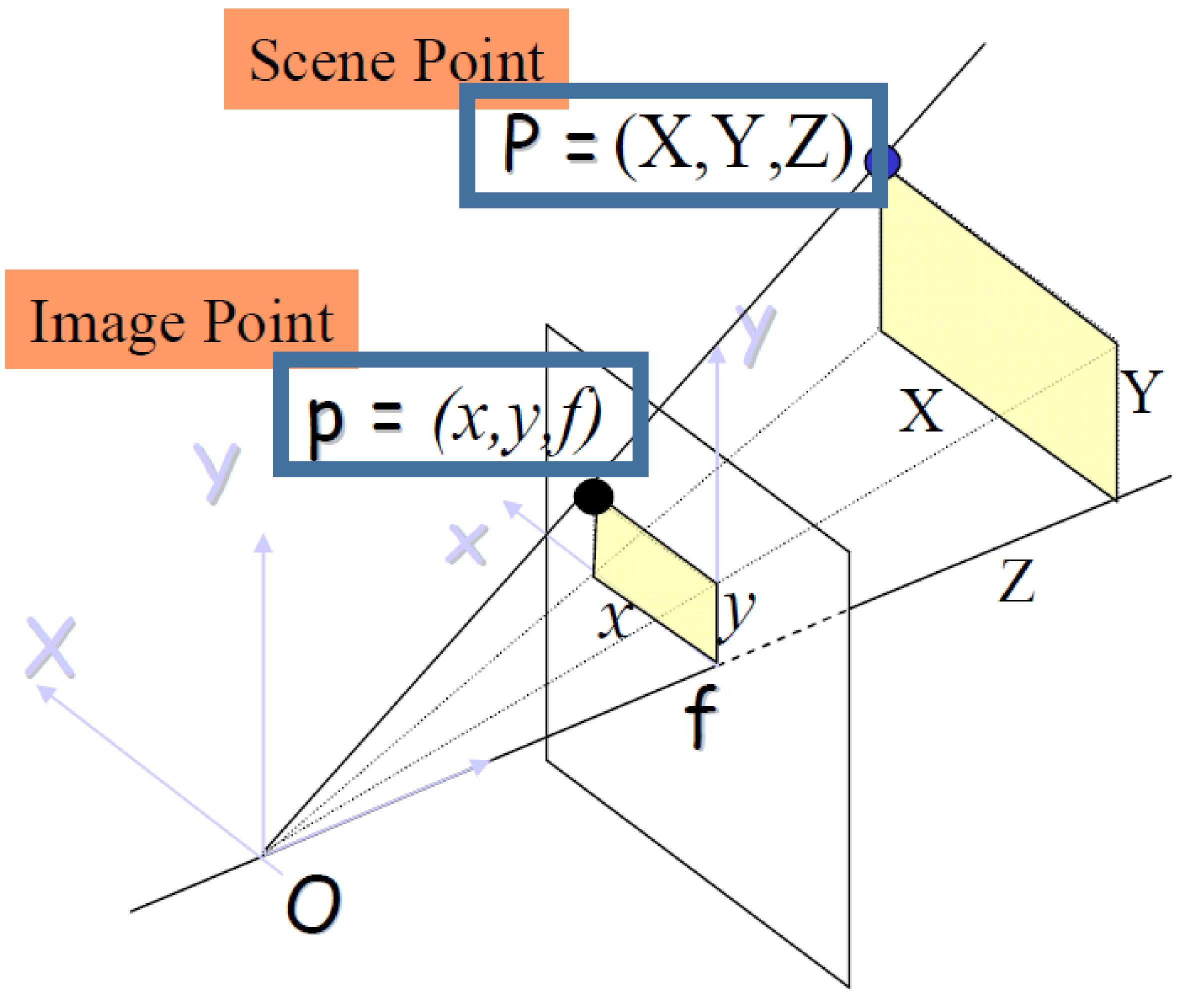
Perspective Projection

- Image has a coordinate system

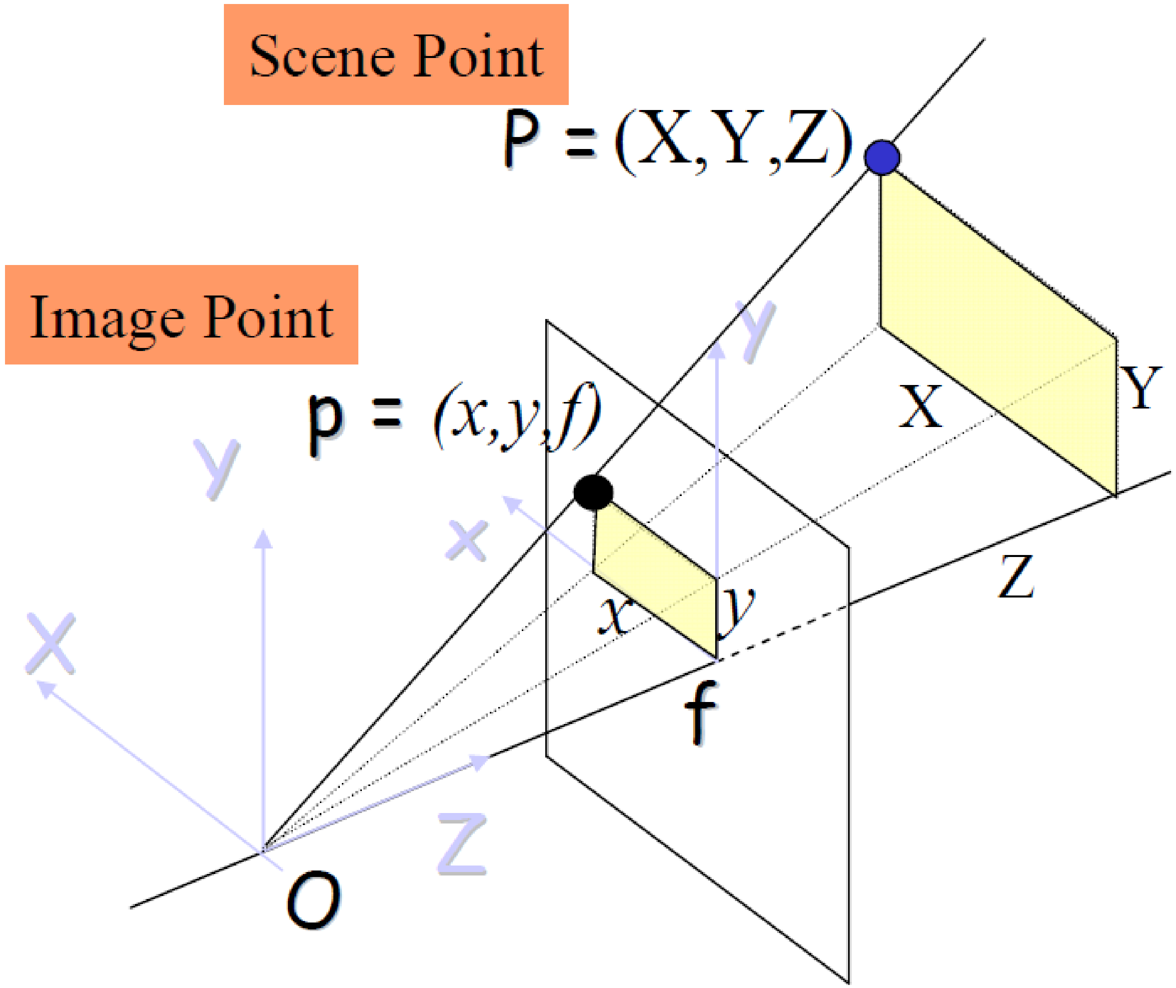


Perspective Projection

- How to connect one point in one coordinate system to another?



Perspective Projection



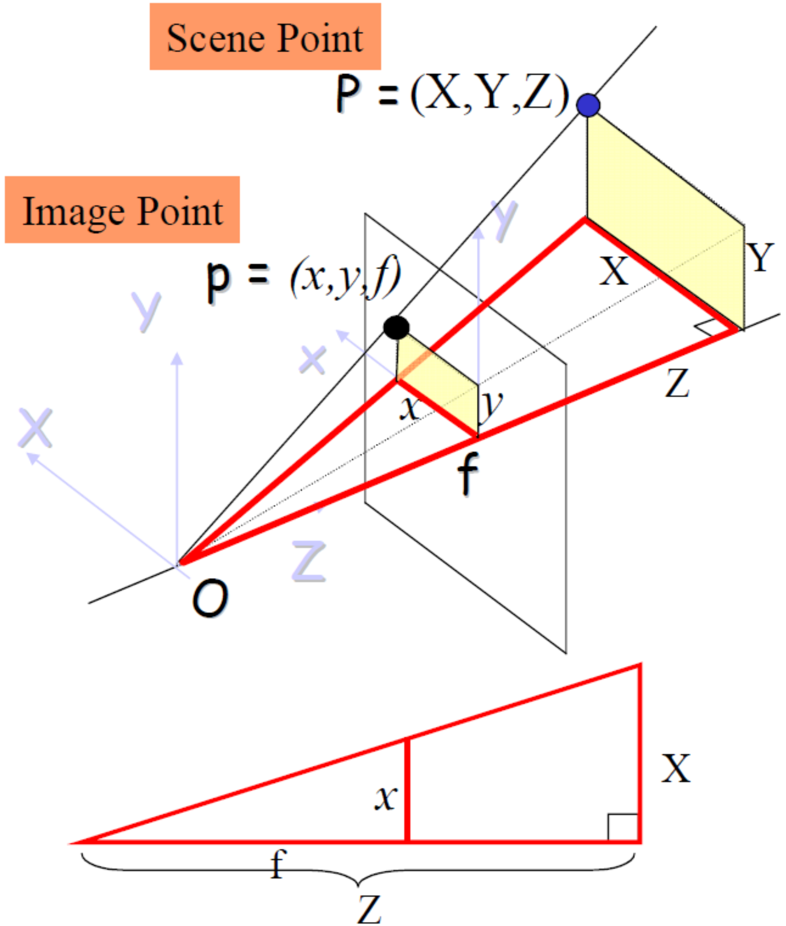
Perspective Projection Eqns

$$x = f\frac{X}{Z}$$

$$y = f\frac{Y}{Z}$$

Perspective Projection

- The basic property comes from the idea of triangle similarity.



Perspective Projection Eqns

$$x = f\,\frac{X}{Z}$$

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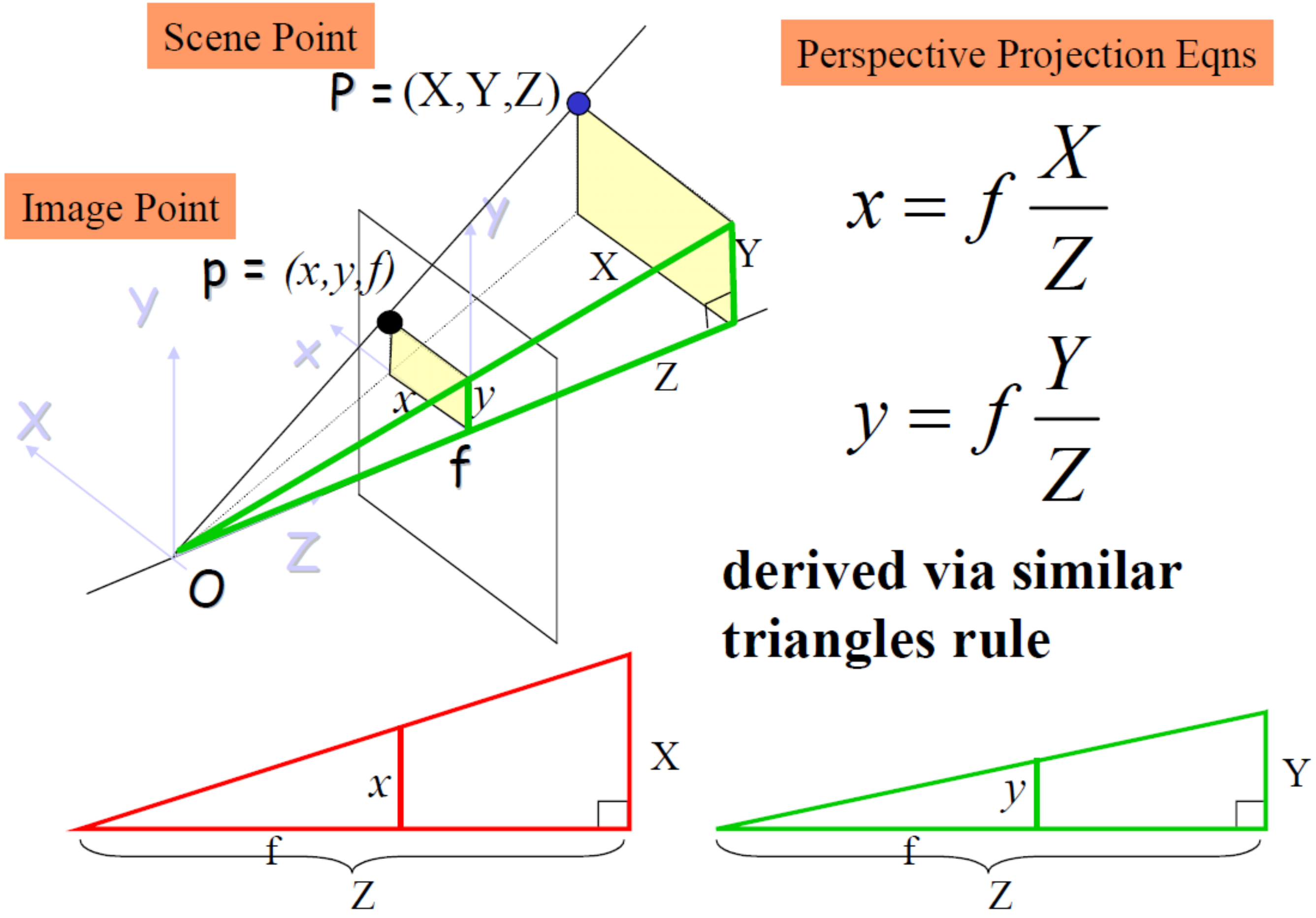
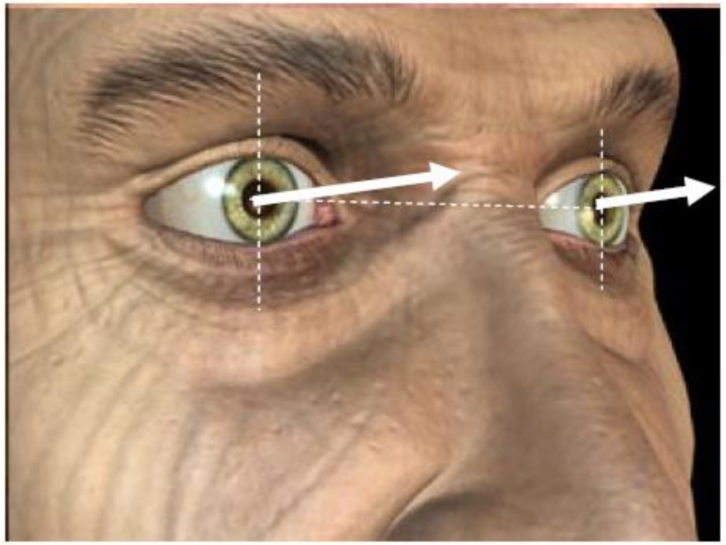
Perspective Projection

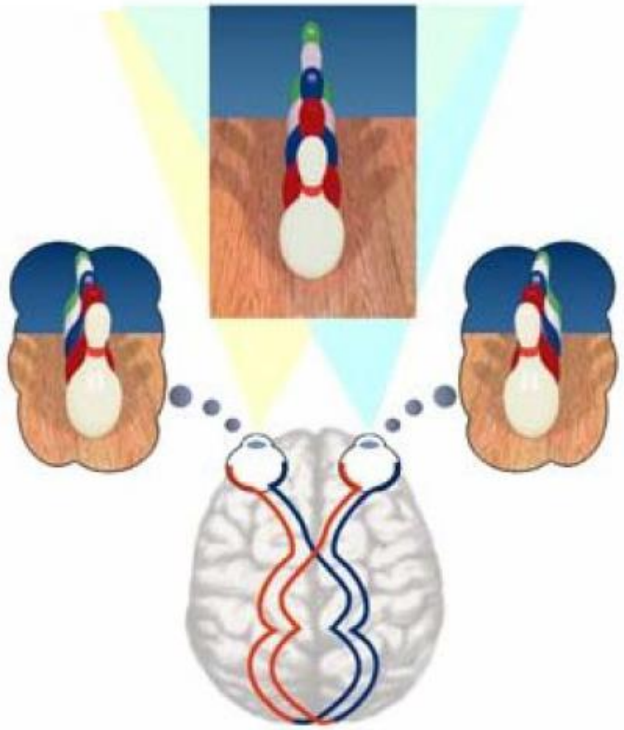
- Similarly for y.



Stereo Vision

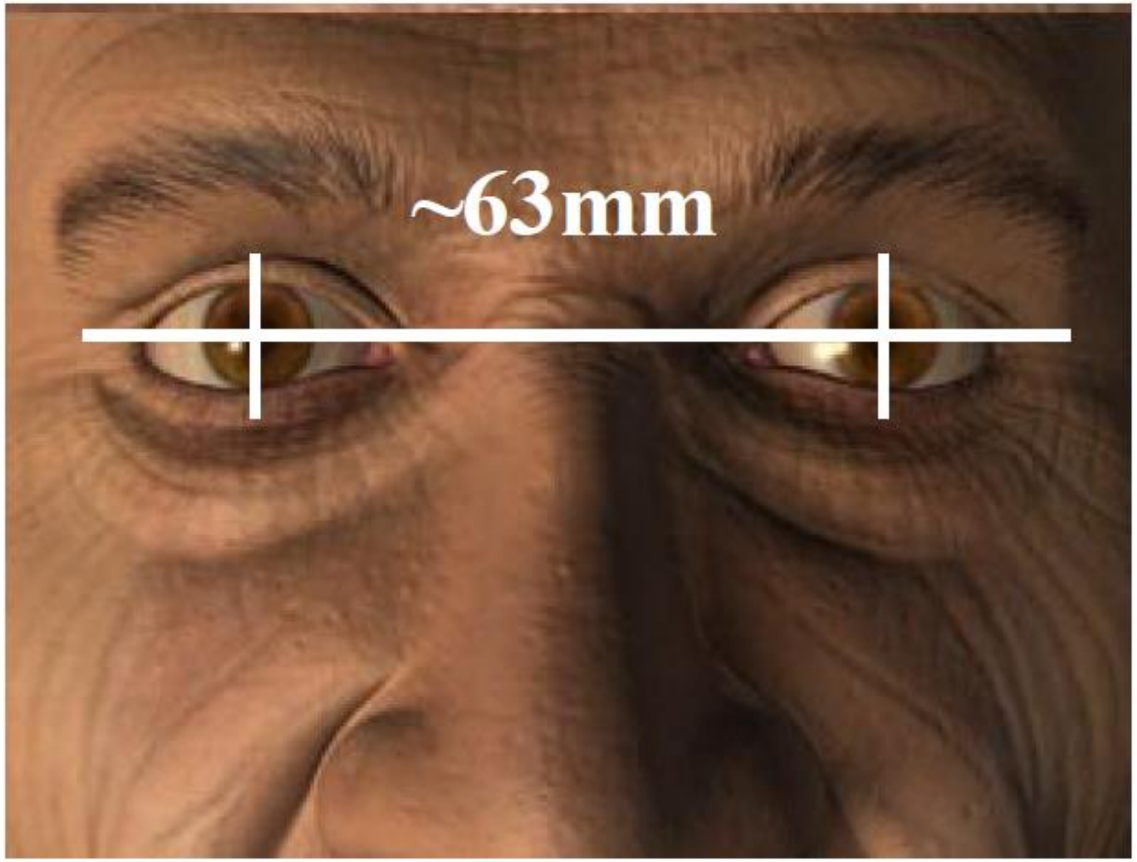
- Let's go back to stereo-views.





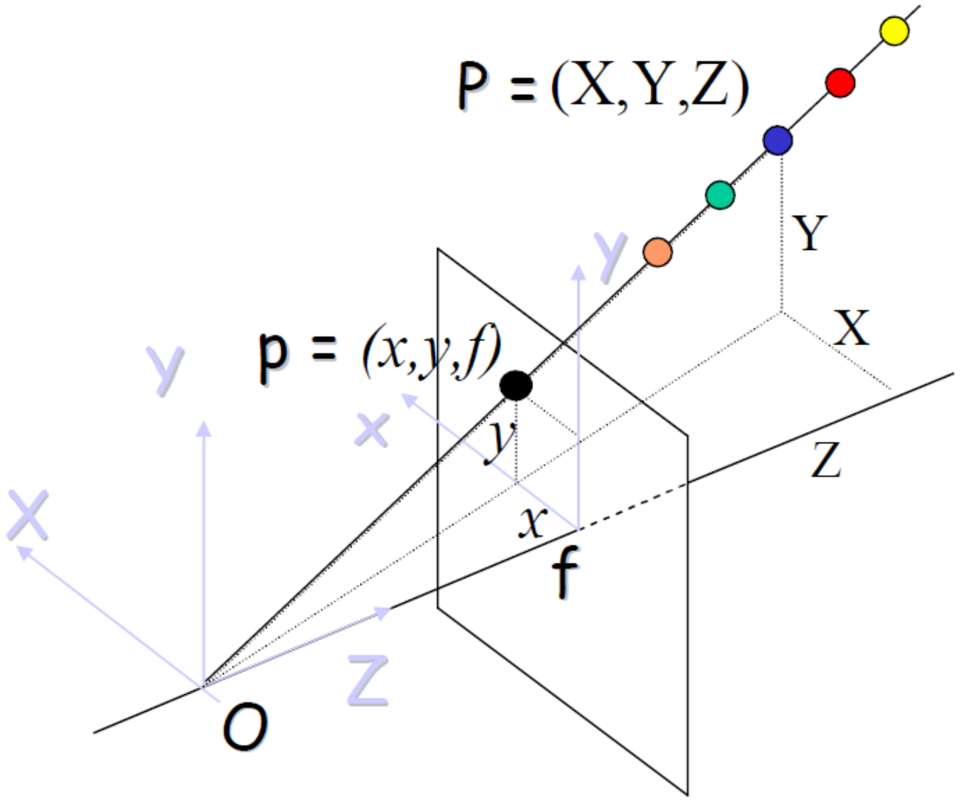
Why two views?

- Our eyes form a stereo system having two perspective views.



Why two views?

- Any point on the ray OP has image p.

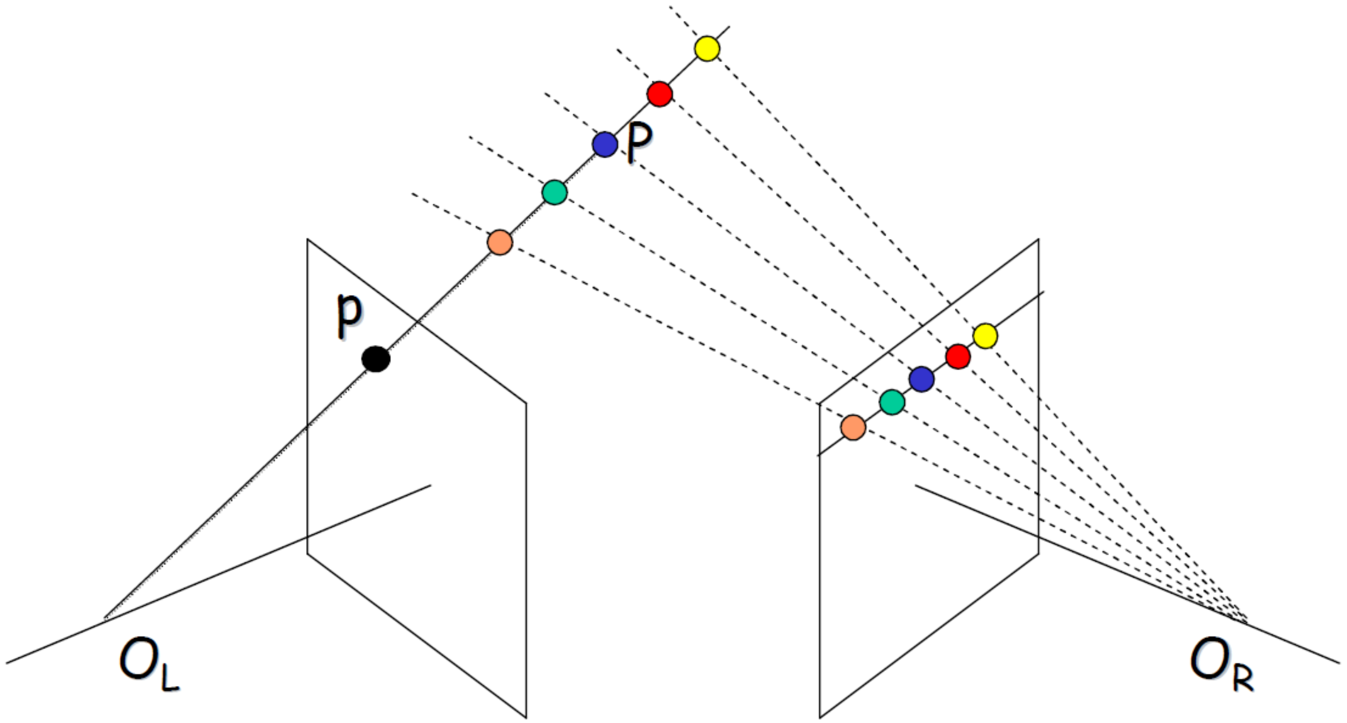


$$x = f\frac{X}{Z} = f\frac{kX}{kZ}$$

$$y = f\frac{Y}{Z} = f\frac{kY}{kZ}$$

Why two views?

- The second camera/eye resolves this issue and provides the depth of P.



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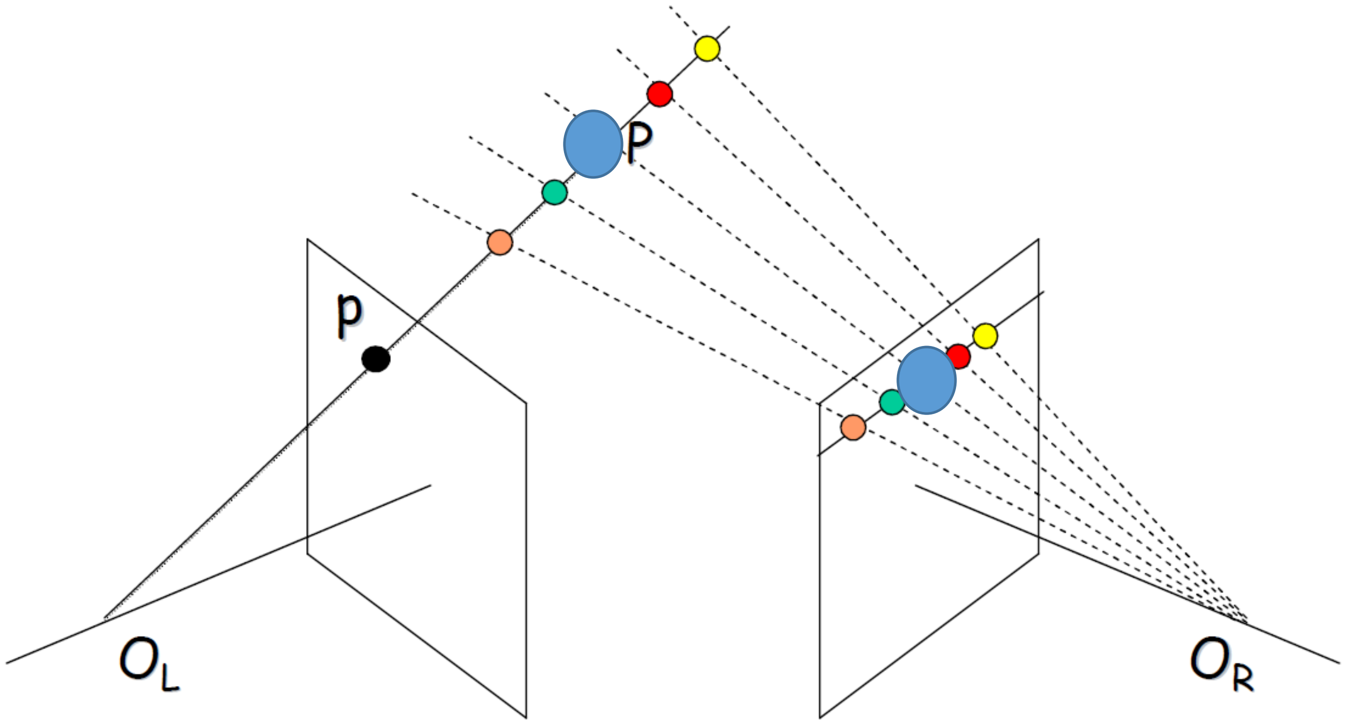
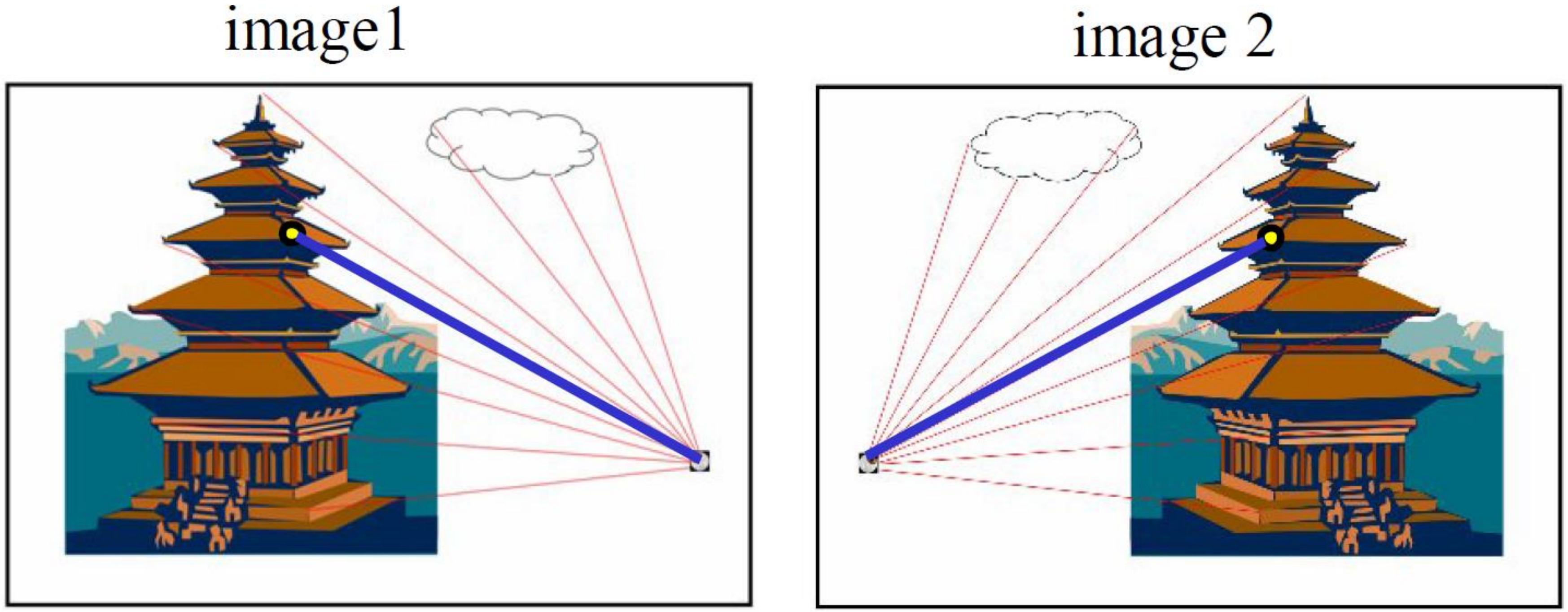


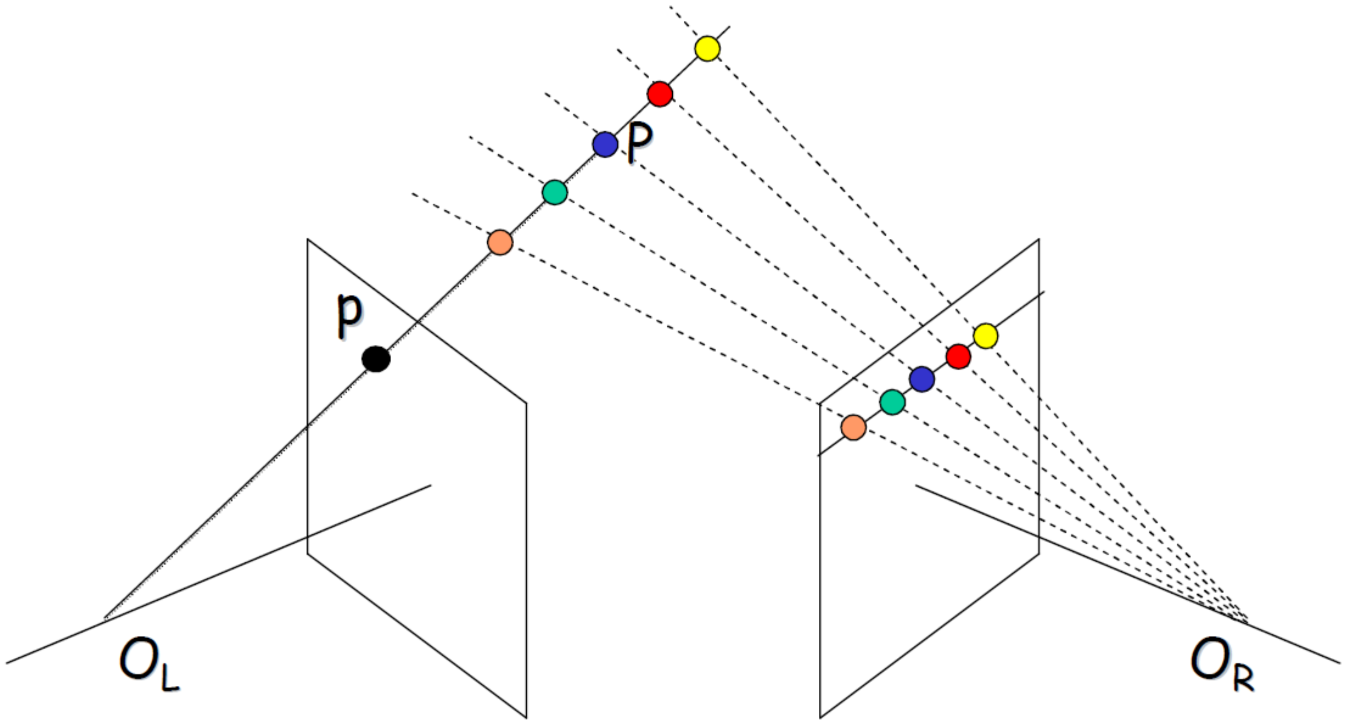
Eight Point Algorithm

- Given a point in the left image, we want to find its corresponding point in the right image.



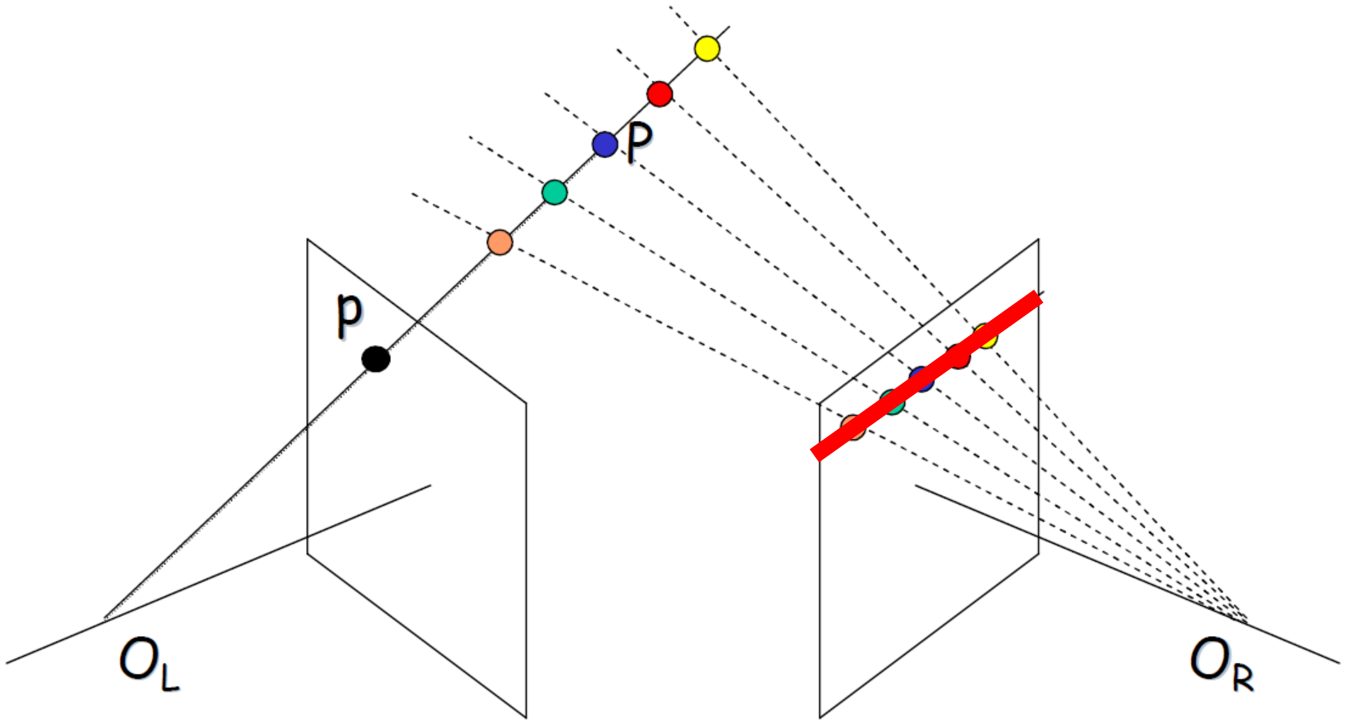
Eight Point Algorithm

- Based on the properties of perspective projections, given a point in the left image, we don't have to search the entire image to find the corresponding point.



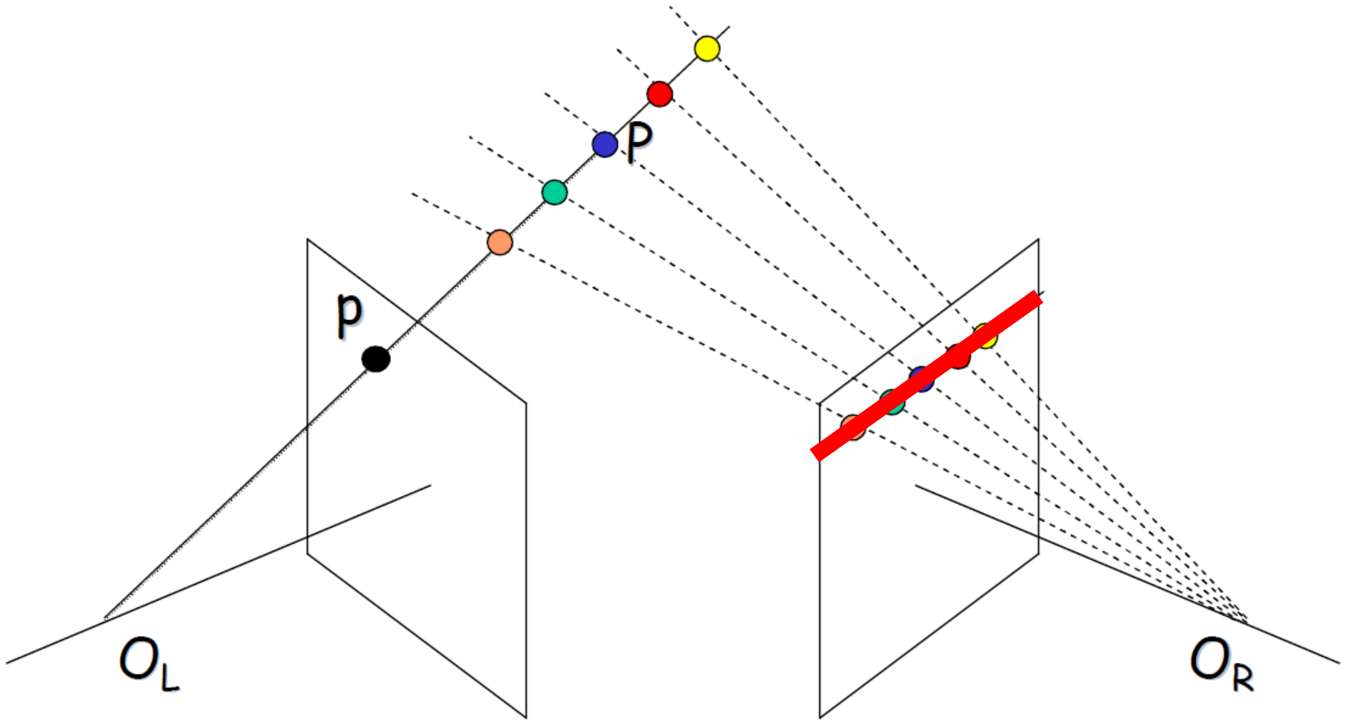
Eight Point Algorithm

- Only a single line is enough.



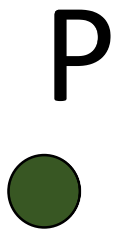
Eight Point Algorithm

- How can we find this line?



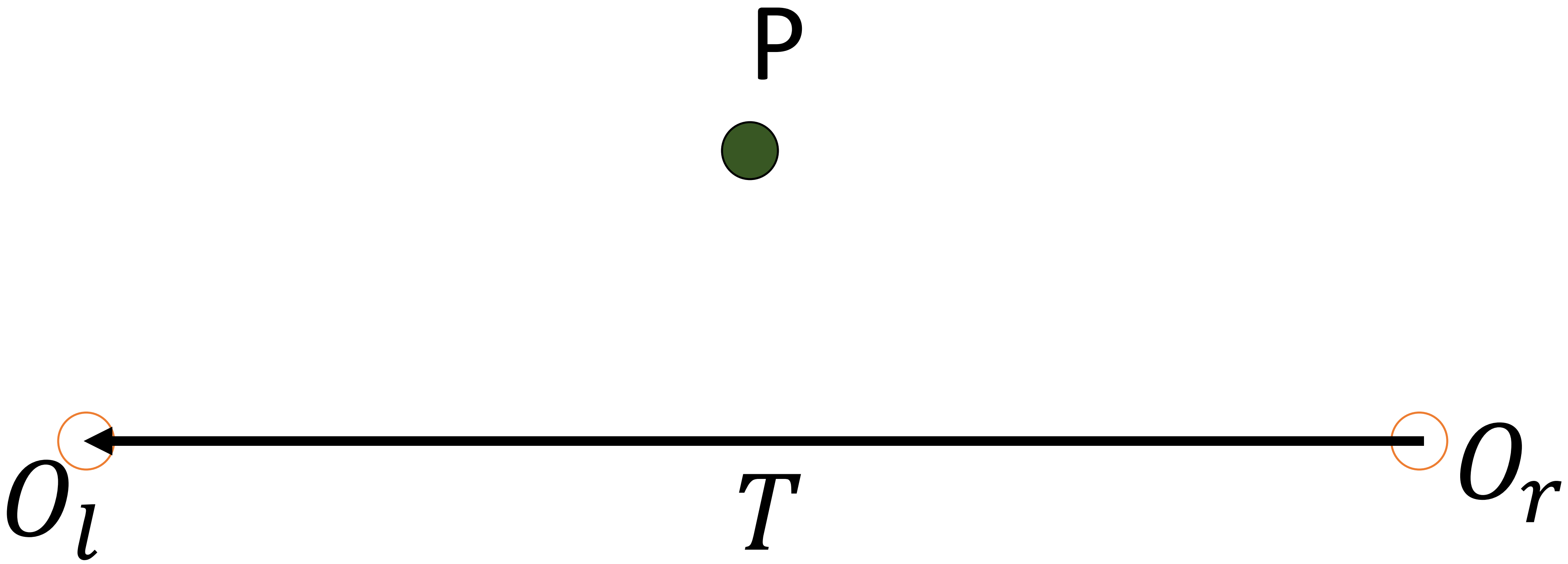
Epipolar Geometry

- Lets define our epipolar geometry:
- Point P in the world coordinate is visible from two point of views



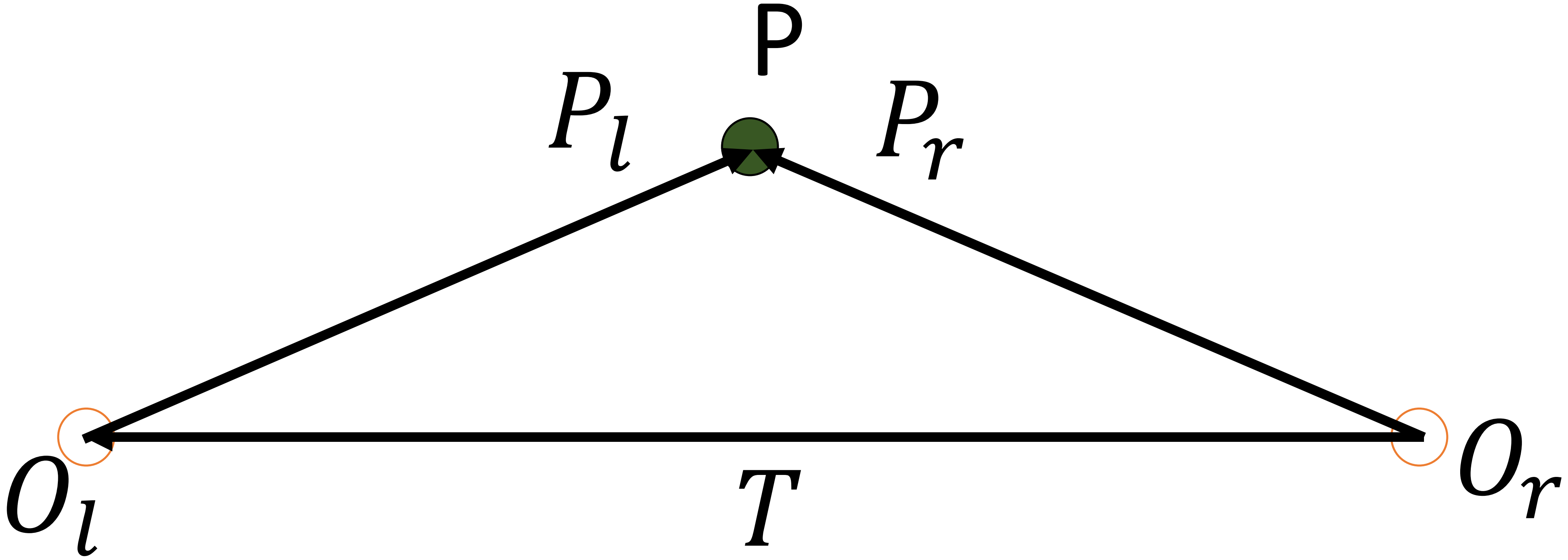
Epipolar Geometry

- Lets define our epipolar geometry:
- The two origin can be translated by vector T



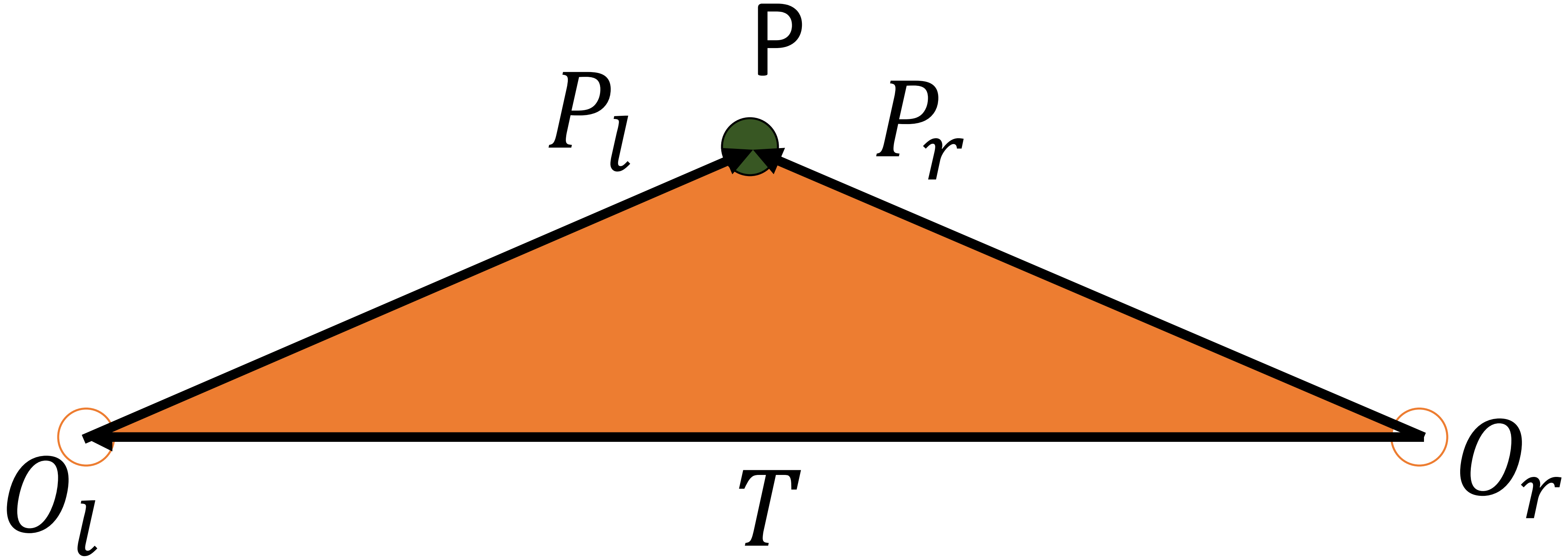
Epipolar Geometry

- Lets define our epipolar geometry:
- We can also form two vectors connecting the origins to the visible 3D point P.



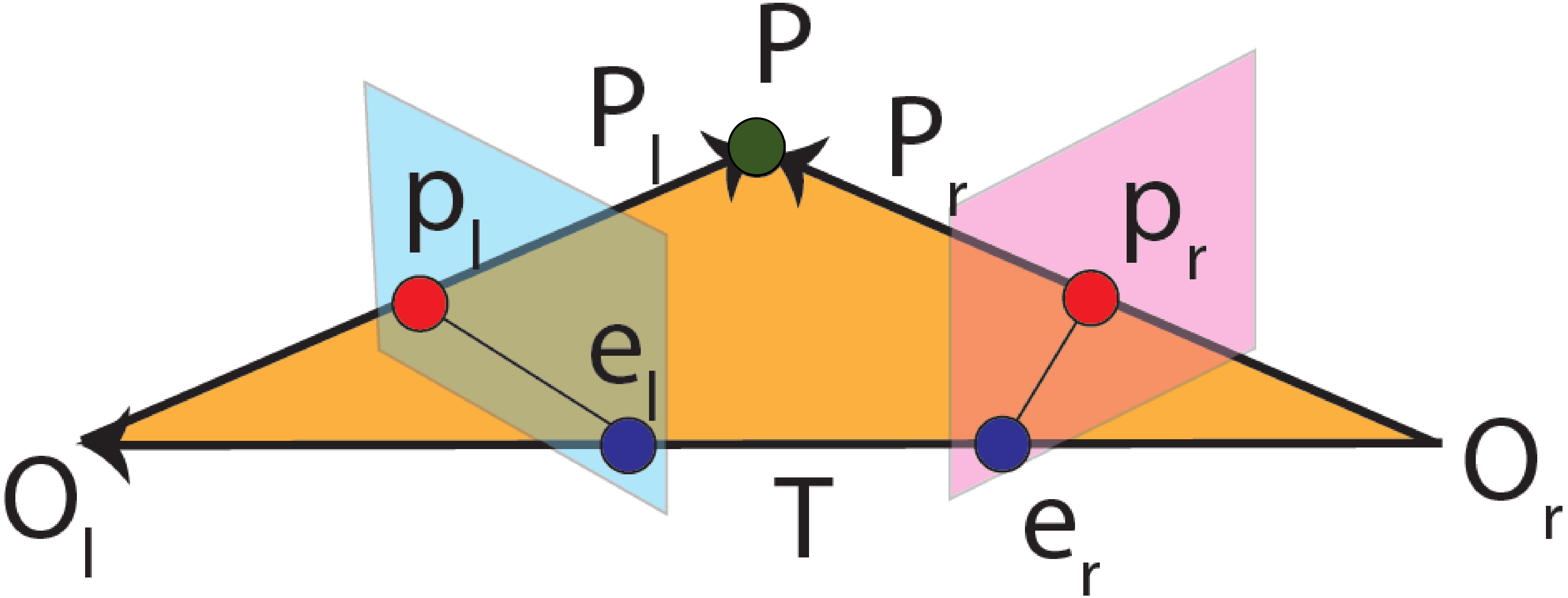
Epipolar Geometry

- Lets define our epipolar geometry:
- These three vectors form a plane called epipolar plane.



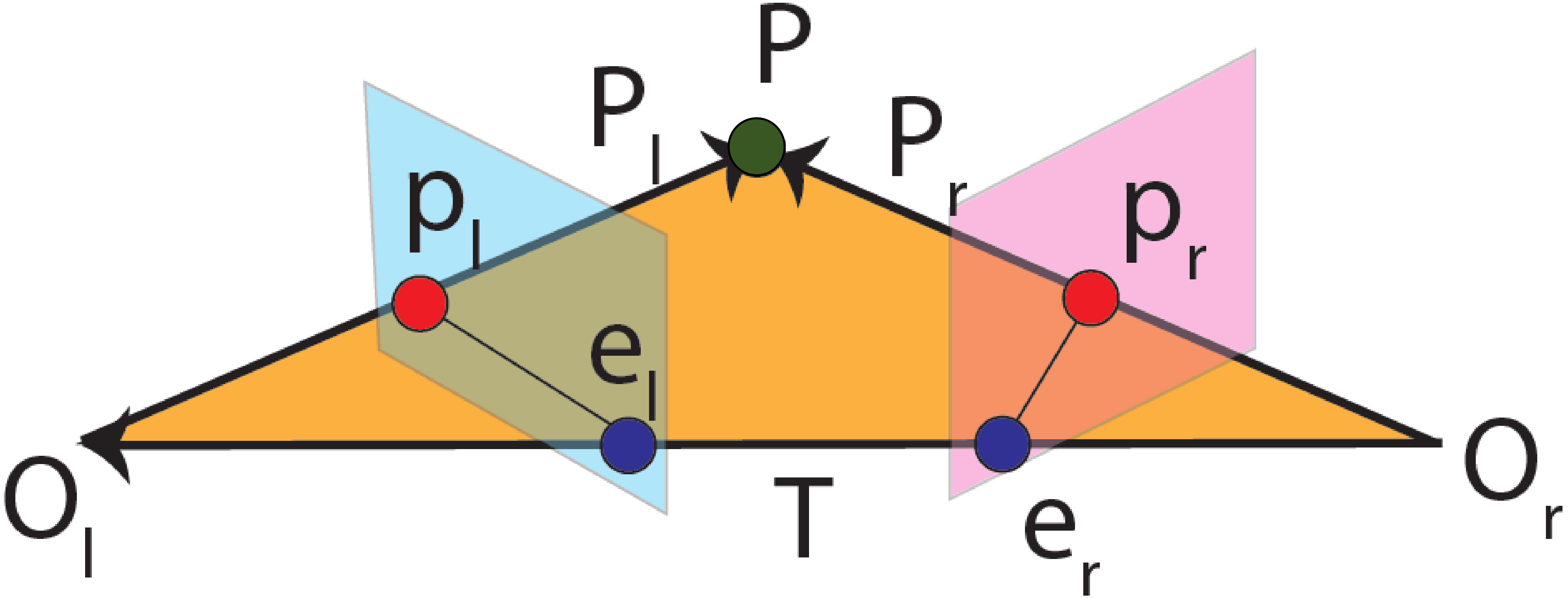
Epipolar Geometry

- Lets define our epipolar geometry:
- On the other hand, each view has an image plane on which point P is projected (p_l and p_r).



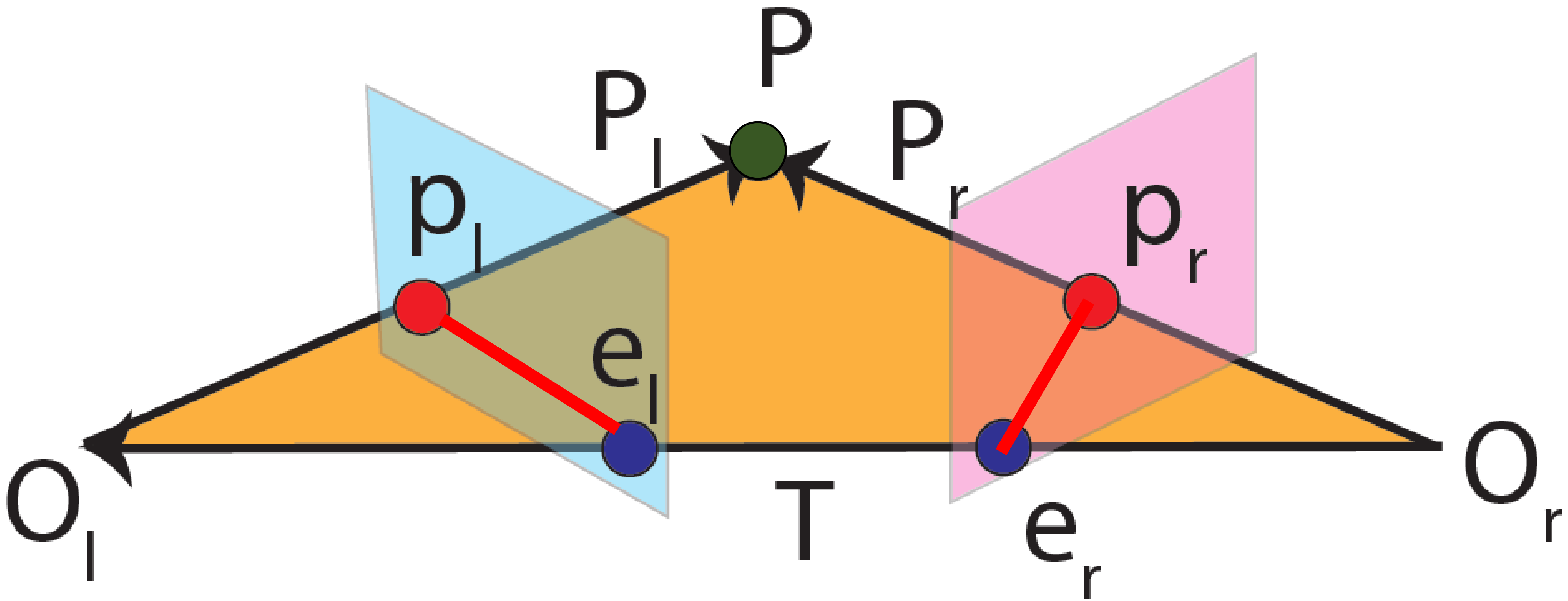
Epipolar Geometry

- Lets define our epipolar geometry:
- The intersection of these planes with vector T are called epipoles (e_l and e_r).



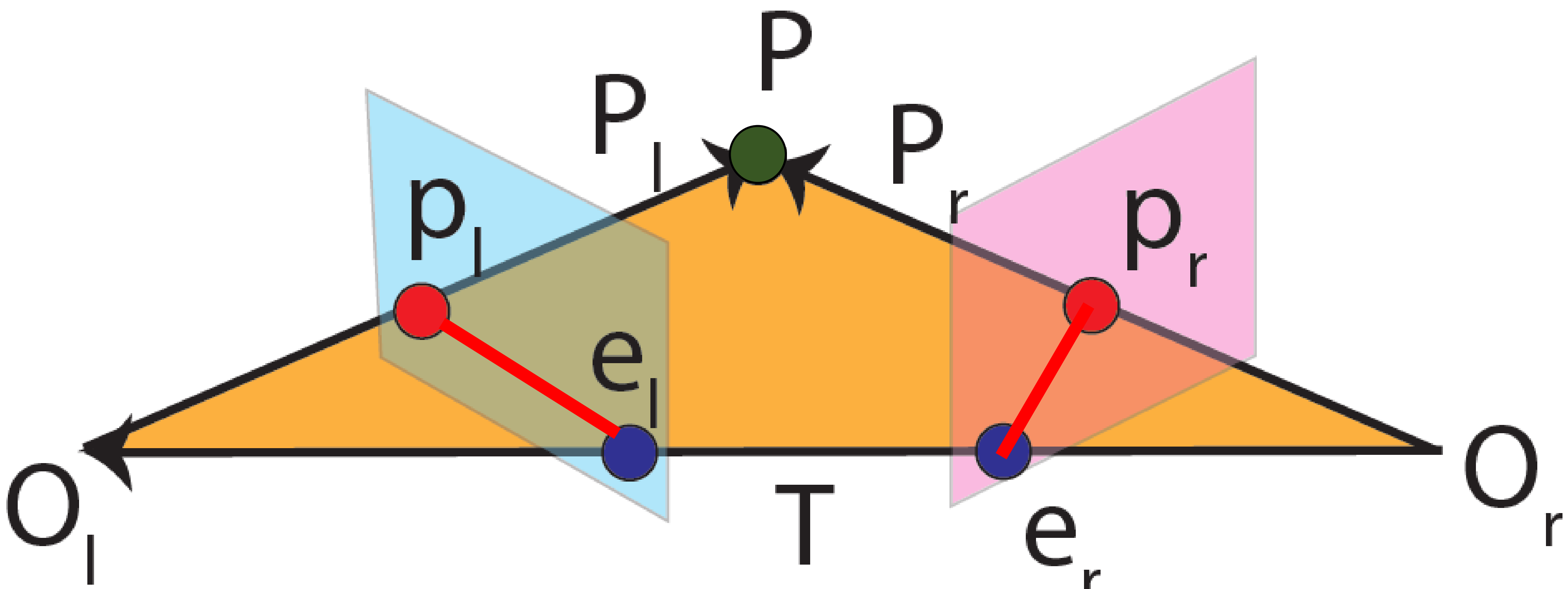
Epipolar Geometry

- Lets define our epipolar geometry:
- The intersection between each image plane and the epipolar plane are called conjugate epipolar lines.



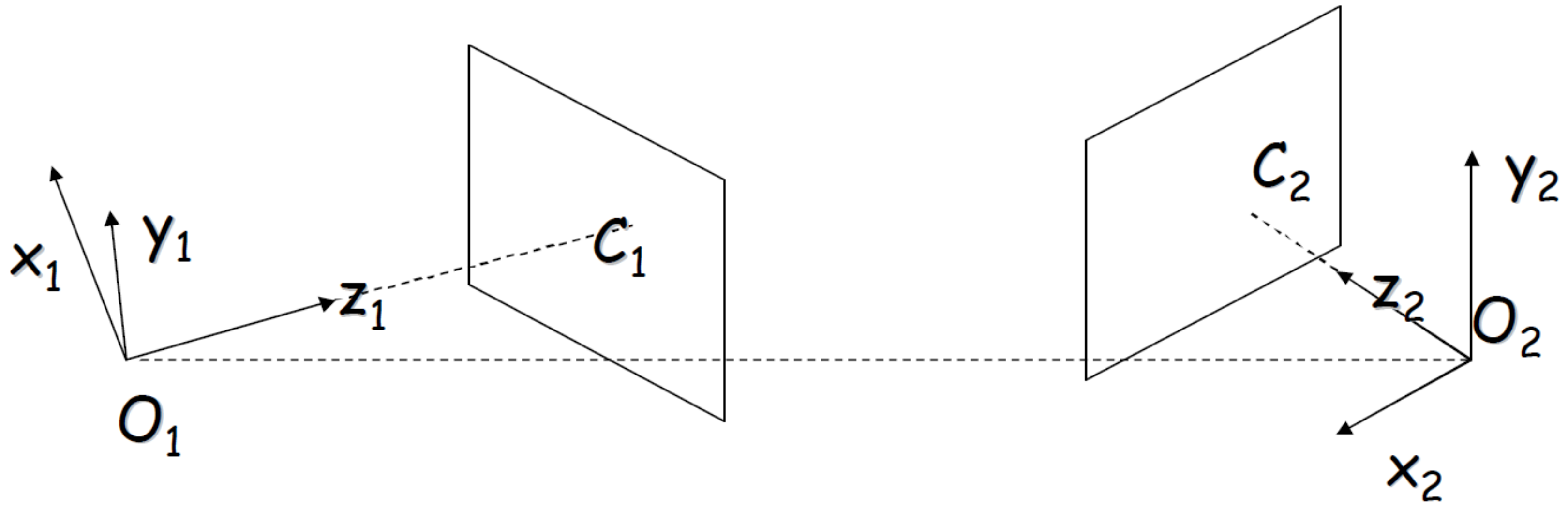
Epipolar Geometry

- We want to know if we can map points on one image plane into another.



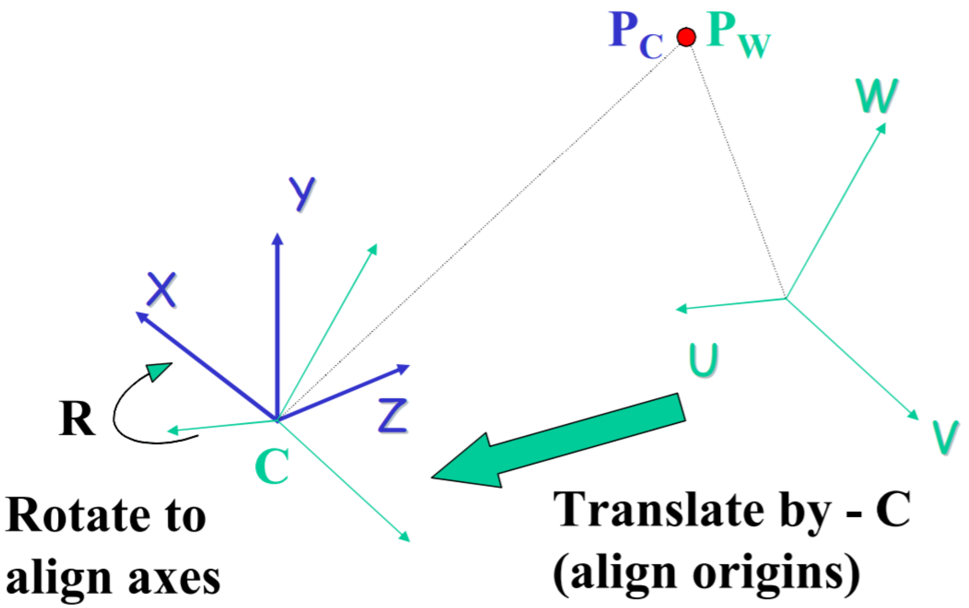
Eight Point Algorithm

- Since each camera has its own coordinate system, we need to connect them somehow (their coordinate system).



Eight Point Algorithm

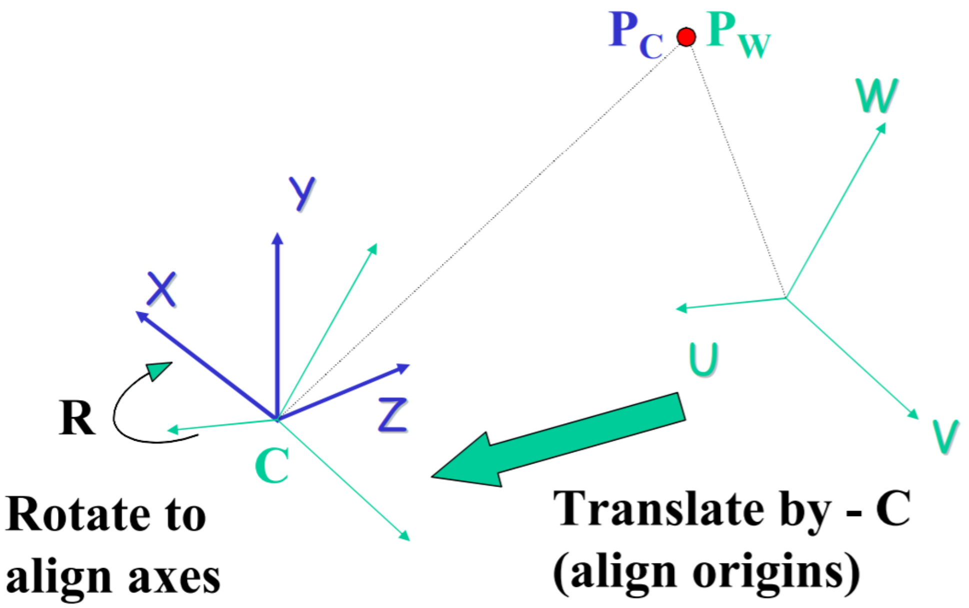
- Two coordinate systems can be aligned by a simple rotation and translation. (Recall world to camera transformation)



$$P_C = R (P_W - C)$$

Eight Point Algorithm

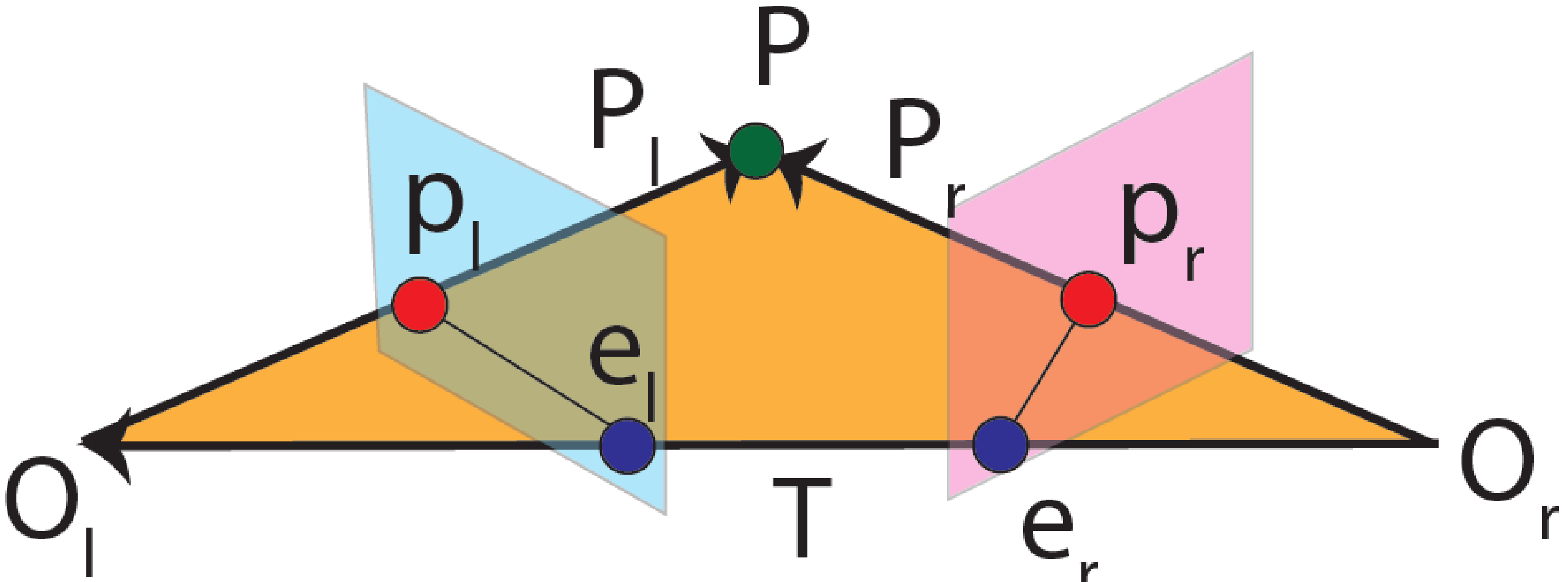
- We can use the same concept to find a connection between the two images.



$$P_C = R\,(P_W - C)$$

Epipolar Geometry

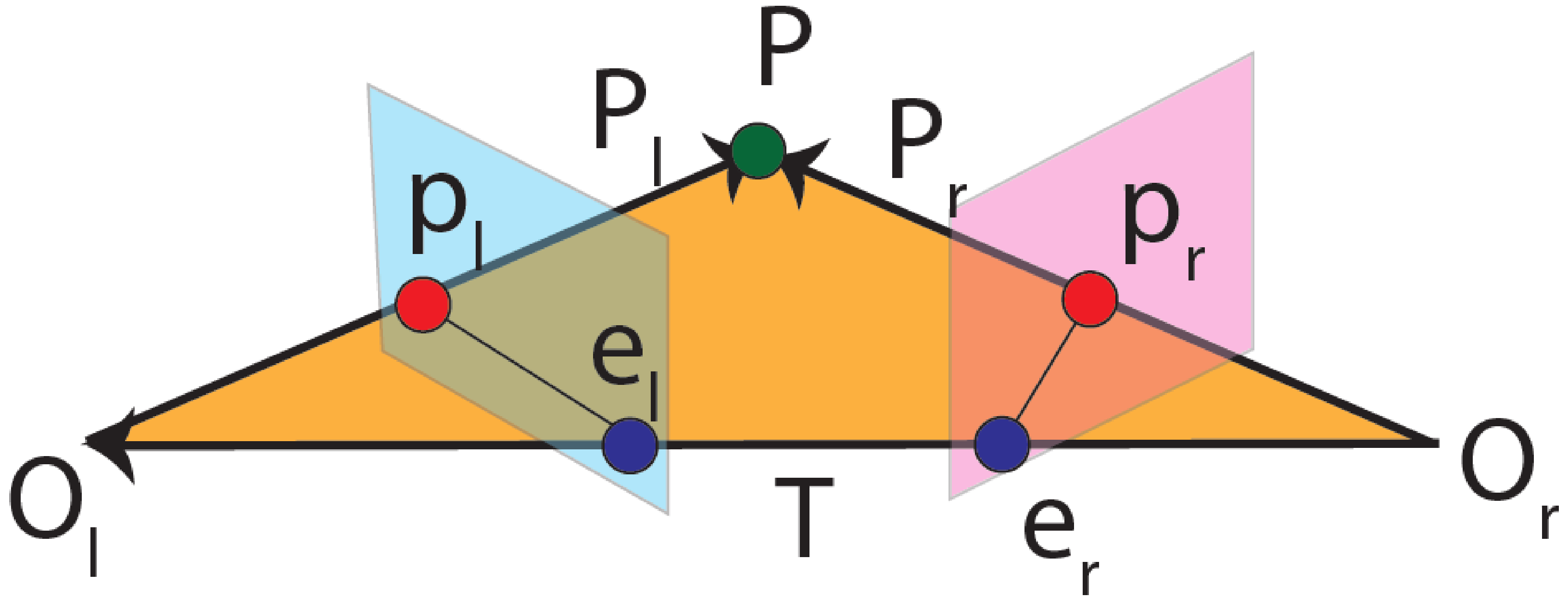
$$P_r = R(P_l - T)$$



Epipolar Geometry

$$P_r = R(P_l - T)$$

$$P_l - T = R^{-1}P_r = R^T P_r$$



Epipolar Geometry

$$(P_l - T)^T . (T \times P_l) = 0$$

Epipolar Geometry

$$(P_l - T)^T . (T \times P_l) = 0$$

$$\boxed{P_l - T = R^T P_r} \Rightarrow (R^T P_r)^T . (T \times P_l) = 0$$

From previous slide

Epipolar Geometry

$$(P_l - T)^T . (T \times P_l) = 0$$

$$P_l - T = R^T P_r \Rightarrow (R^T P_r)^T . (T \times P_l) = 0$$

$$\Rightarrow {P_r}^T R. (T \times P_l) = 0$$

Epipolar Geometry

$$(P_l - T)^T . (T \times P_l) = 0$$

$$P_l - T = R^T P_r \Rightarrow (R^T P_r)^T . (T \times P_l) = 0$$

$$\Rightarrow {P_r}^T R. \boxed{(T \times P_l)} = 0$$

Let’s discuss it a bit more.

Epipolar Geometry

$$T = (T_x, T_y, T_z)$$

$$P_l = (P_{lx}, P_{ly}, P_{lz})$$

$T \times P_l$

Epipolar Geometry

$$T = (T_x, T_y, T_z)$$

$$P_l = (P_{lx}, P_{ly}, P_{lz})$$

$$T \times P_l = (T_y P_{lz} - T_z P_{ly})\vec{i} + (T_z P_{lx} - T_x P_{lz})\vec{j} + (T_x P_{ly} - T_y P_{lx})\vec{k}$$

Epipolar Geometry

$$T = (T_x, T_y, T_z)$$

$$P_l = (P_{lx}, P_{ly}, P_{lz})$$

$$T \times P_l = (T_y P_{lz} - T_z P_{ly})\vec{\imath} + (T_z P_{lx} - T_x P_{lz})\vec{\jmath} + (T_x P_{ly} - T_y P_{lx})\vec{k}$$

$$T \times P_l = SP_l = \begin{bmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & -T_x & 0 \end{bmatrix} \begin{bmatrix} P_{lx} \\ P_{ly} \\ P_{lz} \end{bmatrix} = \begin{bmatrix} T_y P_{lz} - T_z P_{ly} \\ T_z P_{lx} - T_x P_{lz} \\ T_x P_{ly} - T_y P_{lx} \end{bmatrix}$$

Epipolar Geometry

$$T = (T_x, T_y, T_z)$$

$$P_l = (P_{lx}, P_{ly}, P_{lz})$$

$$T \times P_l = (T_y P_{lz} - T_z P_{ly})\vec{i} + (T_z P_{lx} - T_x P_{lz})\vec{j} + (T_x P_{ly} - T_y P_{lx})\vec{k}$$

$$T \times P_l = SP_l = \underbrace{\boxed{\begin{bmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & -T_x & 0 \end{bmatrix}}}_{S} \begin{bmatrix} P_{lx} \\ P_{ly} \\ P_{lz} \end{bmatrix} = \begin{bmatrix} T_y P_{lz} - T_z P_{ly} \\ T_z P_{lx} - T_x P_{lz} \\ T_x P_{ly} - T_y P_{lx} \end{bmatrix}$$

Epipolar Geometry

$$(P_l - T)^T . (T \times P_l) = 0$$

$$P_l - T = R^T P_r \Rightarrow (R^T P_r)^T . (T \times P_l) = 0$$

$$\Rightarrow {P_r}^T R. \underbrace{\boxed{(T \times P_l)}}_{SP_l} = 0$$

Epipolar Geometry

$$(P_l - T)^T . (T \times P_l) = 0$$

$$P_l - T = R^T P_r \Rightarrow (R^T P_r)^T . (T \times P_l) = 0$$

$$\Rightarrow {P_r}^T R. (T \times P_l) = 0$$

$$\Rightarrow {P_r}^T R S P_l = 0$$

Epipolar Geometry

- Essential Matrix

$$(P_l - T)^T . (T \times P_l) = 0$$

$$P_l - T = R^T P_r \Rightarrow (R^T P_r)^T . (T \times P_l) = 0$$

$$\Rightarrow {P_r}^T R . (T \times P_l) = 0$$

$$\Rightarrow {P_r}^T \underbrace{\boxed{RS}}_{E} P_l = 0$$

Epipolar Geometry

- Essential Matrix

$$(P_l - T)^T . (T \times P_l) = 0$$

$$P_l - T = R^T P_r \Rightarrow (R^T P_r)^T . (T \times P_l) = 0$$

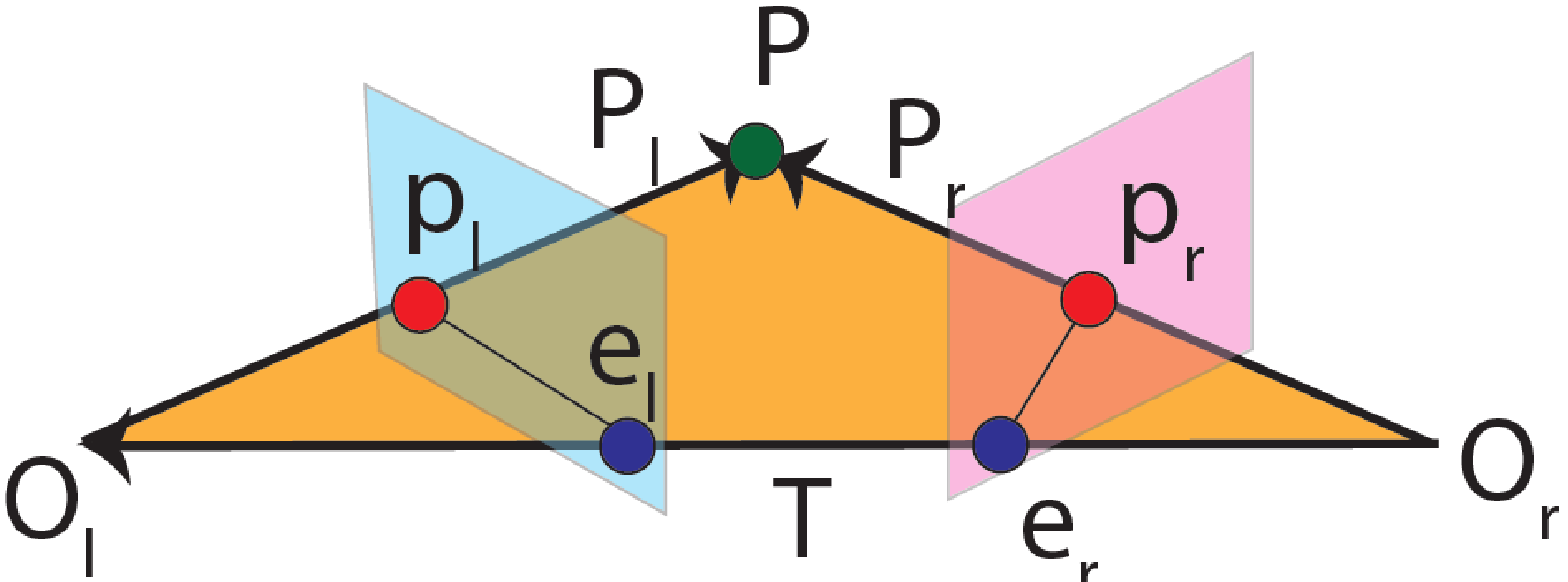
$$\Rightarrow {P_r}^T R. (T \times P_l) = 0$$

$$\Rightarrow {P_r}^T E P_l = 0$$

Epipolar Geometry

- Essential Matrix

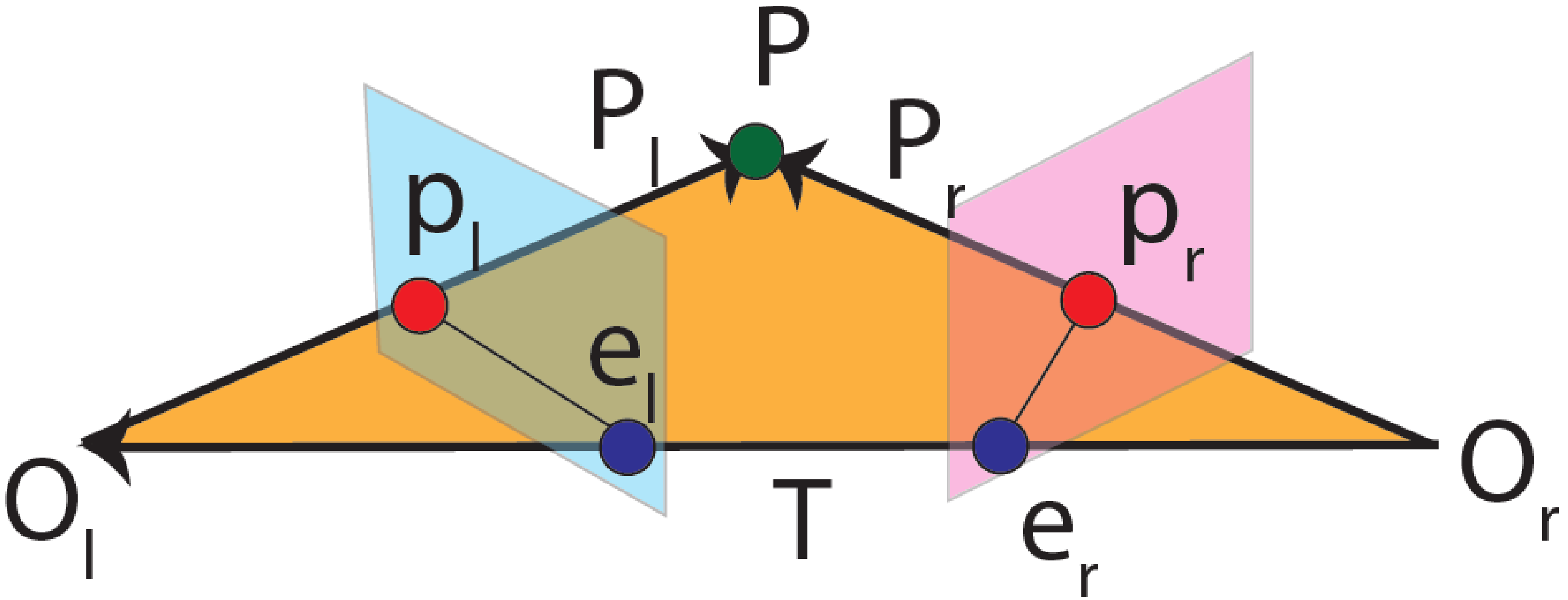
$$P_r{}^T E P_l = 0$$



Epipolar Geometry

- We are looking to map points p_r and p_l.

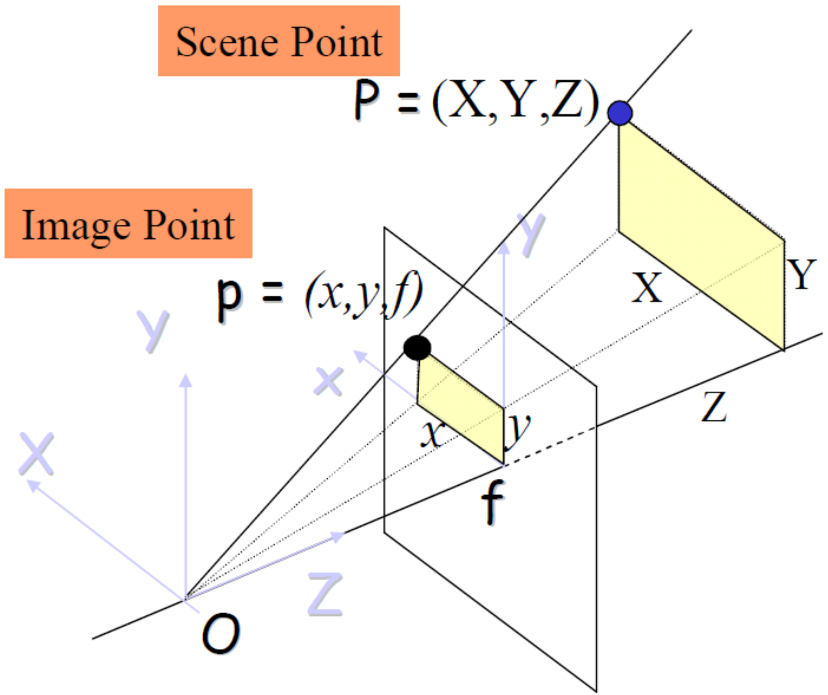
$$P_r{}^T E P_l = 0$$



Epipolar Geometry

- Extend to point on image plane

$$P_r^{\,T} E P_l = 0$$



Perspective Projection Eqns

$$x = f\frac{X}{Z}$$

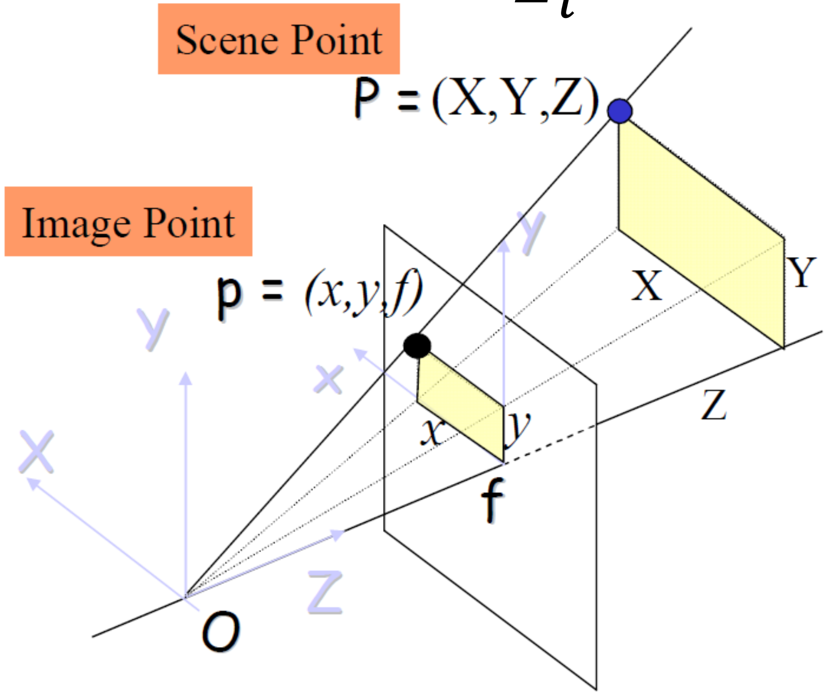
$$y = f\frac{Y}{Z}$$

Epipolar Geometry

- Extend to point on image plane

$$\boxed{P_r}^T E \boxed{P_l} = 0$$

$$p_r = \frac{f_r}{Z_r} P_r \qquad p_l = \frac{f_l}{Z_l} P_l$$



Perspective Projection Eqns

$$x = f\frac{X}{Z}$$

$$y = f\frac{Y}{Z}$$

Epipolar Geometry

- Extend to point o image plane

$$P_r{}^T E P_l = 0$$

$$\left(\boxed{\frac{Z_r}{f_r}} p_r\right)^T E \left(\boxed{\frac{Z_l}{f_l}} p_l\right) = 0$$

Constant and non zero

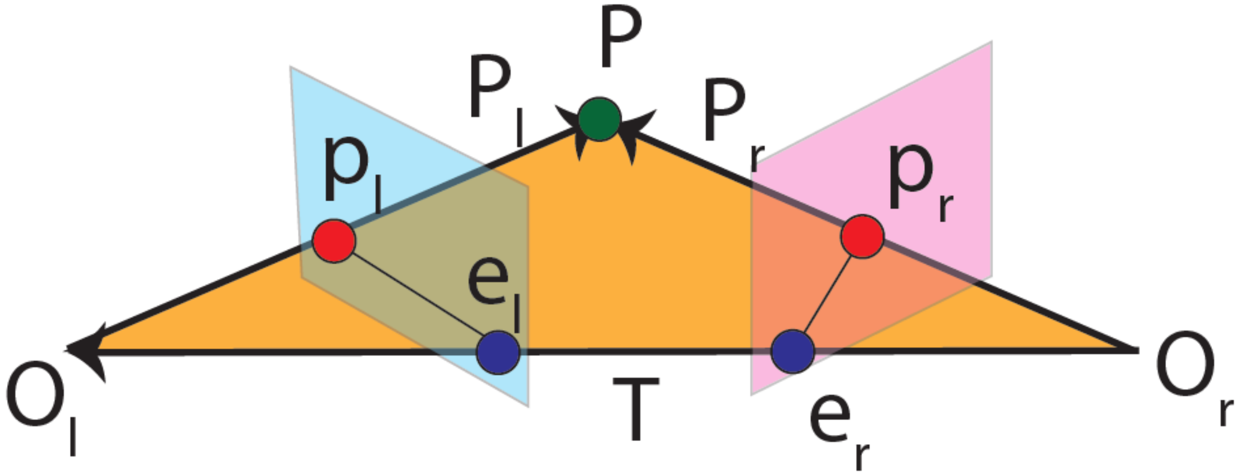
Epipolar Geometry

- Extend to point o image plane

$$P_r{}^T E P_l = 0$$

$$\left(\frac{Z_r}{f_r} p_r\right)^T E \left(\frac{Z_l}{f_l} p_l\right) = 0$$

$$\Rightarrow p_r{}^T E p_l = 0$$

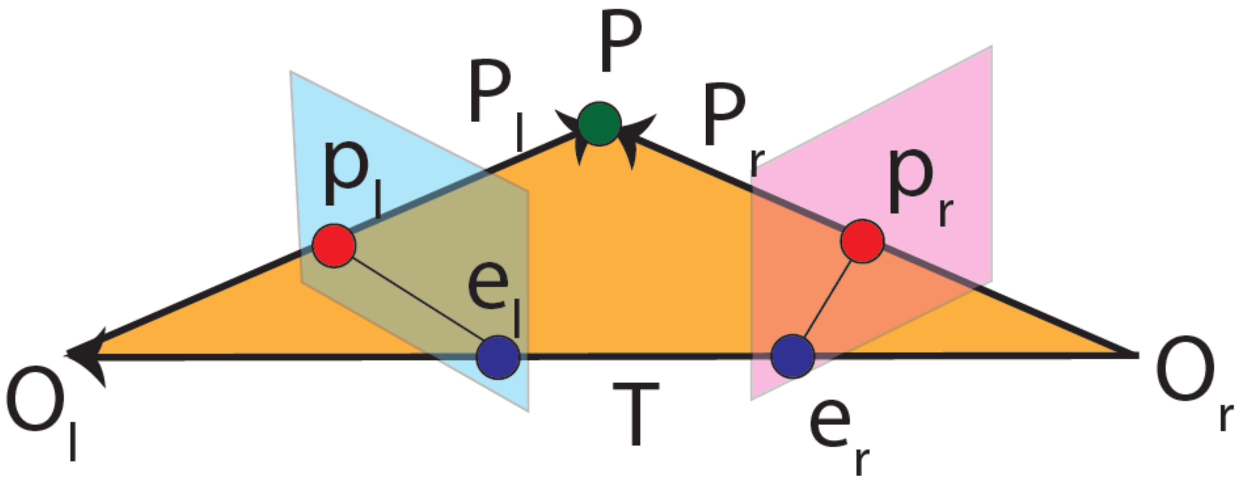


Epipolar Geometry

- Extend to point on image plane

$$p_r{}^T E p_l = 0$$

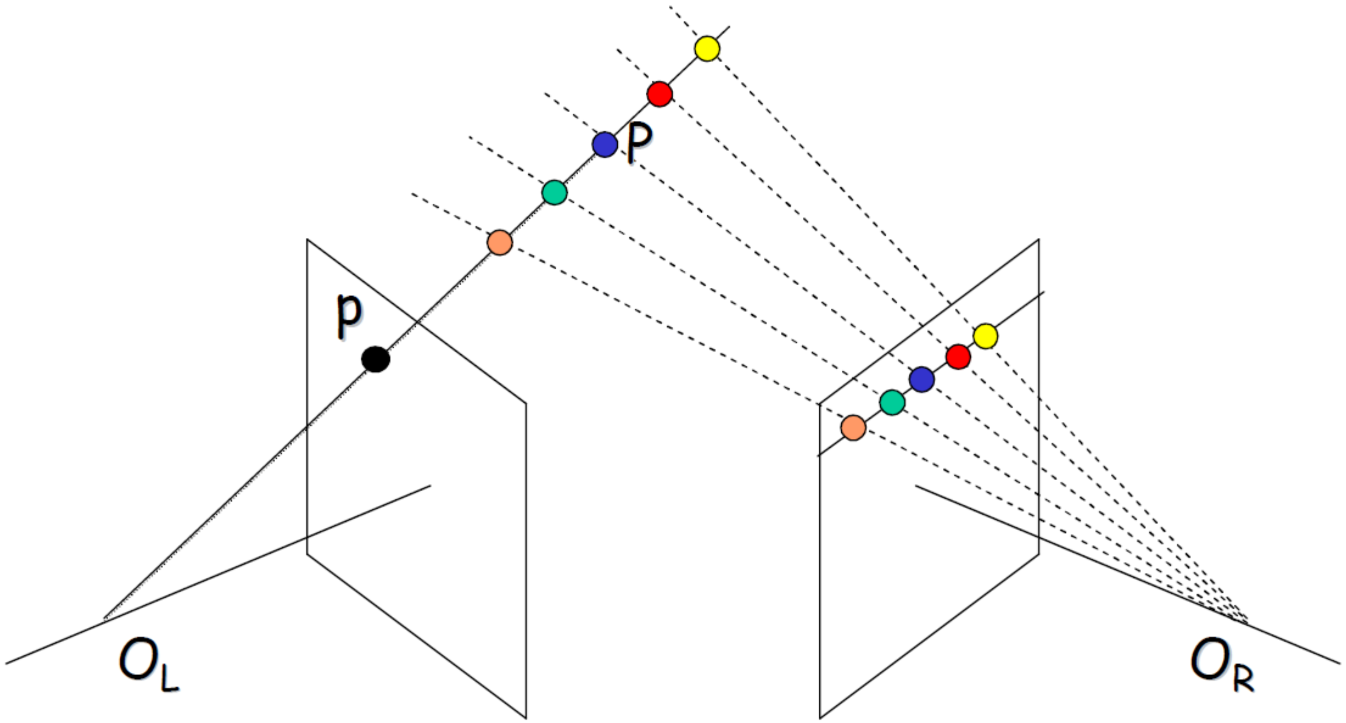
Longuet-Higgins Equation



Epipolar Geometry

- What a bout the lines?

We said that to find the correspondence between two points, one point in an image plane is mapped to a line in the other image plane.



Epipolar Geometry

- What a bout the lines?
- A line in an image is defined as: $au + bv + c = 0$

Epipolar Geometry

- What a bout the lines?
- A line in an image is defined as: $au + bv + c = 0$

$$p = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \qquad l = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Epipolar Geometry

- What a bout the lines?
- A line in an image is defined as: $au + bv + c = 0 \rightarrow p^T\ l = l^T p = 0$

$$p = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \qquad l = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Epipolar Geometry

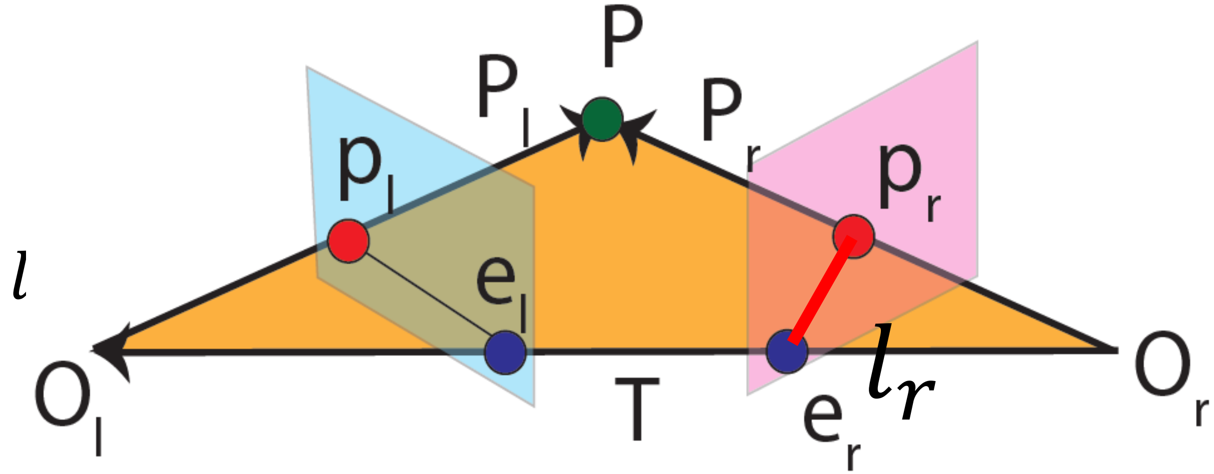
- What a bout the lines?
- A line in an image is defined as:

$$au + bv + c = 0 \rightarrow p^T\ l = l^T p = 0$$

$$p = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \qquad l = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

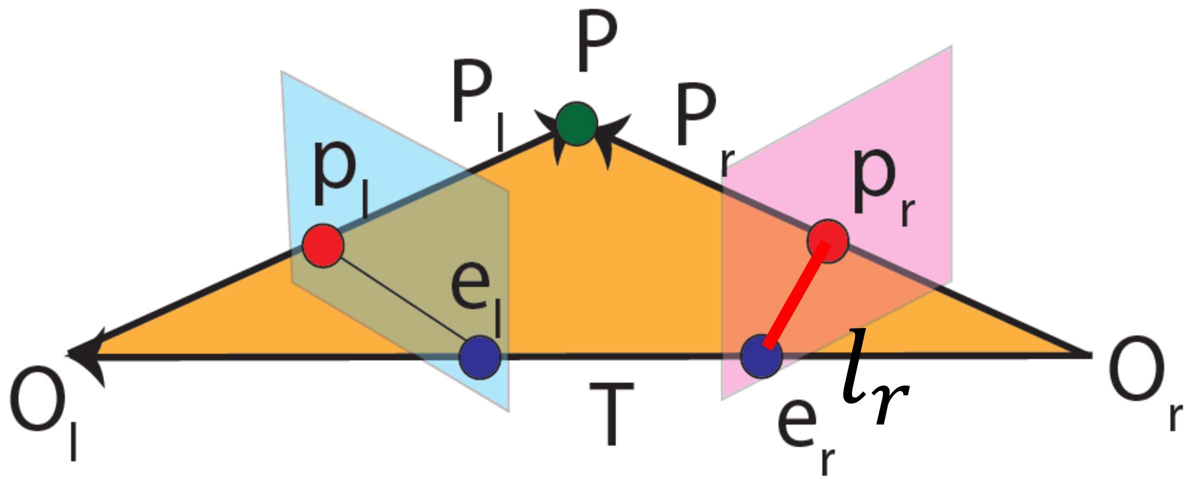
- p_r belongs to epipolar line in right image:

$$p_r{}^T l_r = 0$$



Epipolar Geometry

- p_r belongs to epipolar line in right image: $p_r{}^T l_r = 0$
- On the other hand $p_r{}^T E p_l = 0$ (Longuet-Higgins equation)

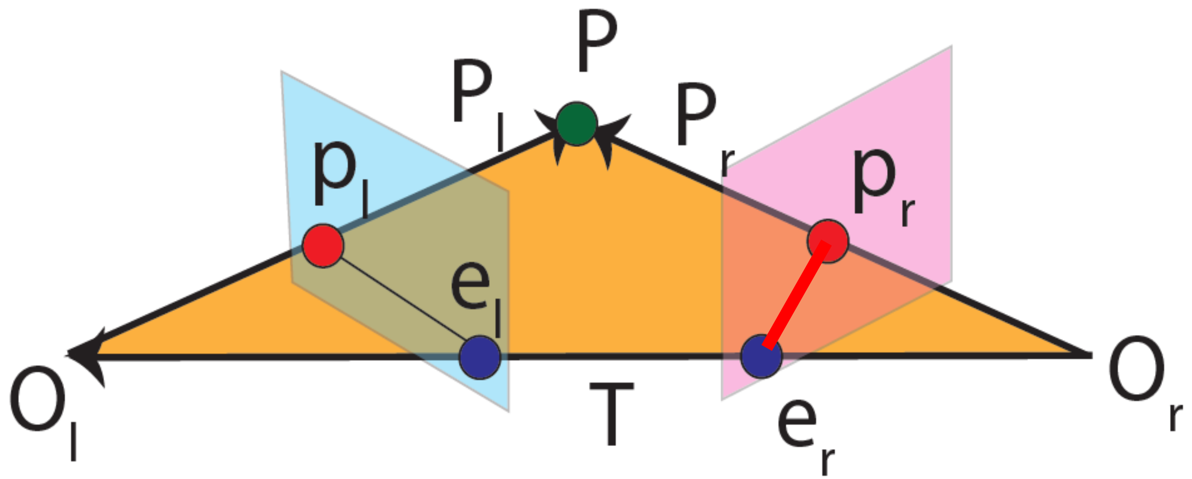


Epipolar Geometry

- p_r belongs to epipolar line in right image: $p_r{}^T \boxed{l_r} = 0$

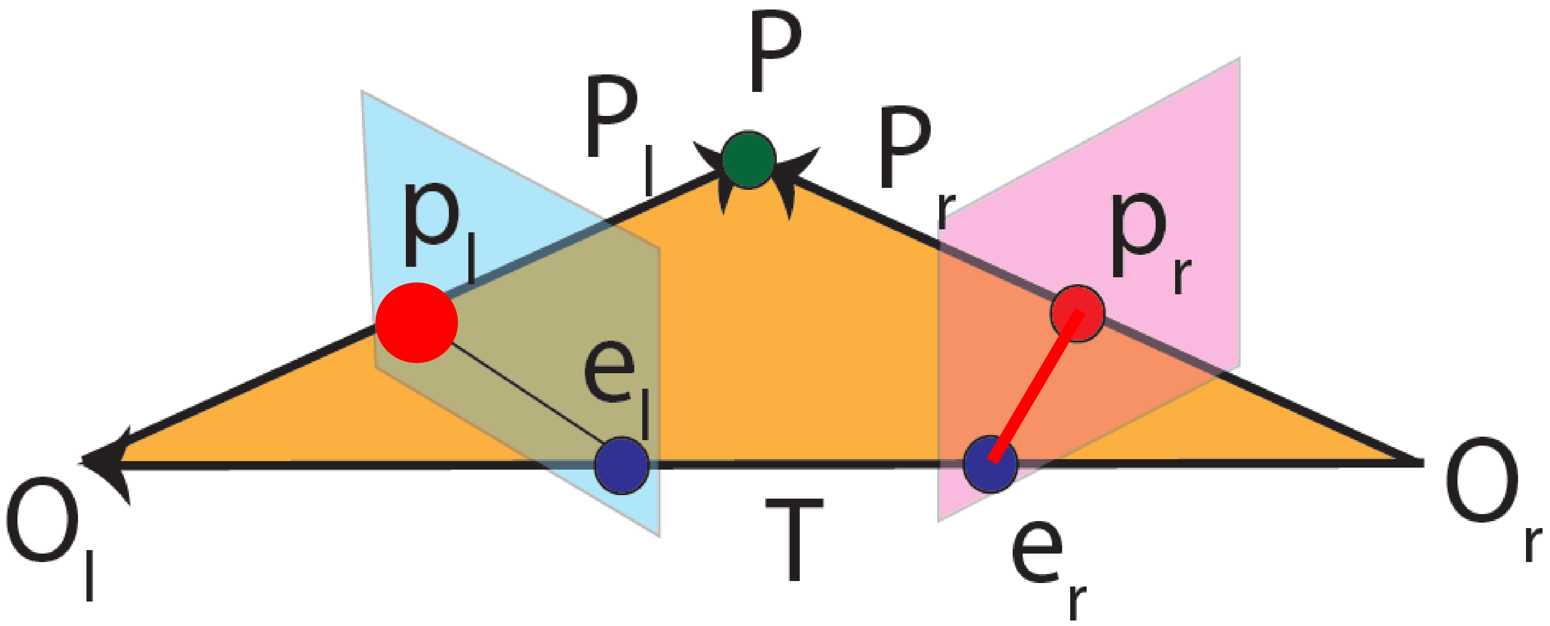
- On the other hand $p_r{}^T \boxed{Ep_l} = 0$

- Therefore $l_r = Ep_l$



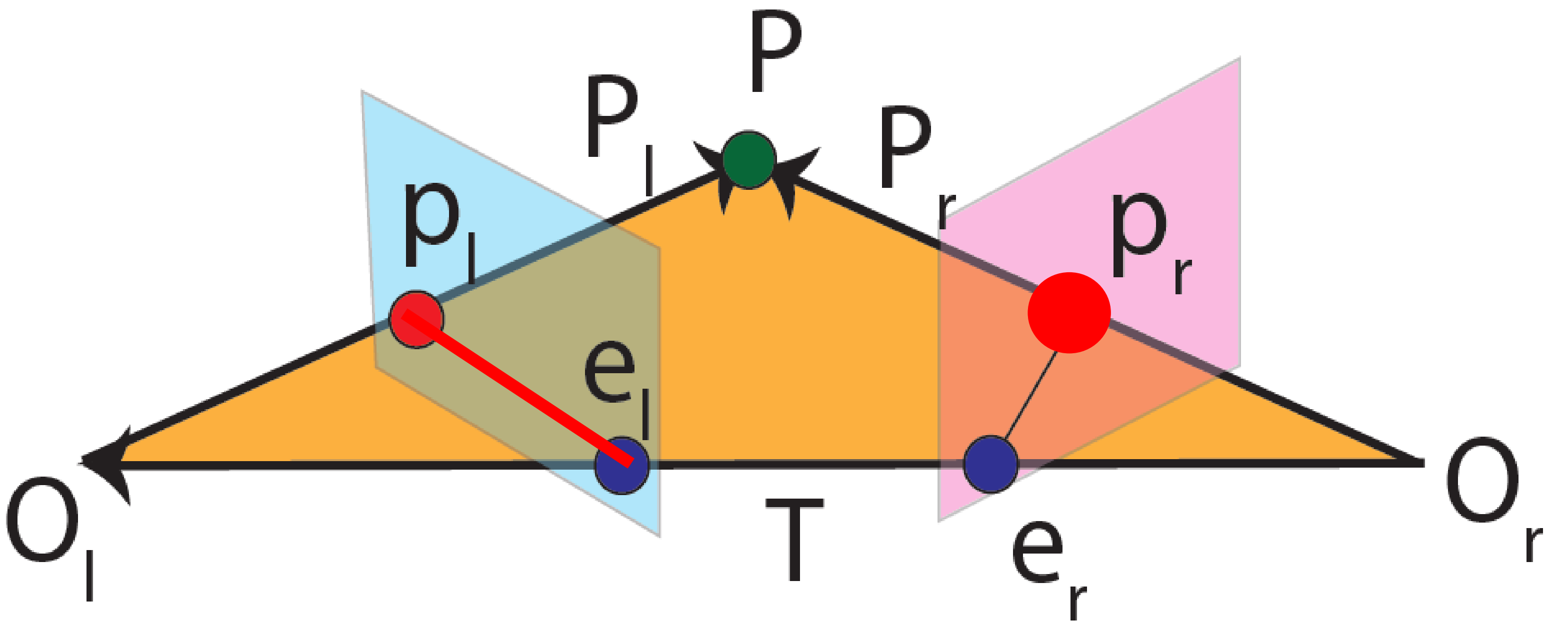
Epipolar Geometry

- Therefore $l_r = Ep_l$
- Therefore, a point in the left image is mapped to a line in the right image



Epipolar Geometry

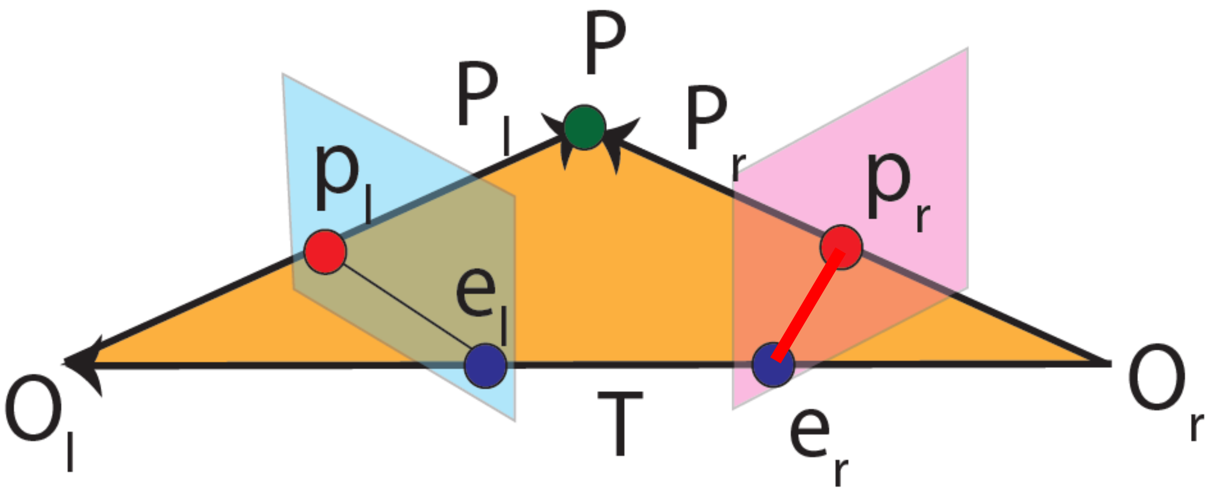
- Similarly $p_l{}^T l_l = 0$ and $l_l = E^T p_r$



Epipolar Geometry

- Epipoles belong to epipolar lines

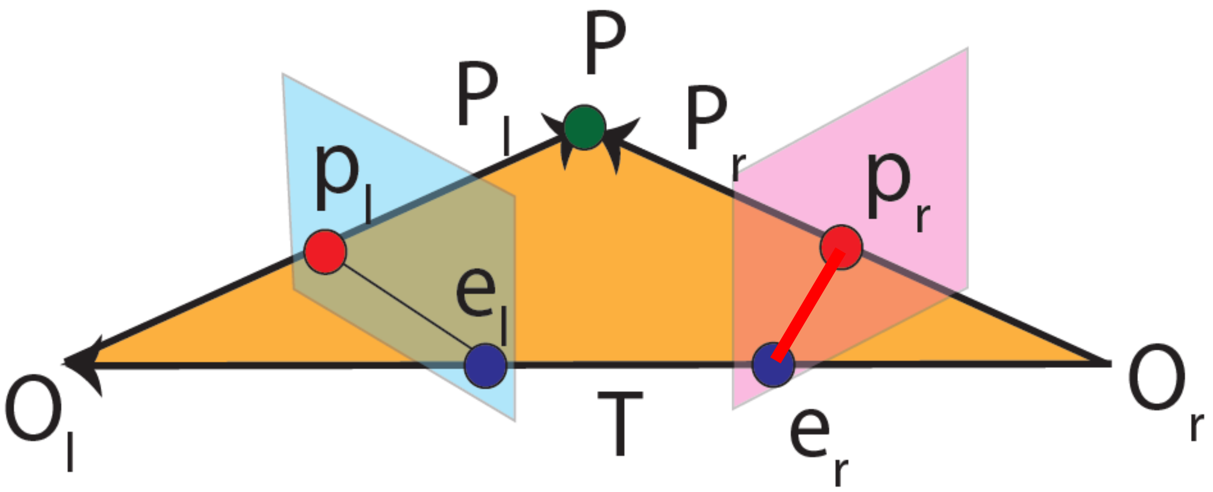
$$e_r^T E p_l = 0$$



Epipolar Geometry

- Epipoles belong to epipolar lines

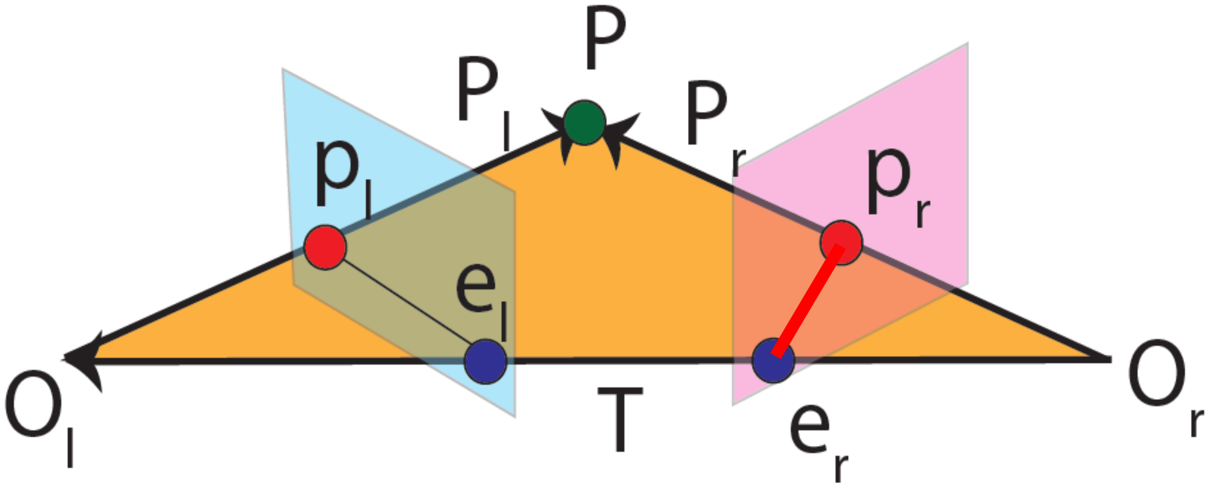
$$e_r^T \underbrace{\boxed{Ep_l}}_{l_r} = 0$$



Epipolar Geometry

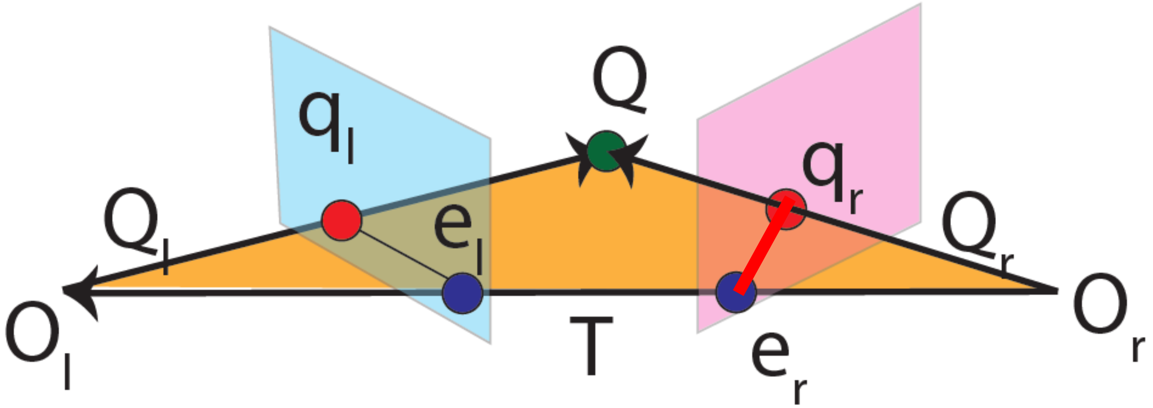
- Epipoles belong to epipolar lines

$$e_r^T E p_l = 0$$



- Since epipoles belong to all epipolar lines:

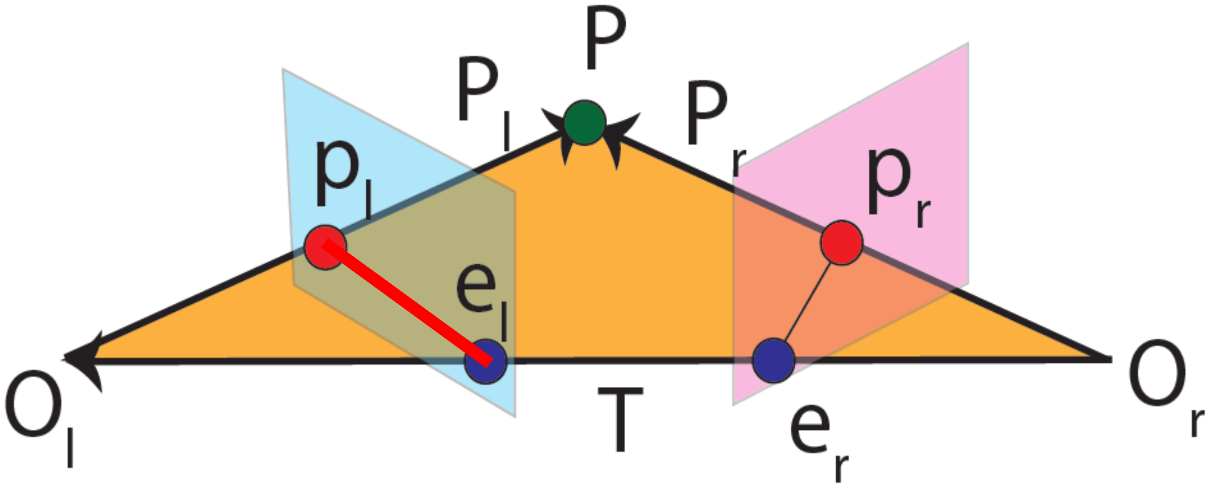
$$e_r^T E q_l = 0 \rightarrow e_r^T E = 0$$



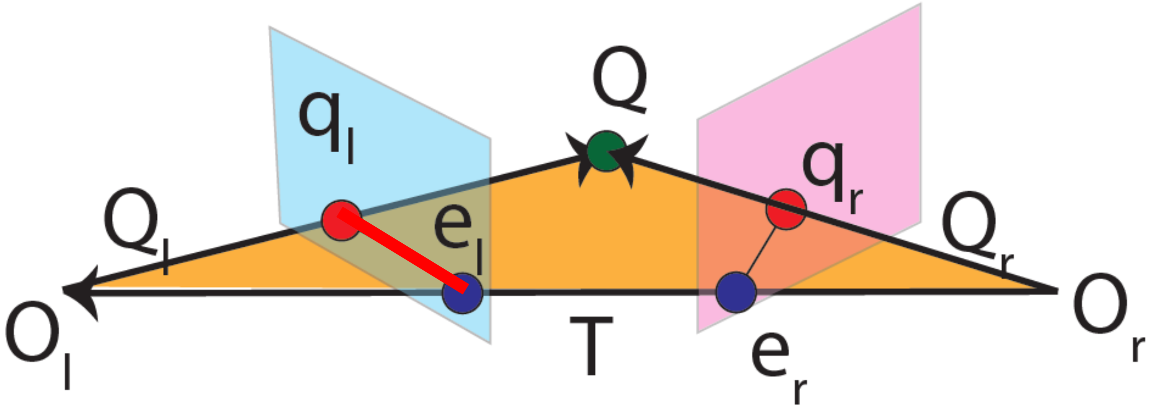
Epipolar Geometry

- Also True for lines in left image

$$e_r^T E p_l = 0 \quad p_r^T E e_l = 0$$



$$e_r^T E = 0 \;\; E e_l = 0$$



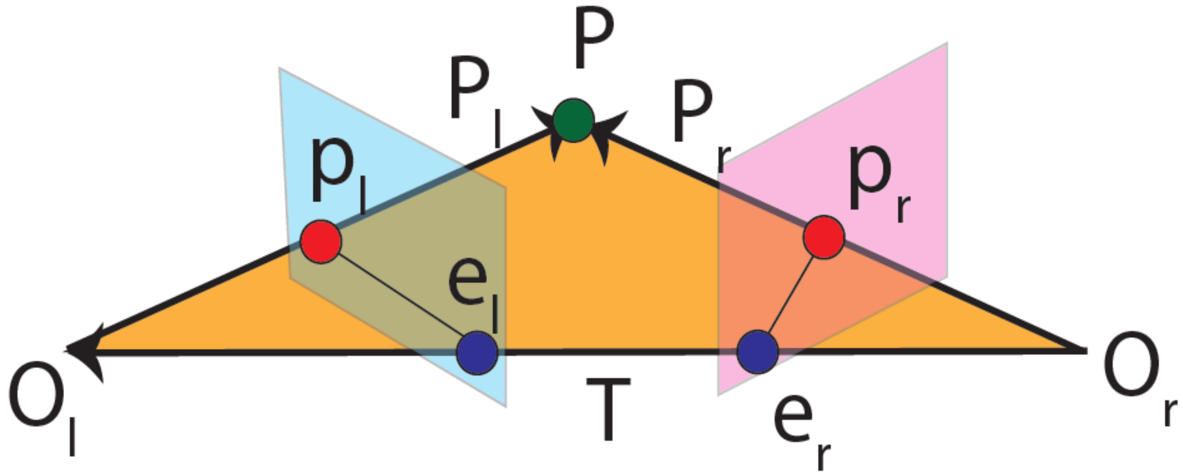
Essential Matrix Summary

- Longuet-Higgins Equation $p_r{}^T E p_l = 0$

- Epipolar lines: $p_r{}^T l_r = 0$ $\quad$ $p_l{}^T l_l = 0$

  $l_r = E p_l$ $\quad$ $l_l = E^T p_r$

- Epipoles: $e_r^T E = 0 \;\; E e_l = 0$



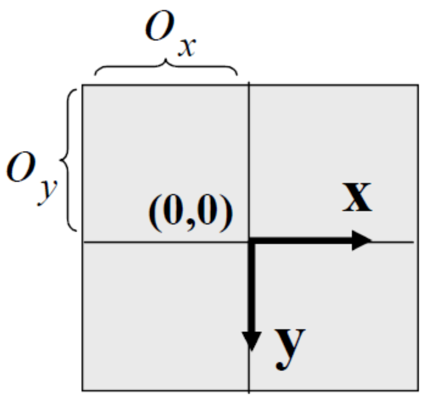
Pixel Coordinate

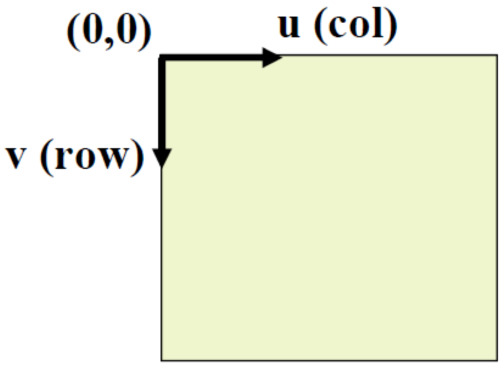
- So far we have mapped a point from a film/image coordinate system to a line in another film/image coordinate system.

- Everything has been continuous.

Pixel Coordinate

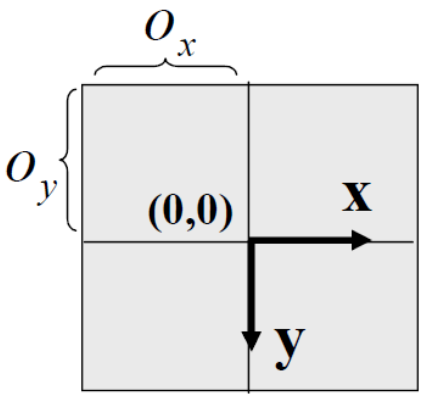
- What about pixel coordinates?

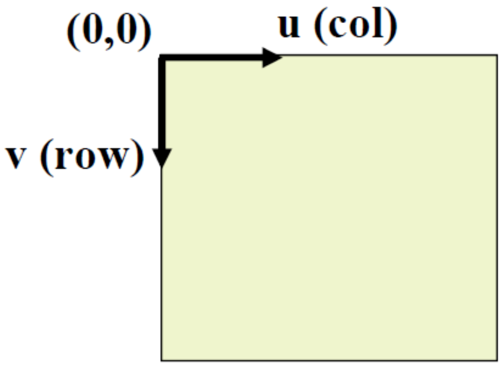




Pixel Coordinate

- What about pixel coordinates?
- We mentioned that there exist a matrix M mapping these two coordinates into each other.





Pixel Coordinate

- What about pixel coordinates?
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$$\bar{p}_l = M_l p_l$$

Pixel coordinates

Pixel Coordinate

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Pixel coordinates

Image coordinates

Pixel Coordinate

- What about pixel coordinates?
- We mentioned that there exist a matrix M mapping these two coordinates into each other.

$$\bar{p}_l = M_l p_l \qquad p_l = M_l^{-1} \bar{p}_l$$

Pixel coordinates

Image coordinates

Pixel Coordinate

- What about pixel coordinates?
- We mentioned that there exist a matrix M mapping these two coordinates into each other.

$$\bar{p}_l = M_l p_l \qquad p_l = M_l^{-1} \bar{p}_l$$

$$\bar{p}_r = M_r p_r \qquad p_r = M_r^{-1} \bar{p}_r$$

Pixel Coordinate

- We can map pixel coordinates having the following equations:

$$p_l = M_l^{-1}\bar{p}_l$$
$$p_r = M_r^{-1}\bar{p}_r$$

$$p_r{}^T E p_l = 0$$

Pixel Coordinate

- Plug in p_l and p_r:

$$p_l = M_l^{-1}\bar{p}_l \qquad {p_r}^T E p_l = 0$$
$$p_r = M_r^{-1}\bar{p}_r$$

$$\Rightarrow (M_r^{-1}\bar{p}_r)^T \, E(M_l^{-1}\bar{p}_l) = 0$$

Pixel Coordinate

- Simplify:

$$p_l = M_l^{-1}\bar{p}_l \qquad p_r{}^T E p_l = 0$$

$$p_r = M_r^{-1}\bar{p}_r$$

$$\Rightarrow (M_r^{-1}\bar{p}_r)^T \; E(M_l^{-1}\bar{p}_l) = 0$$

$$\Rightarrow \bar{p}_r{}^T \; (M_r^{-T} E M_l^{-1})\bar{p}_l = 0$$

Pixel Coordinate

- Form the Fundamental Matrix:

$$p_l = M_l^{-1}\bar{p}_l \qquad {p_r}^T E p_l = 0$$

$$p_r = M_r^{-1}\bar{p}_r$$

$$\Rightarrow (M_r^{-1}\bar{p}_r)^T\ E(M_l^{-1}\bar{p}_l) = 0$$

$$\Rightarrow {\bar{p}_r}^T\ (M_r^{-T} E M_l^{-1})\bar{p}_l = 0$$

$$\Rightarrow {\bar{p}_r}^T\ F\bar{p}_l = 0$$

Pixel Coordinate

- Fundamental matrix maps a pixel in one image to pixel in another image

$$F = M_r^{-T} E M_l^{-1}$$

Example

- Having a fundamental matrix, a point on the pixel coordinate of an image is mapped to a line in another image viewing the same point from another point of view.

Example

- Having a fundamental matrix, a point on the pixel coordinate of an image is mapped to a line in another image viewing the same point from another point of view.

$$F = \begin{bmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{bmatrix}$$

Example

- Having a fundamental matrix, a point on the pixel coordinate of an image is mapped to a line in another image viewing the same point from another point of view.

x = 343.5300 y = 221.7005

$$F = \begin{bmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{bmatrix}$$

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$$\begin{bmatrix} 0.0001 \\ 0.0045 \\ -1.1942 \end{bmatrix} \rightarrow \begin{bmatrix} 0.0295 \\ 0.9996 \\ -265.1531 \end{bmatrix}$$

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$$\begin{bmatrix} 0.0001 \\ 0.0045 \\ -1.1942 \end{bmatrix} \rightarrow \begin{bmatrix} 0.0295 \\ 0.9996 \\ -265.1531 \end{bmatrix}$$

Normalize sum to one for the first two elements

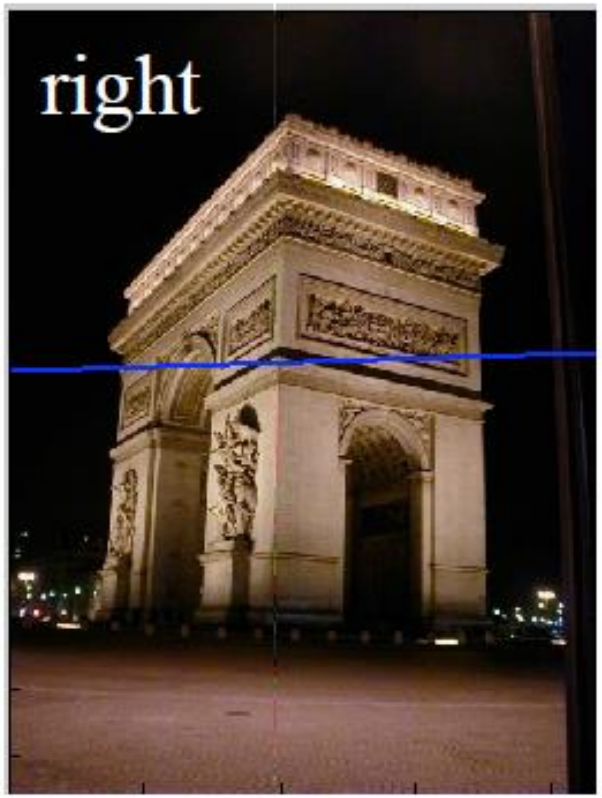
Example

- Having a fundamental matrix, a point on the pixel coordinate of an image is mapped to a line in another image viewing the same point from another point of view.

x = 343.5300 y = 221.7005

$$\begin{bmatrix} 0.0001 \\ 0.0045 \\ -1.1942 \end{bmatrix} \rightarrow \begin{bmatrix} 0.0295 \\ 0.9996 \\ -265.1531 \end{bmatrix}$$

0.0295
0.9996
-265.1531

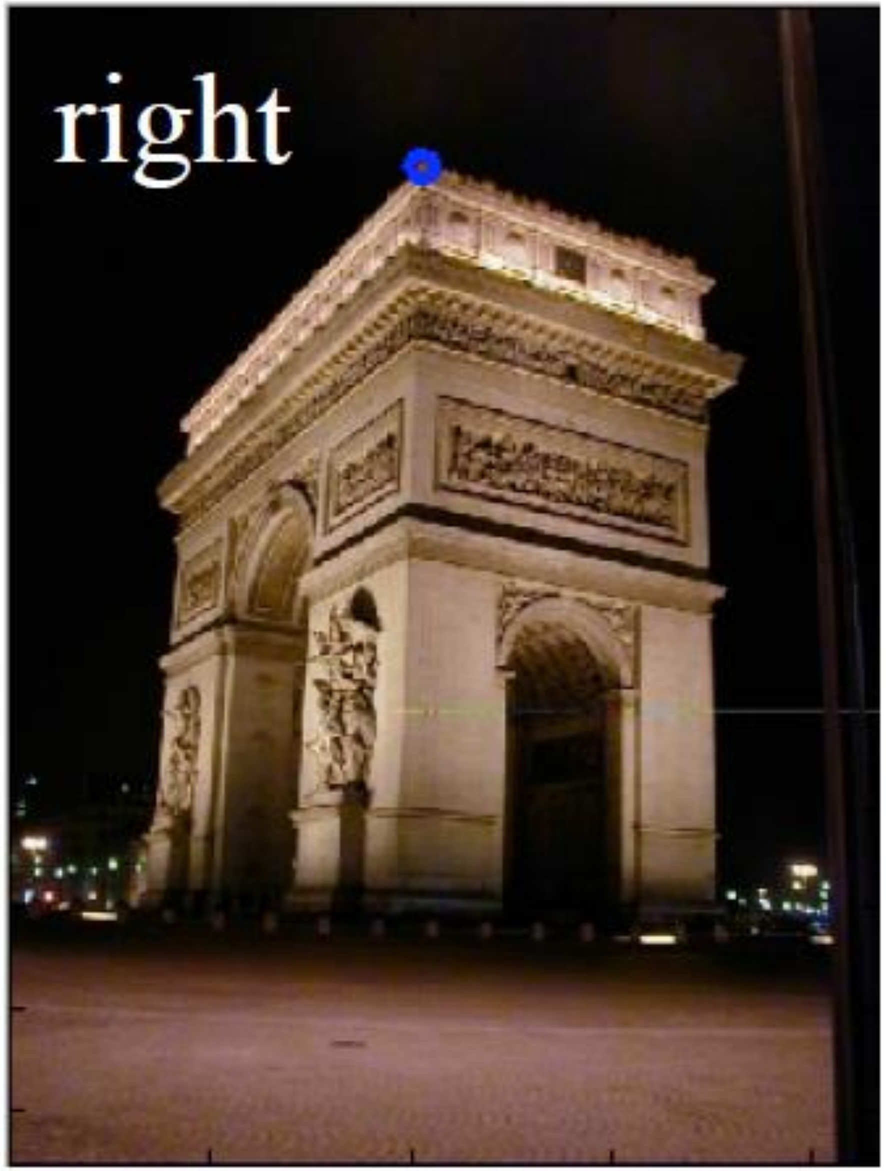


Example

- Right to left is similar

$$\boxed{\begin{bmatrix} 205.5526 & 80.5 & 1.0 \end{bmatrix}} \begin{bmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{bmatrix}$$

F



$\boxed{x = 205.5526 \quad y = 80.5000}$

Example

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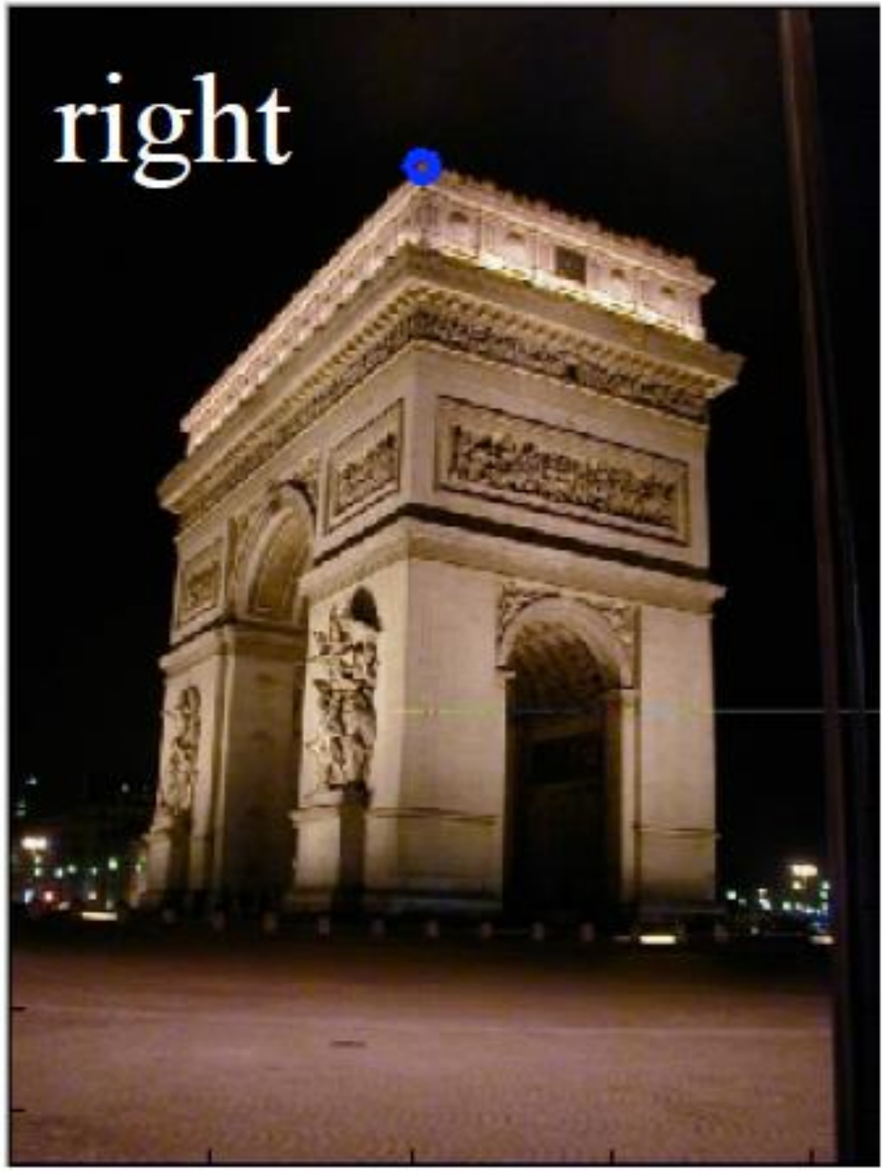
$$[205.5526 \quad 80.5 \quad 1.0]\begin{bmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{bmatrix}$$

$$= [0.0010 \quad -0.0030 \quad -0.4851]$$



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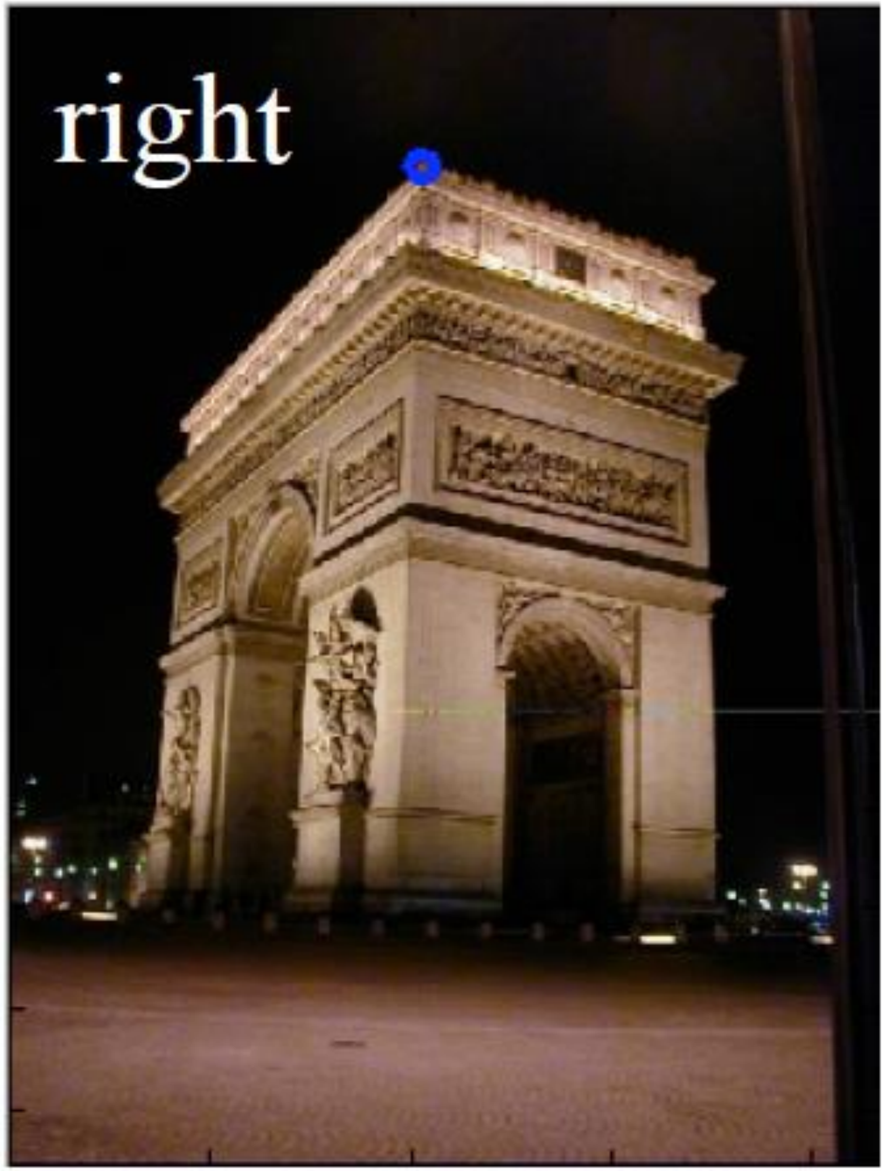
$$= [0.0010 \quad -0.0030 \quad -0.4851]$$

$$\rightarrow [0.3211 \quad -0.9470 \quad -151.39]$$

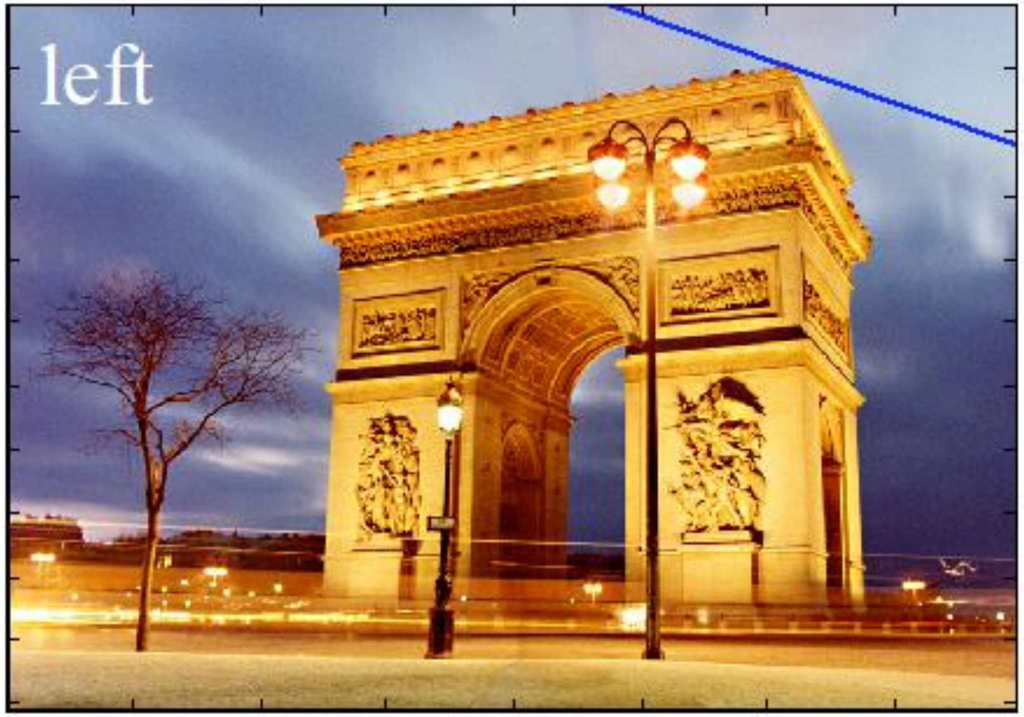


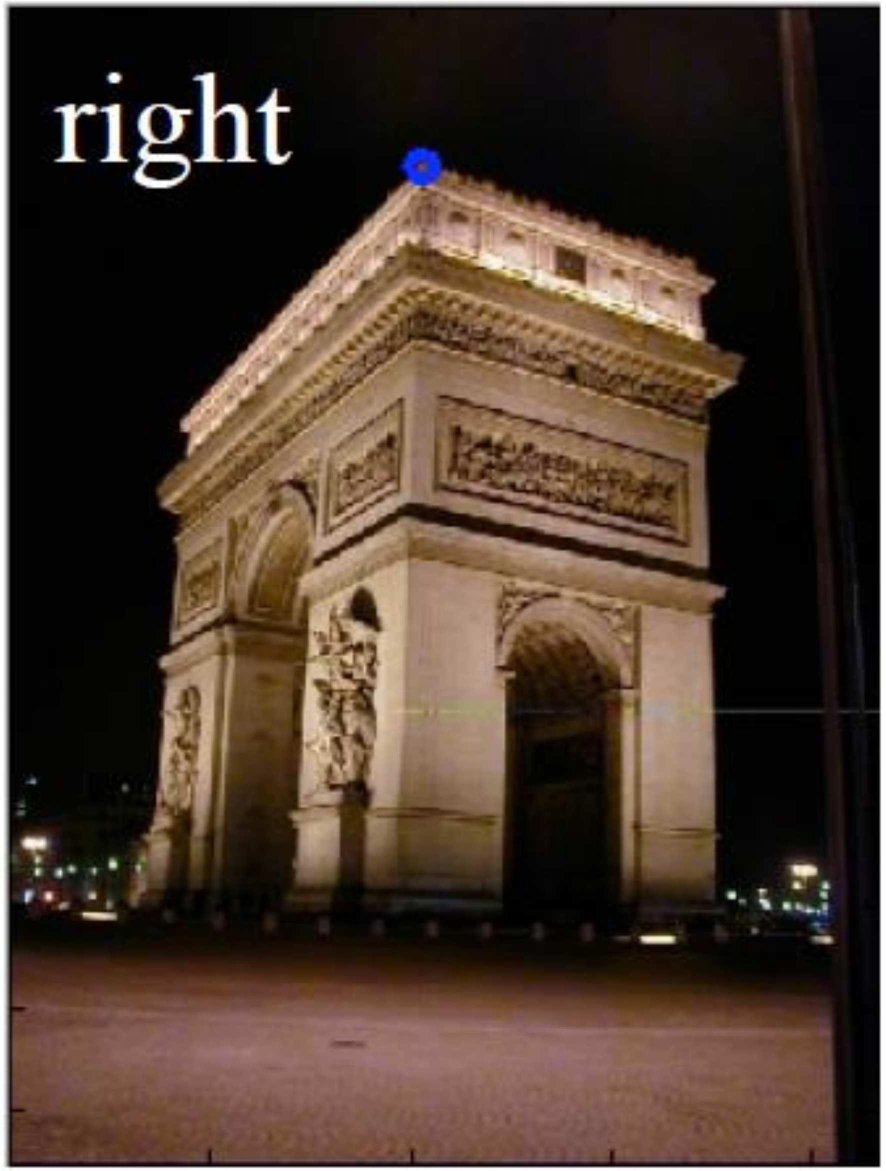
x = 205.5526 y = 80.5000

Example

- Right to left is similar

L= (0.3211   -0.9470 -151.39)





x =   205.5526   y = 80.5000

Fundamental Matrix

- In our example, fundamental matrix was given.

$$F = \begin{bmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{bmatrix}$$

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- The main question is how to find F.

How to compute F?

- Having two points in the pixel coordinates of left and right images:

$$\bar{p}_{li} = (x_i, y_i, 1)^T \quad \bar{p}_{ri} = (x'_i, y'_i, 1)^T$$

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$$\bar{p}_{li} = (x_i, y_i, 1)^T \quad \bar{p}_{ri} = ({x'}_i, {y'}_i, 1)^T$$

- We can connect them using fundamental matrix:

$$\bar{p}_{ri}{}^T F \bar{p}_{li} = 0$$

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$$[{x'}_i \quad {y'}_i \quad 1] \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = 0$$

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- We can open up the equation

$$x_i x'_i f_{11} + x_i y'_i f_{21} + x_i f_{31} + y_i x'_i f_{12} + y_i y'_i f_{22} + y_i f_{32} + x'_i f_{13} + y'_i f_{23} + f_{33} = 0$$

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$$\begin{bmatrix} x_1 x'_1 & x_1 y'_1 & x_1 & y_1 x'_1 & y_1 y'_1 & y_1 & x'_1 & y'_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_m x'_m & x_m y'_m & x_m & y_m x'_m & y_m y'_m & y_m & x'_m & y'_m & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{21} \\ f_{31} \\ f_{12} \\ f_{22} \\ f_{32} \\ f_{13} \\ f_{23} \\ f_{33} \end{bmatrix} = 0$$

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How many points do we need?

How to compute F?

- Having 8 corresponding points:

$$\begin{bmatrix} x_1x'_1 & x_1y'_1 & x_1 & y_1x'_1 & y_1y'_1 & y_1 & x'_1 & y'_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_8x'_8 & x_8y'_8 & x_8 & y_8x'_8 & y_8y'_8 & y_8 & x'_8 & y'_8 & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{21} \\ f_{31} \\ f_{12} \\ f_{22} \\ f_{32} \\ f_{13} \\ f_{23} \\ f_{33} \end{bmatrix} = 0$$

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- Solve $Ax = 0$

How to compute F?

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What should we do?

How to compute F?

- Solve $Ax = 0$, what is the trivial answer?
- To avoid $x = 0$, we consider constraint $\|x\|^2 = 1$.
- Use the Lagrangian multiplier:

$$\mathcal{L}(x) = x^T A^T A x - \lambda(x^T x - 1)$$

How to compute F?

- Solve $Ax = 0$, what is the trivial answer?

- To avoid $x = 0$, we consider constraint $\|x\|^2 = 1$.

- Use the Lagrangian multiplier:

$$\mathcal{L}(x) = x^T A^T A x - \lambda(x^T x - 1)$$

- We want its minimum

How to compute F?

- Use the Lagrangian multiplier:

$$\mathcal{L}(x) = x^T A^T A x - \lambda(x^T x - 1)$$

- We want its minimum

- Take the derivative

$$A^T A x - \lambda x = 0$$

$$A^T A x = \lambda x$$

What is λ what is x?

How to compute F?

- Use the Lagrangian multiplier:

$$\mathcal{L}(x) = x^T A^T A x - \lambda(x^T x - 1)$$

- We want its minimum

- Take the derivative

$$A^T A x - \lambda x = 0$$
$$A^T A x = \lambda x$$

What is λ what is x?

We want the eigen-vector with the smallest eigen-value since it is a minimization problem

How to compute F?

- Use SVD on matrix A to find the smallest eigen-vector of the matrix $A^T A$.

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- More details on the algorithm in tutorials.